PART 1 · LOCAL KUBERNETES

Every database, every monitoring tool, every scope

A recorded walkthrough of RunShore deploying PostgreSQL, MySQL, MongoDB and SAP ASE, registering them with PMM, New Relic and Splunk, and running failover, replication, workload and point-in-time-recovery tests: first on local Kubernetes, then on AWS, Azure and Google Cloud. Each chapter is shown in the order it was captured, with the log of every action.

VERSION	DATE	EVIDENCE	ORDER
1.0	November 2026	98 screenshots · 62 logs	Capture time, oldest first

CHAPTERS

CHAPTER	SCOPE	DATABASES	OBSERVED WITH	STEPS	SCREENS
01 · Deploy local Kubernetes	Local Kubernetes	—	—	1	3
02 · PostgreSQL	Local Kubernetes	PostgreSQL	New Relic, PMM, Splunk	17	18
03 · SAP ASE / Sybase	Local Kubernetes	SAP ASE	New Relic, Splunk	8	10
04 · MongoDB	Local Kubernetes	MongoDB	New Relic, PMM, Splunk	21	43
05 · MySQL	Local Kubernetes	MySQL	New Relic, PMM, Splunk	19	21
06 · Destroy local Kubernetes	Local Kubernetes	—	—	2	3

EACH STEP IS ONE OF

1. Deploy database
2. Deploy tool
3. Register
4. Failover
5. Replication
6. Workload
7. Assessment
8. Unregister
9. Destroy

Account numbers, subscription IDs, cluster addresses, private IP addresses and usernames have been removed from logs and blurred in screenshots. Warnings in a log are shown as they happened.

01 Deploy local Kubernetes

Local Kubernetes

RunShore starts by creating a local Kubernetes cluster (minikube) on the operator's own Mac. Everything in chapters 2 to 6 runs on this cluster.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Local Kubernetes	—	—	3 screens · 0 logs	—

1.1 Screenshots

What the console and dashboards showed at this point.

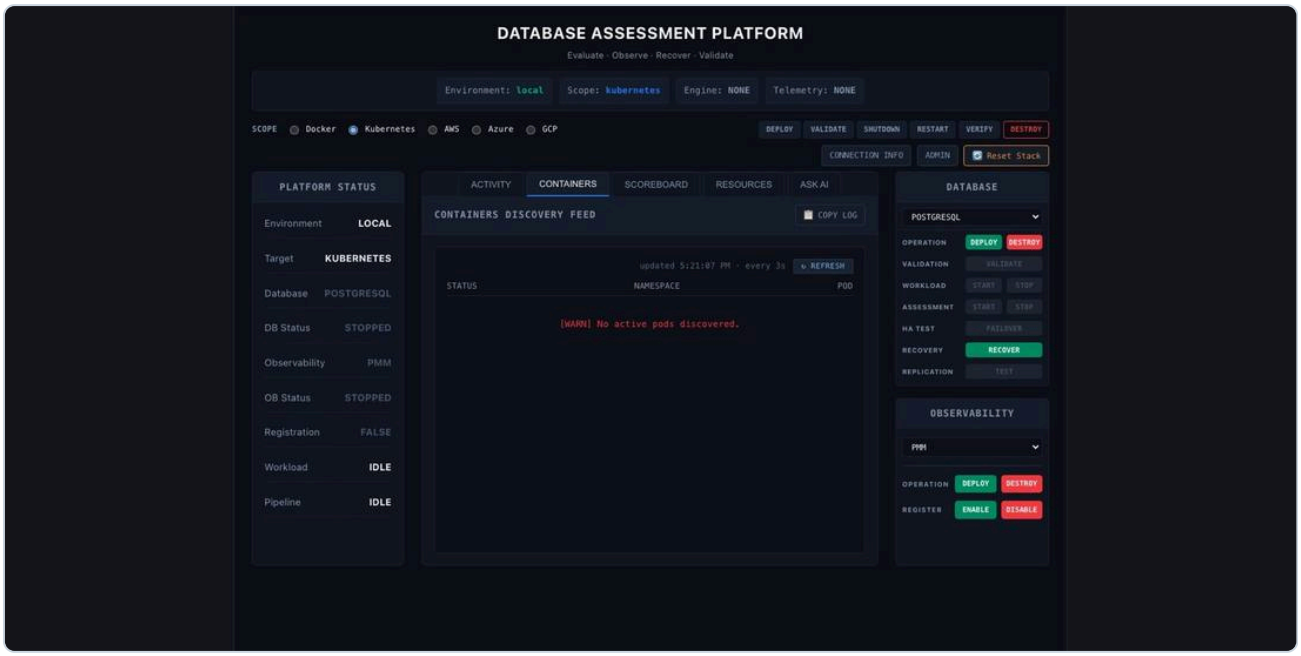


Figure 1.1 Home screen

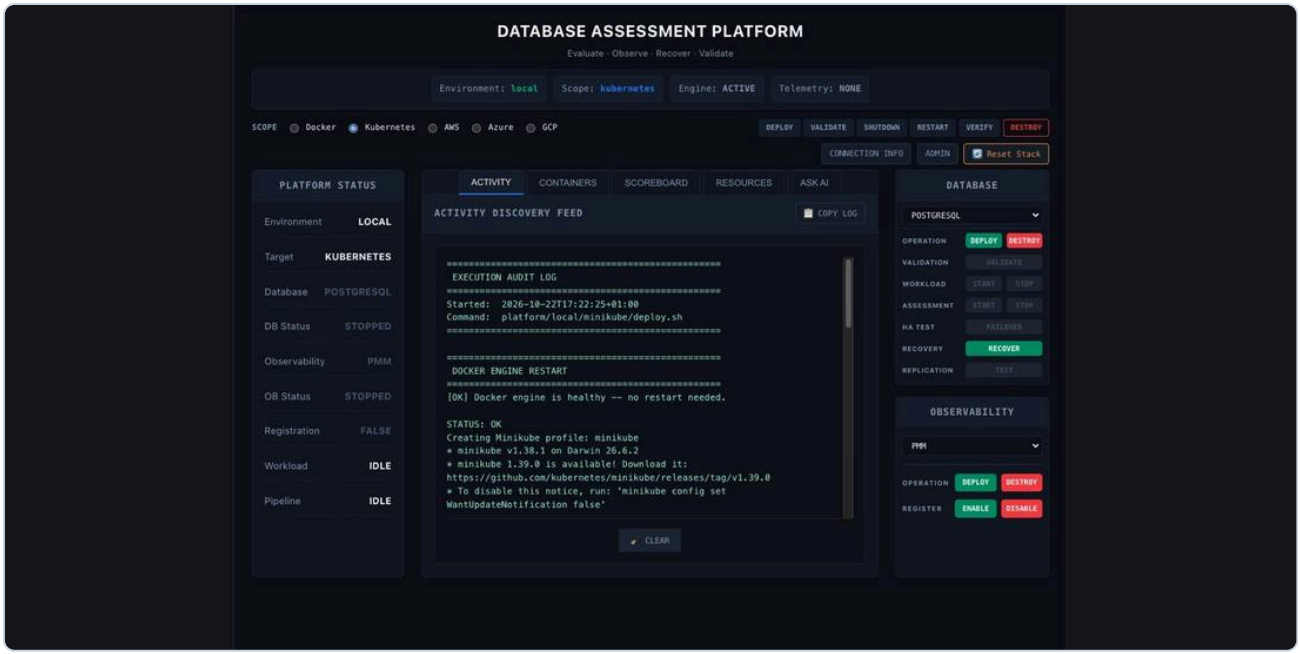


Figure 1.2 Deploy local k8s

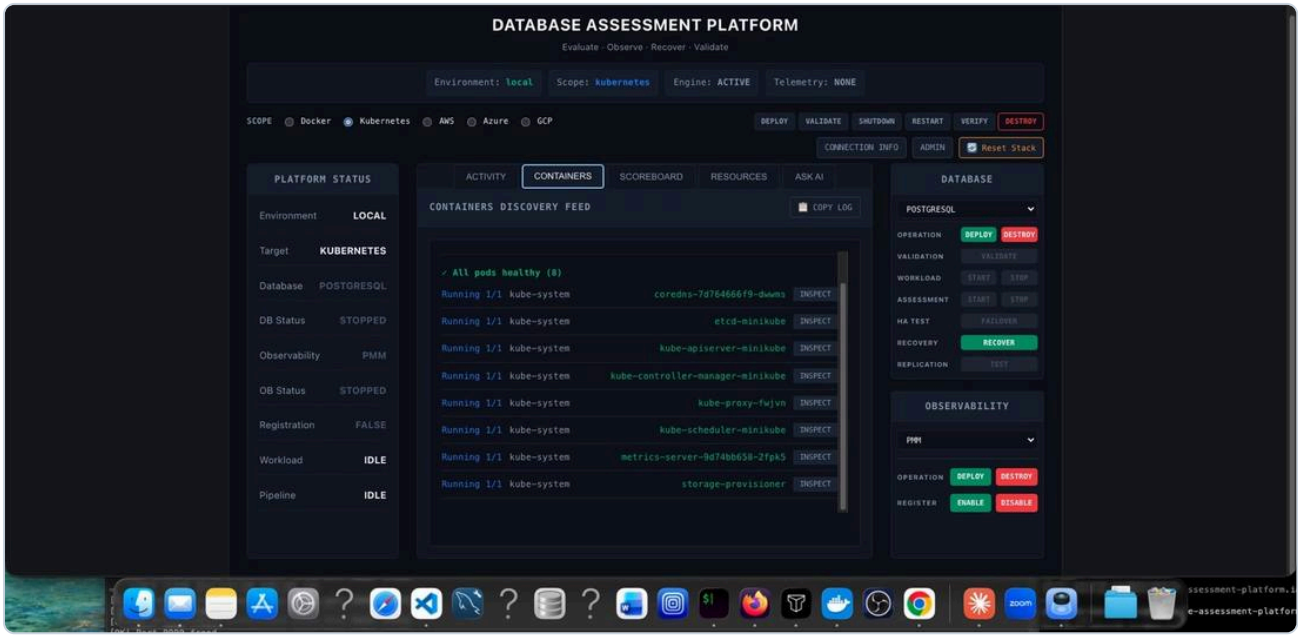


Figure 1.3 K8s pods created

02 PostgreSQL

Local Kubernetes

PostgreSQL is deployed with the CloudNativePG operator, monitored, put under load, and tested for failover, replication and point-in-time recovery.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Local Kubernetes	PostgreSQL	New Relic, PMM, Splunk	18 screens · 17 logs	26 Oct 2026

2.1 Deploy PostgreSQL

Deploy database | PostgreSQL | Completed

RunShore deploys PostgreSQL on Local Kubernetes and waits until every node is ready.

Started 26 Oct 2026, 13:07 · took 2m 25s

COMMAND

```
SCOPE=kubernetes db-platform/postgres/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes available
- ✔ MinIO image available
- ✔ MinIO Client image available
- ✔ Images loaded into Minikube
- ✔ Manifest rendered for local MinIO bucket: postgres-backups
- ✔ PostgreSQL platform deployed

2.2 Failover test · PostgreSQL

Failover | PostgreSQL | Completed

The primary PostgreSQL node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 26 Oct 2026, 13:10 · took 41s

COMMAND

```
SCOPE=kubernetes db-platform/postgres/failover/kubernetes/run_failover.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ Final CNPG phase: Cluster in healthy state
- ✔ Final ready instances: 3

2.3 Replication test · PostgreSQL

Replication | PostgreSQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 26 Oct 2026, 13:13 · took 12s

COMMAND

```
SCOPE=kubernetes db-platform/postgres/replication/kubernetes/run_replication.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.

2.4 Workload · PostgreSQL

Workload | PostgreSQL | Completed

A load generator runs against PostgreSQL, so the monitoring shows real traffic. It stops by itself after a set time.

Started 26 Oct 2026, 13:14 · took 8s

COMMAND

```
bash db-platform/postgres/workload/kubernetes/start.sh
```

- ✔ pgbench schema initialized.
- ✔ pgbench runtime permissions ensured for appuser.
- ✔ pgbench schema validated.

2.5 Install Splunk

Deploy tool | Splunk | PostgreSQL | Completed with warnings

Splunk is installed, ready to monitor the databases in this cluster.

Started 26 Oct 2026, 13:15 · took 4m 05s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available
- ✔ Splunk Operator manifest applied
- ✔ SPLUNK_GENERAL_TERMS configured on Operator
- ✔ No pods in a failed state

✔ Splunk accepted a HEC test event

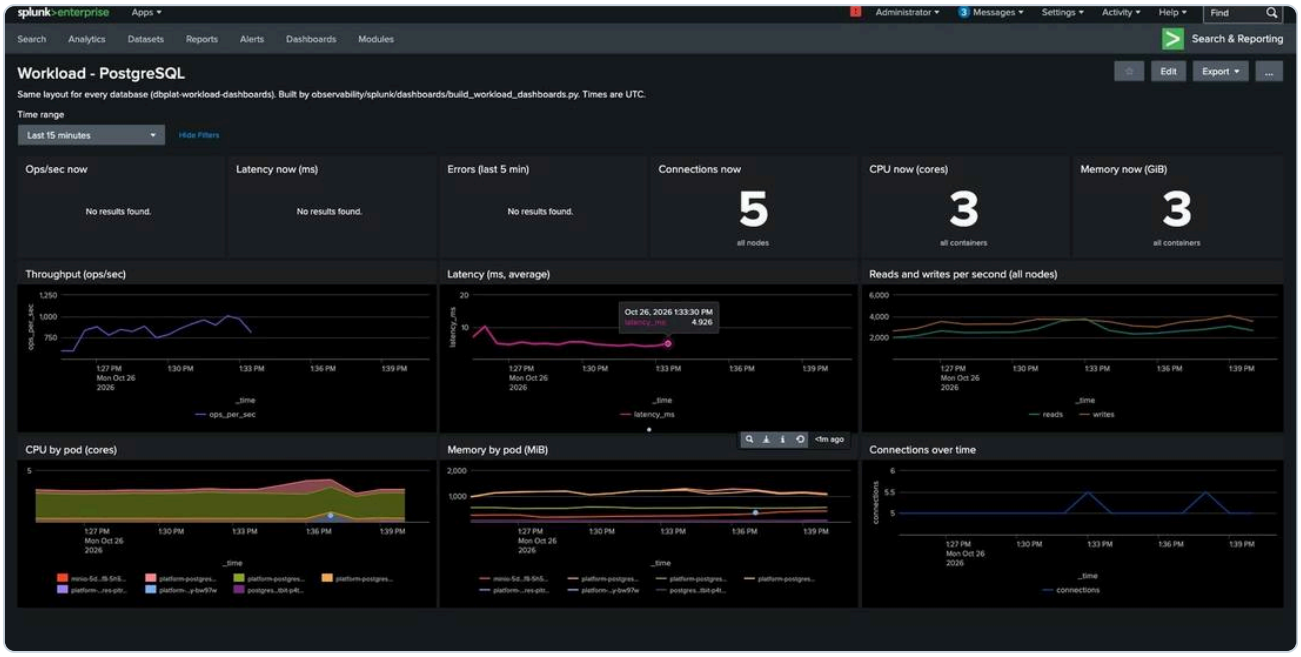


Figure 2.1 26 Oct 2026 · 13:40:49

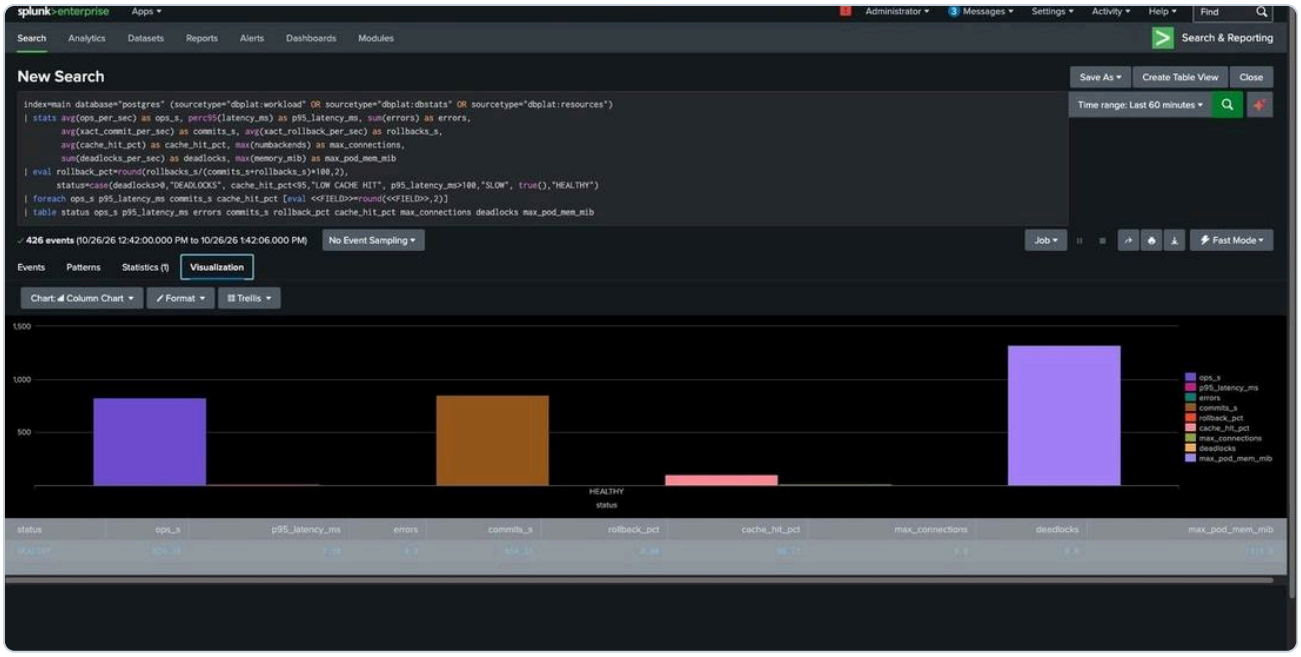


Figure 2.2 26 Oct 2026 · 13:42:23

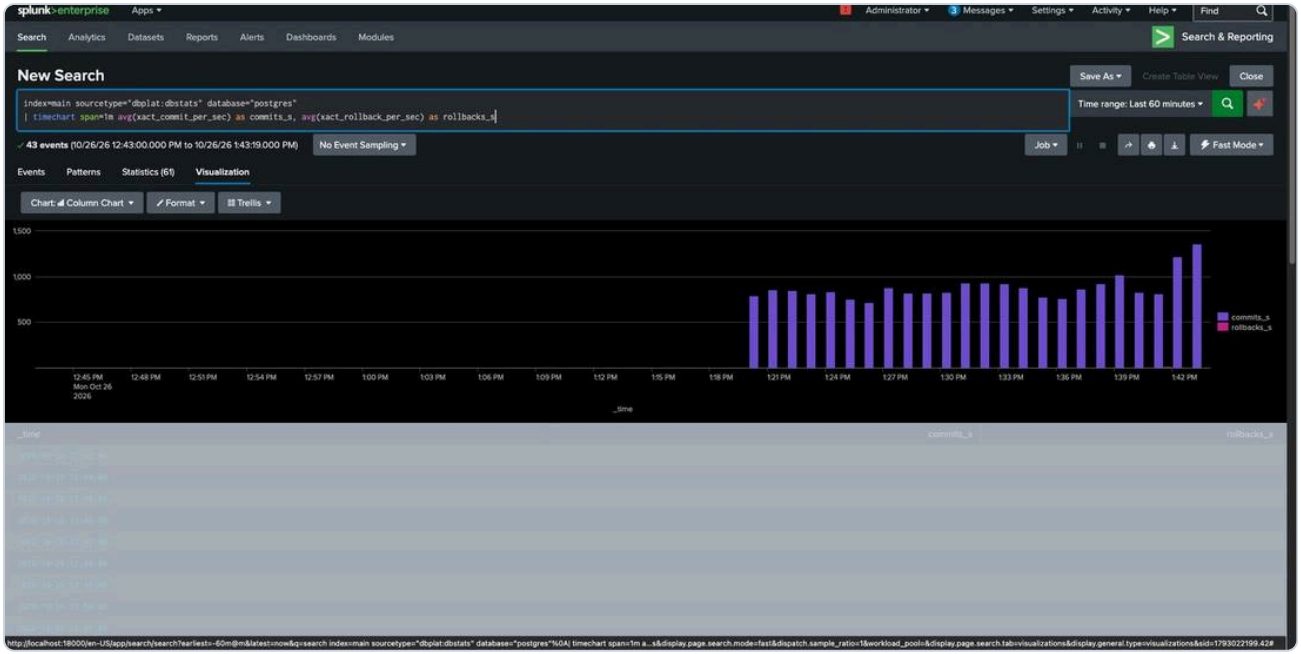


Figure 2.3 26 Oct 2026 · 13:43:41

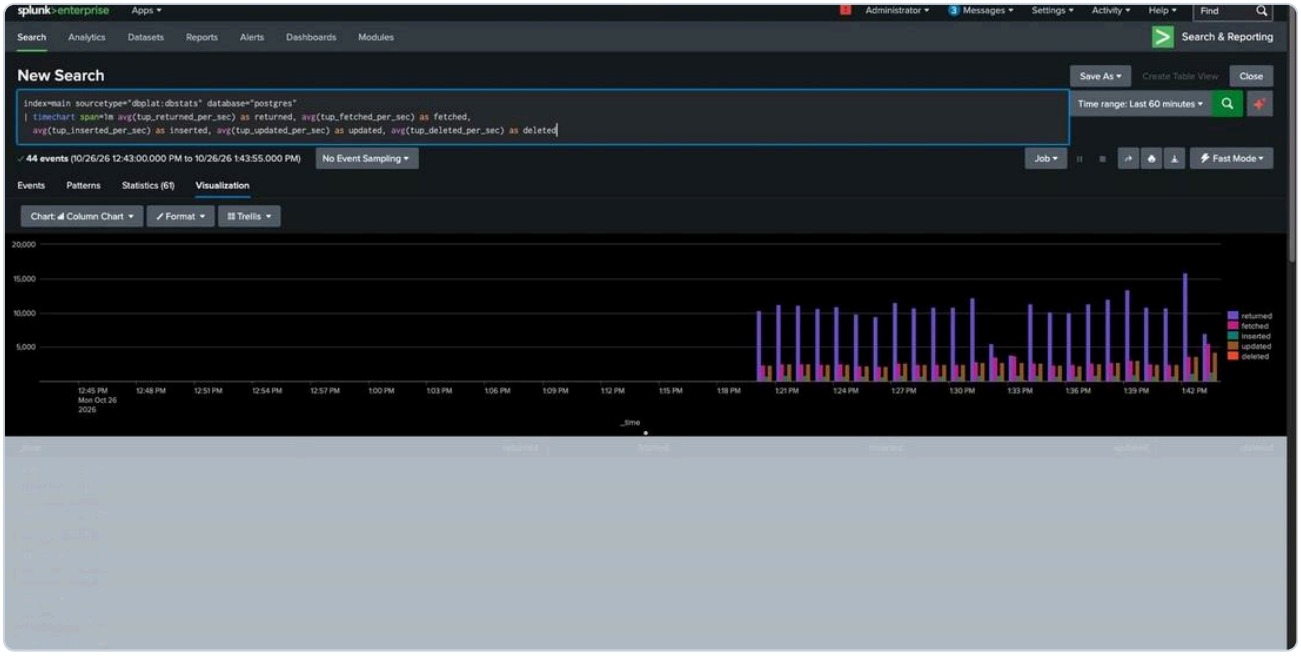


Figure 2.4 26 Oct 2026 · 13:44:07

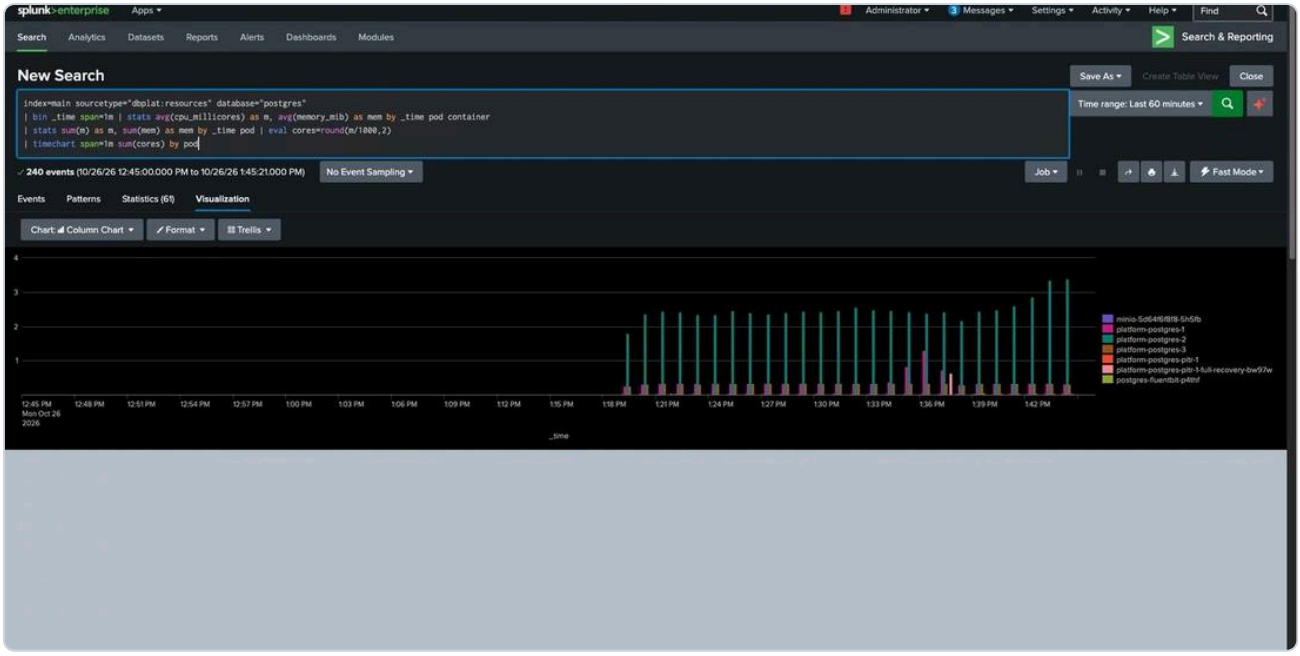


Figure 2.5 26 Oct 2026 · 13:45:27

2.6 Register PostgreSQL with Splunk

Register	Splunk	PostgreSQL	Completed with warnings

PostgreSQL is connected to Splunk, so its metrics and dashboards appear there.

Started 26 Oct 2026, 13:47 · took 12s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available
- ✔ Splunk Operator deployment already installed
- ✔ SPLUNK_GENERAL_TERMS configured on Operator
- ✔ No pods in a failed state
- ✔ Splunk accepted a HEC test event

2.7 Unregister PostgreSQL from Splunk

Unregister | Splunk | PostgreSQL | Completed

PostgreSQL is removed from Splunk; the database itself keeps running.

Started 26 Oct 2026, 13:46 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/postgres/unregister.sh
```

- ✔ Registration state cleared
- ✔ Postgres Splunk log forwarding removed

2.8 Remove Splunk

Unregister | Splunk | PostgreSQL | Completed

Splunk is removed from the cluster once no database is registered with it.

Started 26 Oct 2026, 13:49 · took 1m 17s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/destroy.sh
```

- ✔ Managed Splunk Web port-forward stopped
- ✔ Splunk Web port 18000 released
- ✔ Workload feed to Splunk stopped
- ✔ Kubernetes API reachable
- ✔ Splunk resources removed
- ✔ Splunk Web access removed

2.9 Install PMM

Deploy tool | PMM | PostgreSQL | Completed

PMM is installed, ready to monitor the databases in this cluster.

Started 26 Oct 2026, 14:02 · took 1m 05s

COMMAND

```
SCOPE=kubernetes observability/pmm/deploy/kubernetes/deploy_pmm_stack.kubernetes.sh
```

- ✔ Kubernetes API reachable
- ✔ Required lifecycle scripts found.
- ✔ Kubernetes API reachable
- ✔ Namespace pmm is Active
- ✔ The PMM token in ADMIN is accepted -- kept
- ✔ PMM server accepts the token from ADMIN

2.10 Register PostgreSQL with PMM

Register	PMM	PostgreSQL	Completed

PostgreSQL is connected to PMM, so its metrics and dashboards appear there.

Started 26 Oct 2026, 14:05 · took 24s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/register_cnpg_service.sh
```

- ✔ Credentials loaded
- ✔ pmm-0
- ✔ PostgreSQL successfully registered into PMM

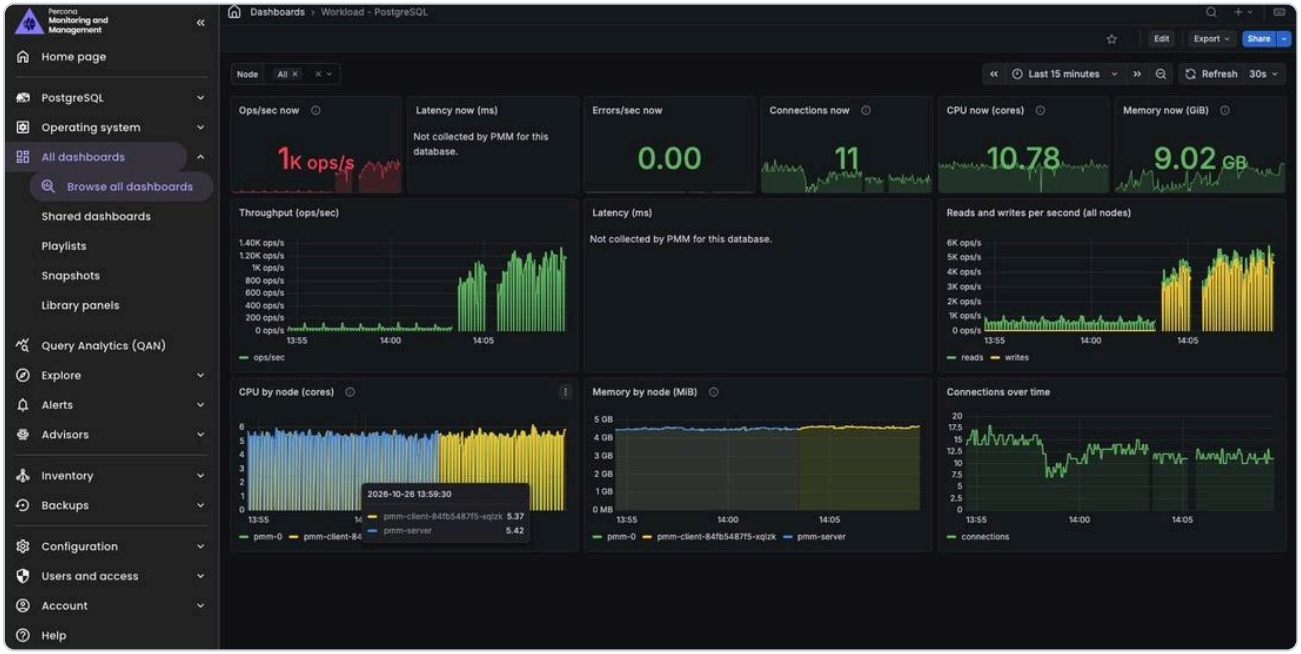


Figure 2.6 26 Oct 2026 · 14:09:36

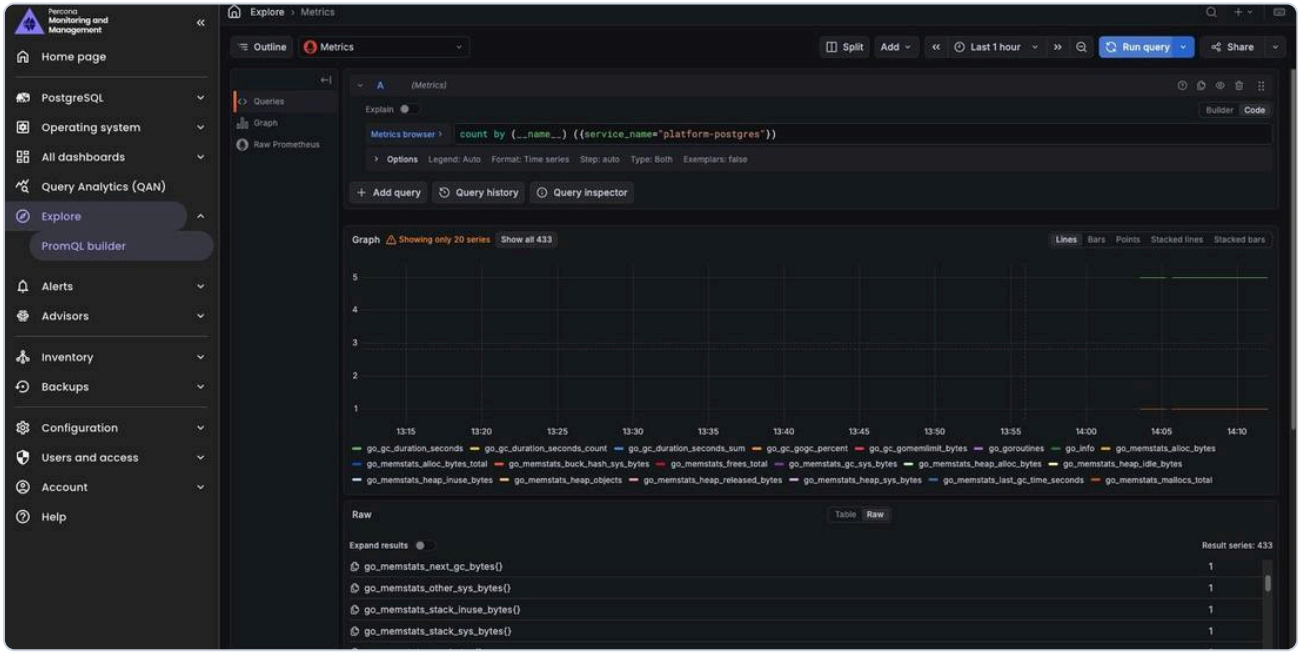


Figure 2.7 26 Oct 2026 · 14:12:20

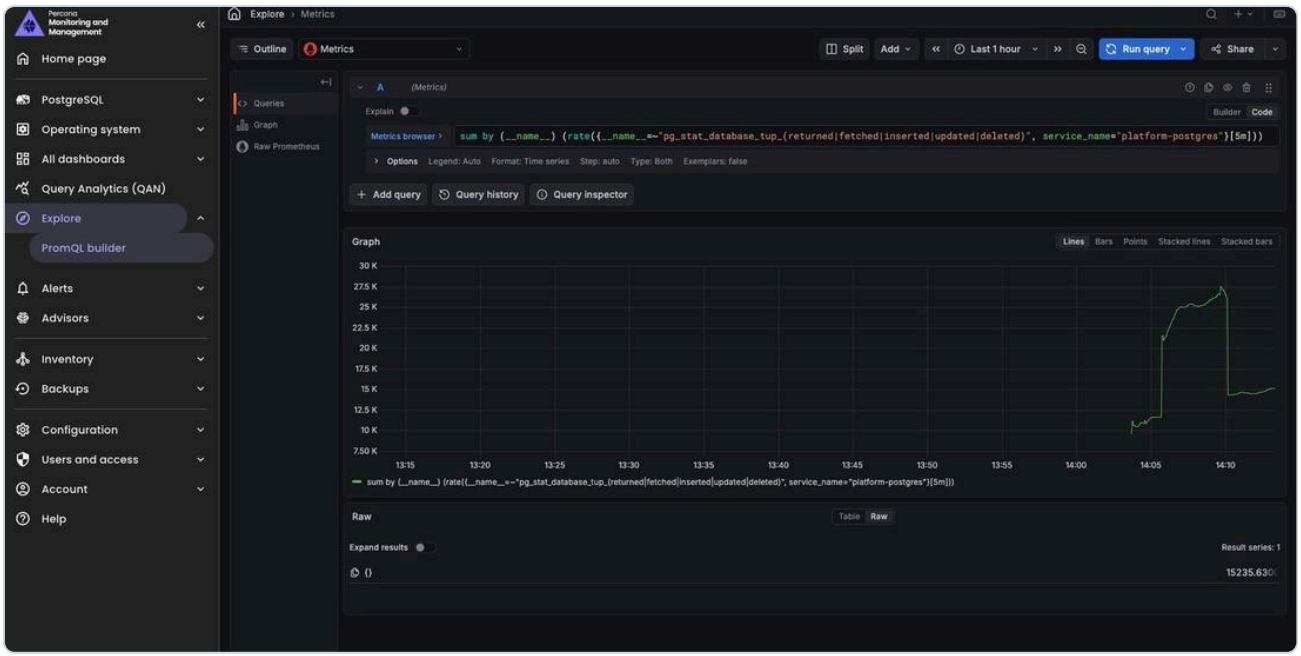


Figure 2.8 26 Oct 2026 · 14:13:55

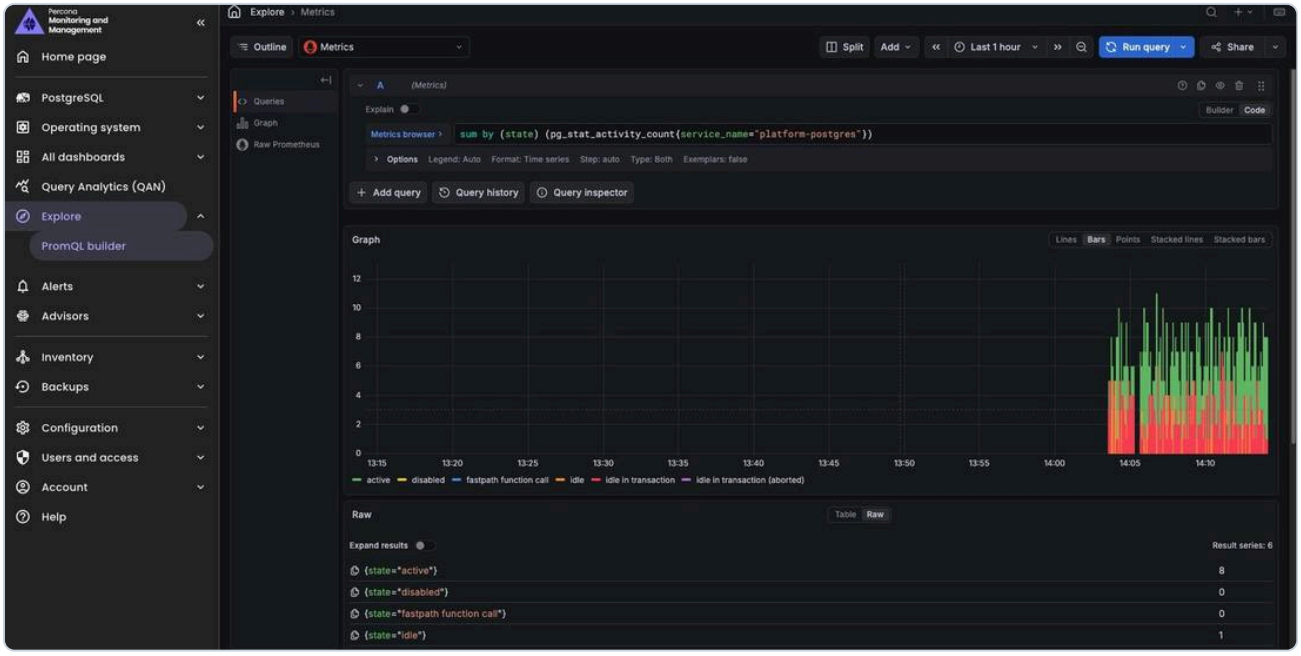


Figure 2.9 26 Oct 2026 · 14:14:41

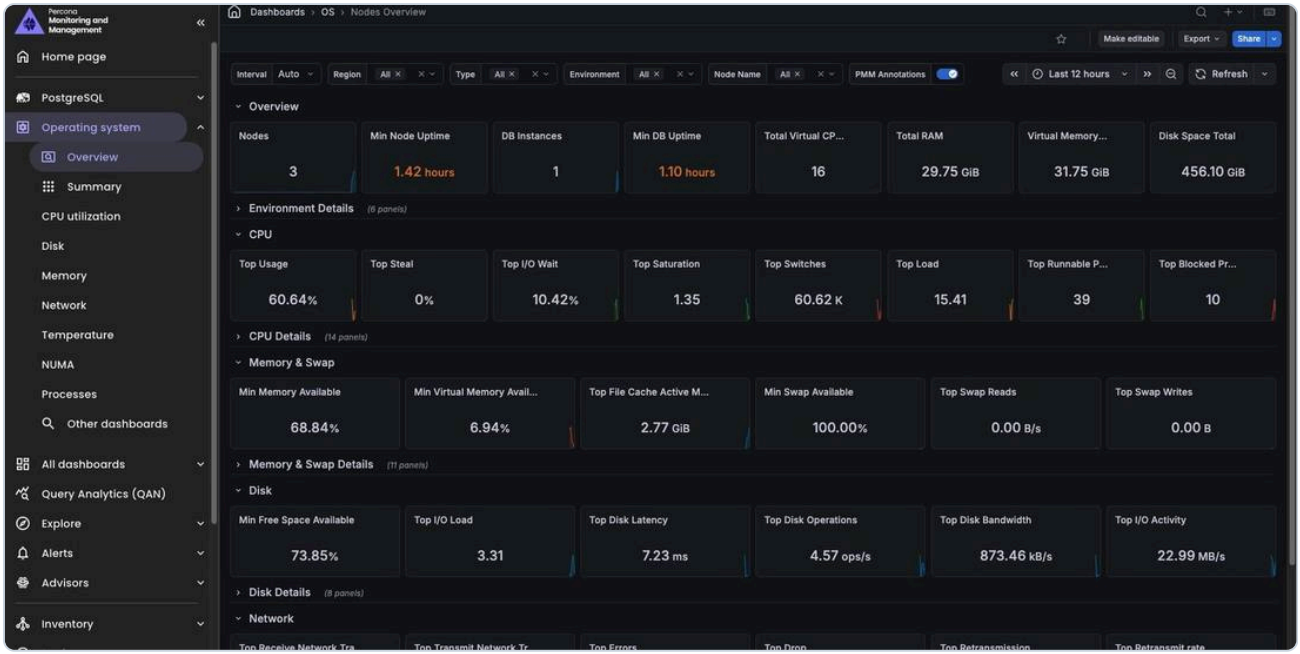


Figure 2.10 26 Oct 2026 · 14:15:41

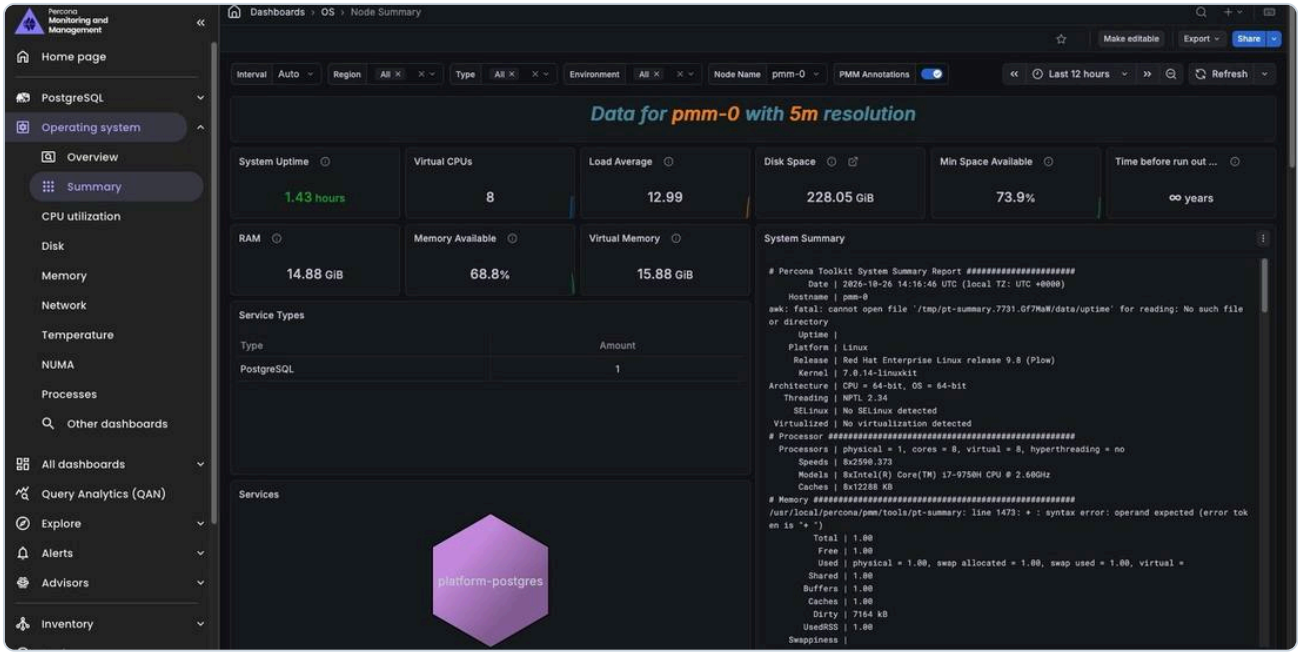


Figure 2.11 26 Oct 2026 · 14:17:09

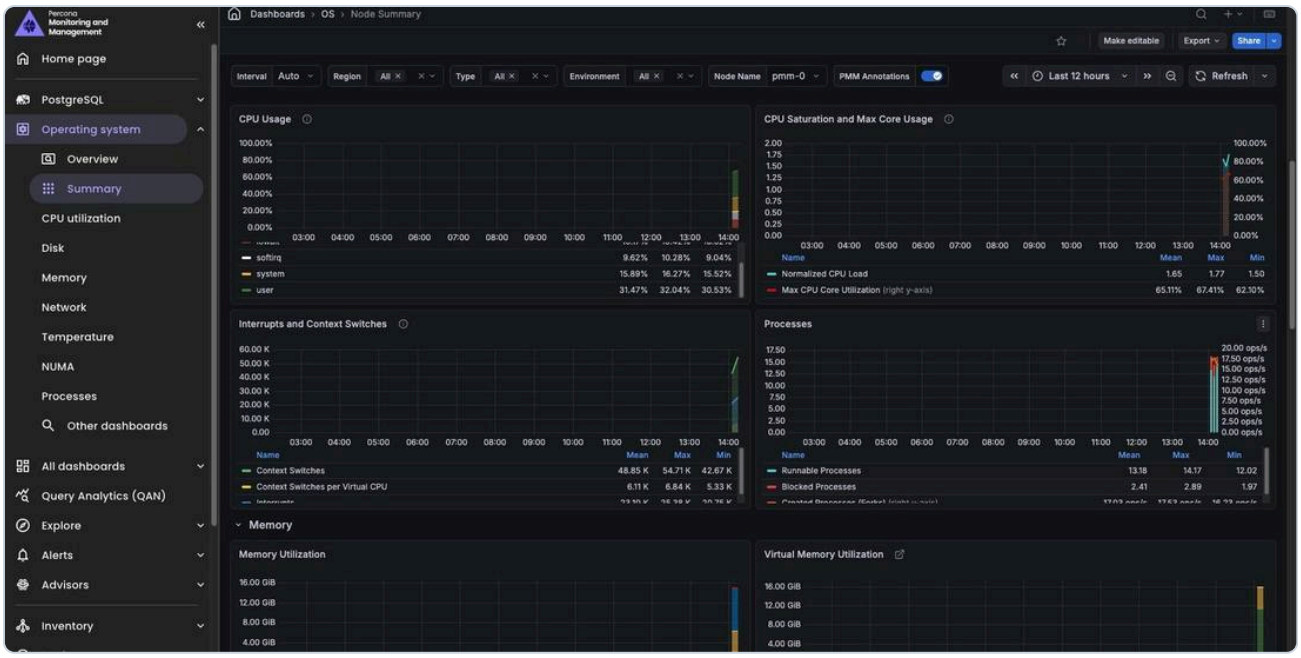


Figure 2.12 26 Oct 2026 · 14:17:17

2.11 Point-in-time recovery test · PostgreSQL

Assessment | PostgreSQL | Completed

RunShore restores PostgreSQL to a chosen moment from its backups and checks the restored data is correct.

Started 26 Oct 2026, 14:10 · took 10m 24s

COMMAND

```
SCOPE=kubernetes db-platform/postgres/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ PITR clone submitted
- ✔ chmod 777 applied to /tmp/hostpath-provisioner/postgres/platform-postgres-pitr-1

2.12 Unregister PostgreSQL from PMM

Unregister	PMM	PostgreSQL	Completed

PostgreSQL is removed from PMM; the database itself keeps running.

Started 26 Oct 2026, 14:21 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE

2.13 Remove PMM

Unregister	PMM	PostgreSQL	Completed

PMM is removed from the cluster once no database is registered with it.

Started 26 Oct 2026, 14:22 · took 24s

COMMAND

```
SCOPE=kubernetes observability/pmm/deploy/kubernetes/destroy.sh
```

- ✔ PMM local web access stopped.
- ✔ PMM local web access stopped.
- ✔ PMM Client deployment removed
- ✔ PMM namespace absent
- ✔ PMM Helm release absent

2.14 Install New Relic

Deploy tool	New Relic	PostgreSQL	Completed with warnings

New Relic is installed, ready to monitor the databases in this cluster.

Started 26 Oct 2026, 14:23 · took 1m 05s

COMMAND

```
SCOPE=kubernetes observability/newrelic/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available
- ✔ Cluster identity: minikube
- ✔ Namespace available: newrelic
- ! Dashboard Workload - Sybase ASE not installed: authentication required
- ✔ Dashboard links for CONNECTION INFO: runtime/state/newrelic/dashboards.json

2.15 Register PostgreSQL with New Relic

Register | New Relic | PostgreSQL | Completed

PostgreSQL is connected to New Relic, so its metrics and dashboards appear there.

Started 26 Oct 2026, 14:25 · took 13s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/register.sh
```

- ✔ Kubernetes, PostgreSQL and New Relic available
- ✔ Prerequisites available
- ✔ Monitoring secret available: postgres-monitoring
- ✔ Monitoring credentials loaded
- ✔ New Relic agents refreshed
- ✔ nri-postgresql configuration present

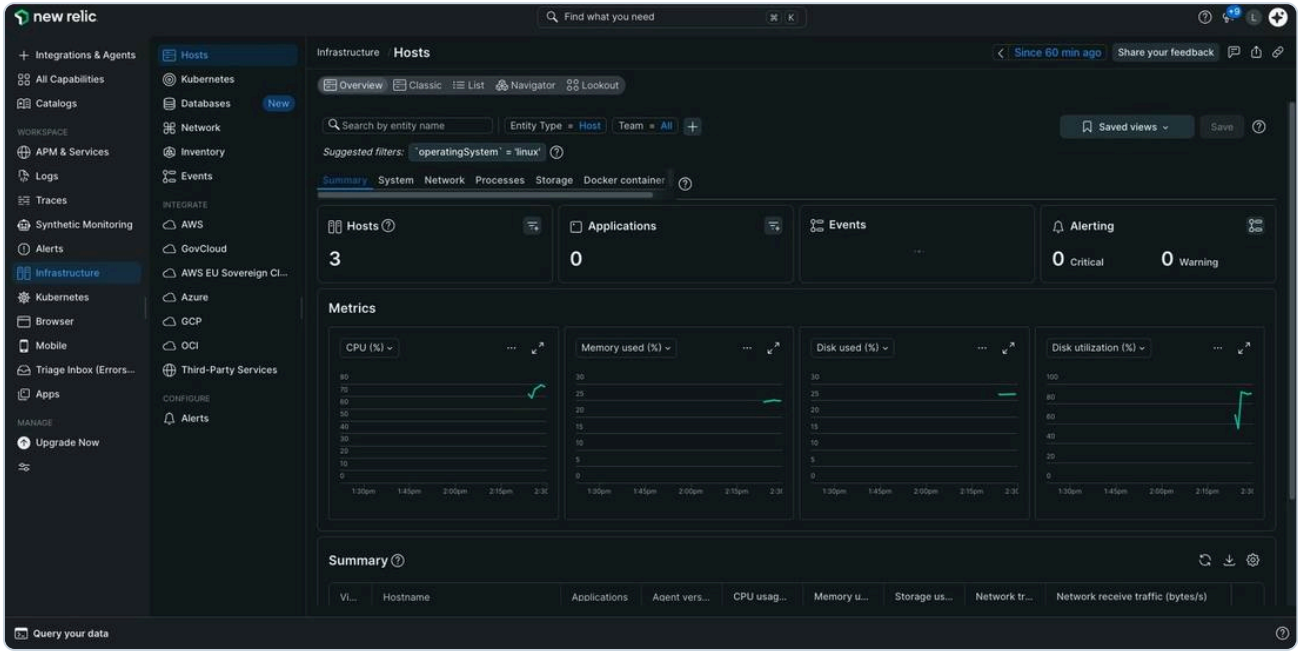


Figure 2.13 26 Oct 2026 · 14:31:30



Figure 2.14 26 Oct 2026 · 14:37:17

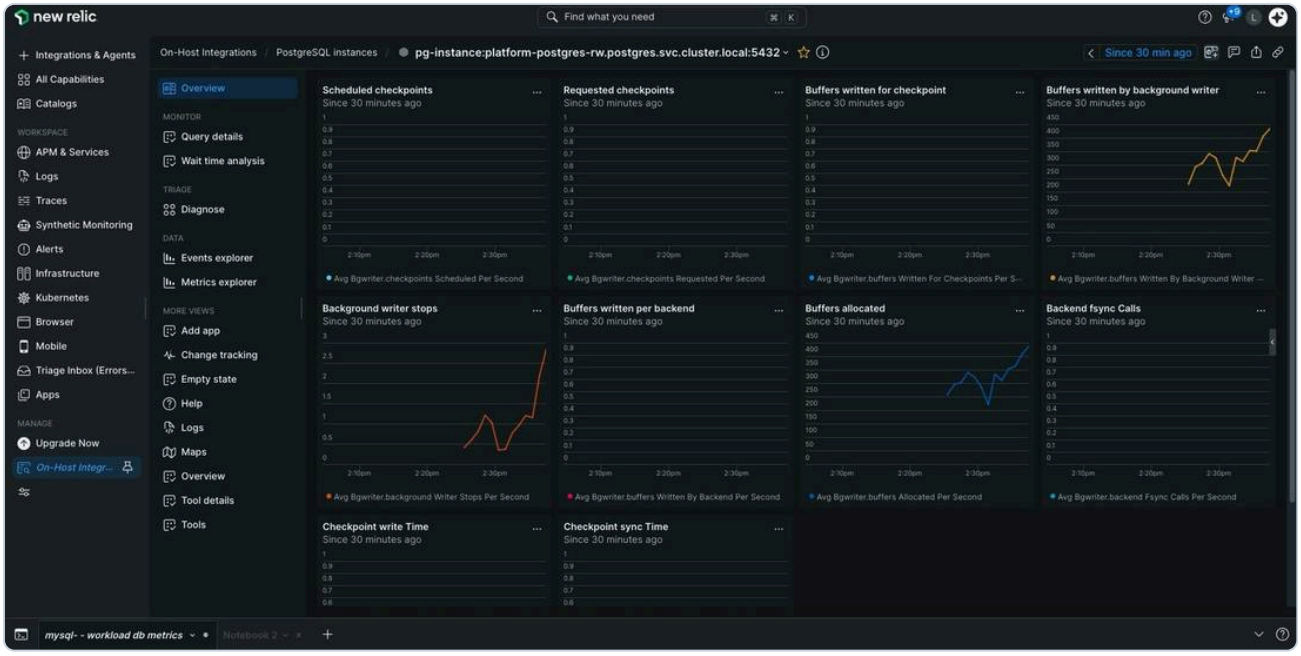


Figure 2.15 26 Oct 2026 · 14:38:44

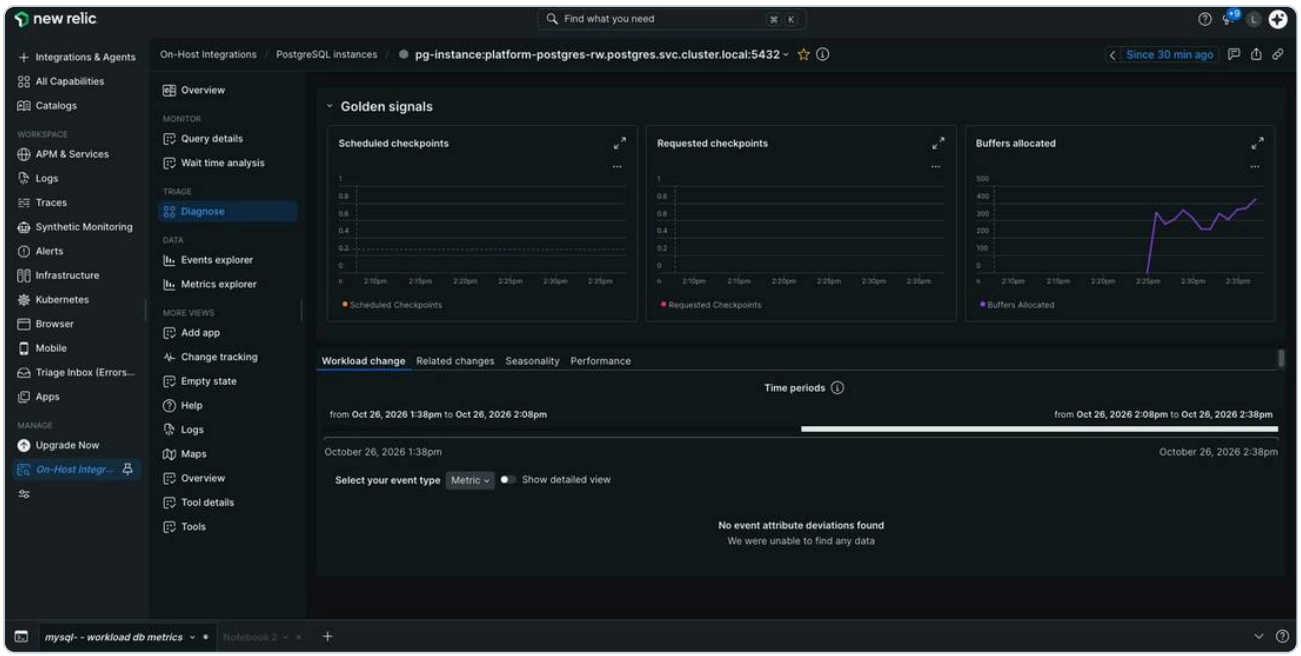


Figure 2.16 26 Oct 2026 · 14:39:07

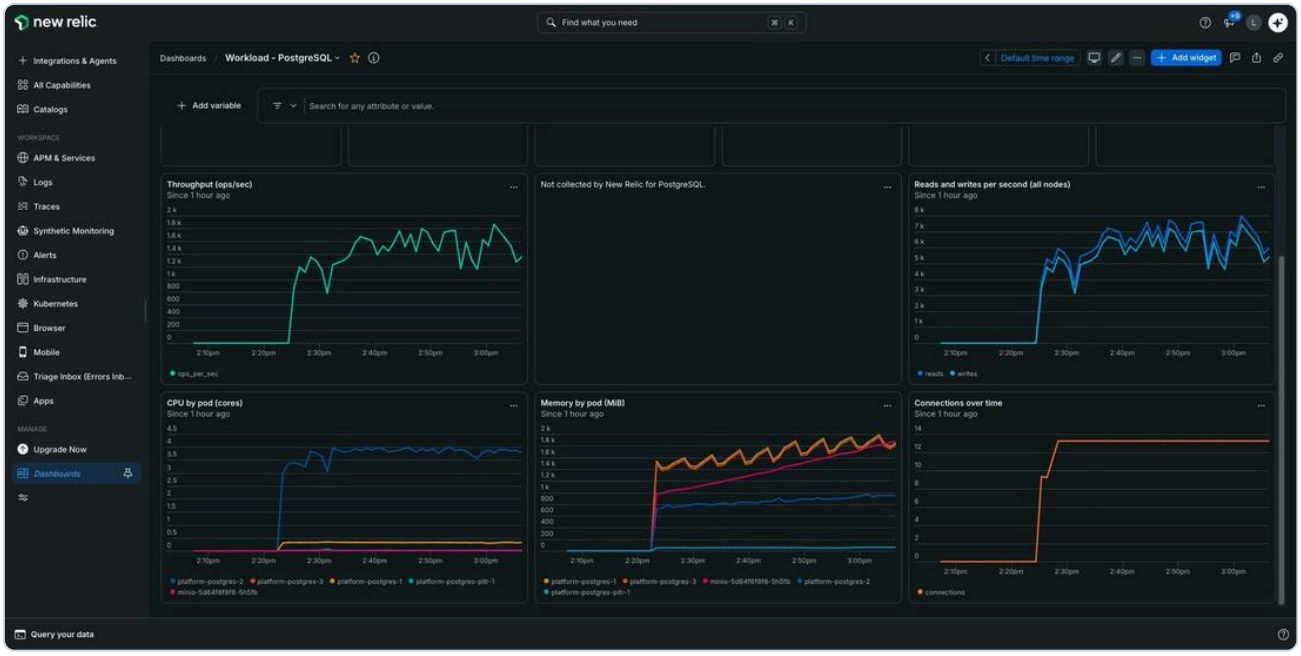


Figure 2.17 26 Oct 2026 · 15:07:36

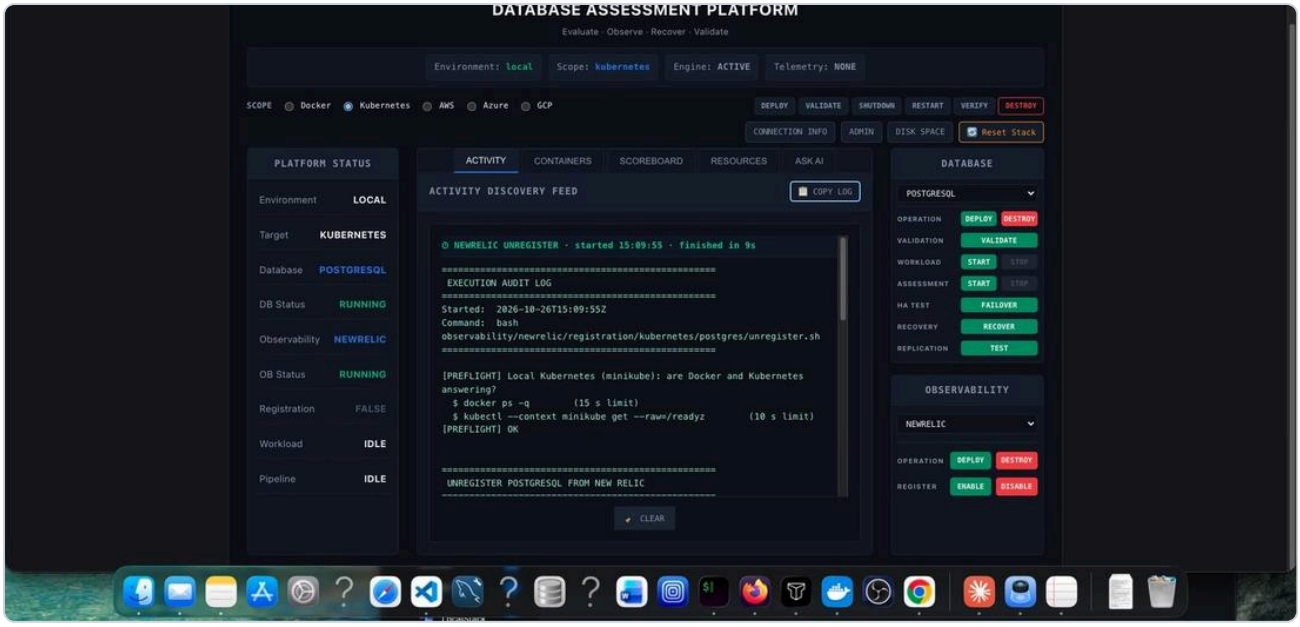


Figure 2.18 26 Oct 2026 · 15:10:26

2.16 Unregister PostgreSQL from New Relic

Unregister | New Relic | PostgreSQL | Completed

PostgreSQL is removed from New Relic; the database itself keeps running.

Started 26 Oct 2026, 15:09 · took 9s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/unregister.sh
```

- ✔ Kubernetes API reachable
- ✔ PostgreSQL integration configuration absent
- ✔ New Relic PostgreSQL telemetry secret absent
- ✔ New Relic agents refreshed
- ✔ postgres-config.yml absent
- ✔ newrelic-postgres-secret absent

2.17 Destroy PostgreSQL

Destroy | PostgreSQL | Completed

PostgreSQL is removed and its storage released.

Started 26 Oct 2026, 15:11 · took 1m 04s

COMMAND

```
SCOPE=kubernetes db-platform/postgres/deploy/kubernetes/destroy.sh
```

- ✔ PostgreSQL Kubernetes workload stopped
- ✔ Kubernetes API reachable
- ✔ All PostgreSQL clusters removed
- ✔ PostgreSQL pods stopped (25s)

- ✔ CloudNativePG operator removed
- ✔ PostgreSQL namespace removed

03 SAP ASE / Sybase

Local Kubernetes

SAP ASE (Sybase) runs as a single instance from an image built from the client's own SAP installer. It is monitored and its point-in-time recovery is tested.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Local Kubernetes	SAP ASE	New Relic, Splunk	10 screens · 7 logs	22 Oct–25 Oct 2026

3.1 Screenshots

What the console and dashboards showed at this point.

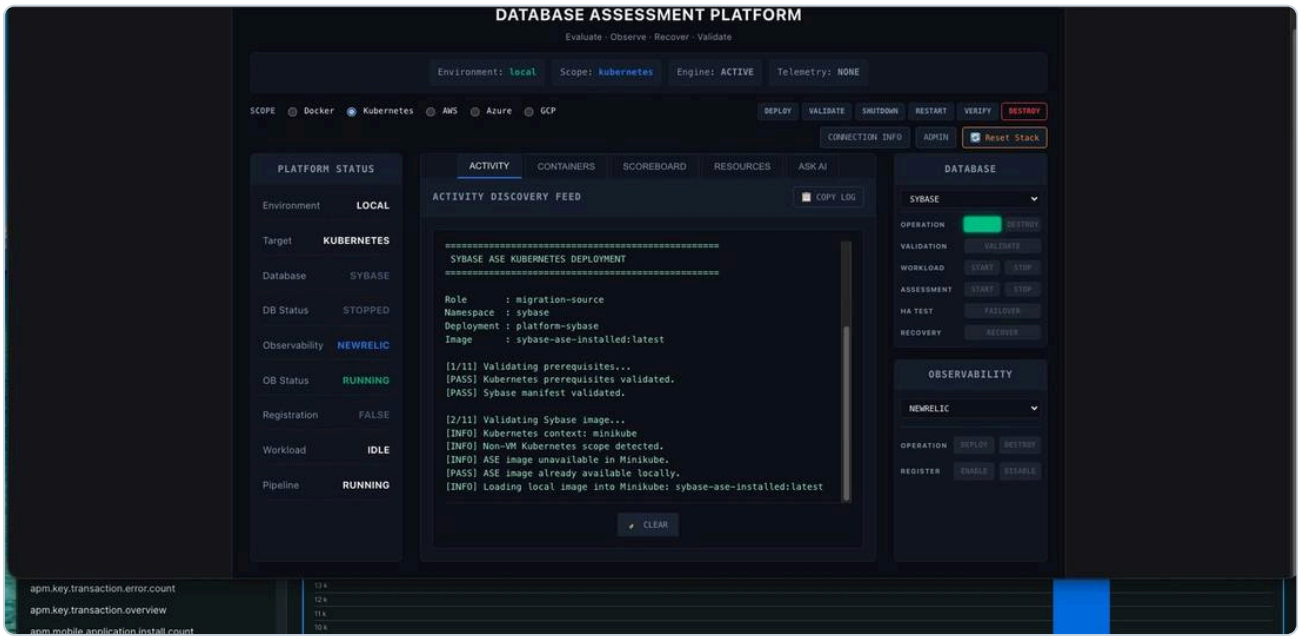


Figure 3.1 22 Oct 2026 · 18:42:29

3.2 Register SAP ASE with New Relic

Register	New Relic	SAP ASE	Completed

SAP ASE is connected to New Relic, so its metrics and dashboards appear there.

Started 22 Oct 2026, 18:59

COMMAND

```
bash observability/newrelic/registration/kubernetes/sybase/register.sh
```

✔ New Relic licence key accepted

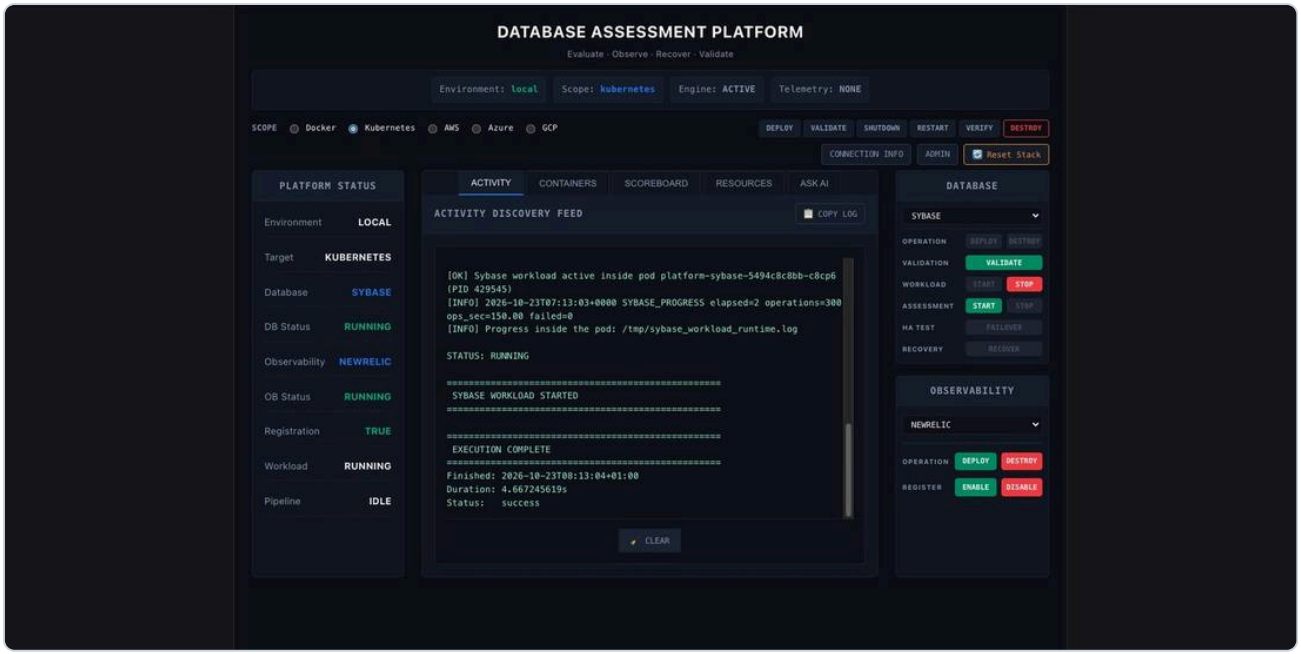


Figure 3.2 23 Oct 2026 · 08:14:23

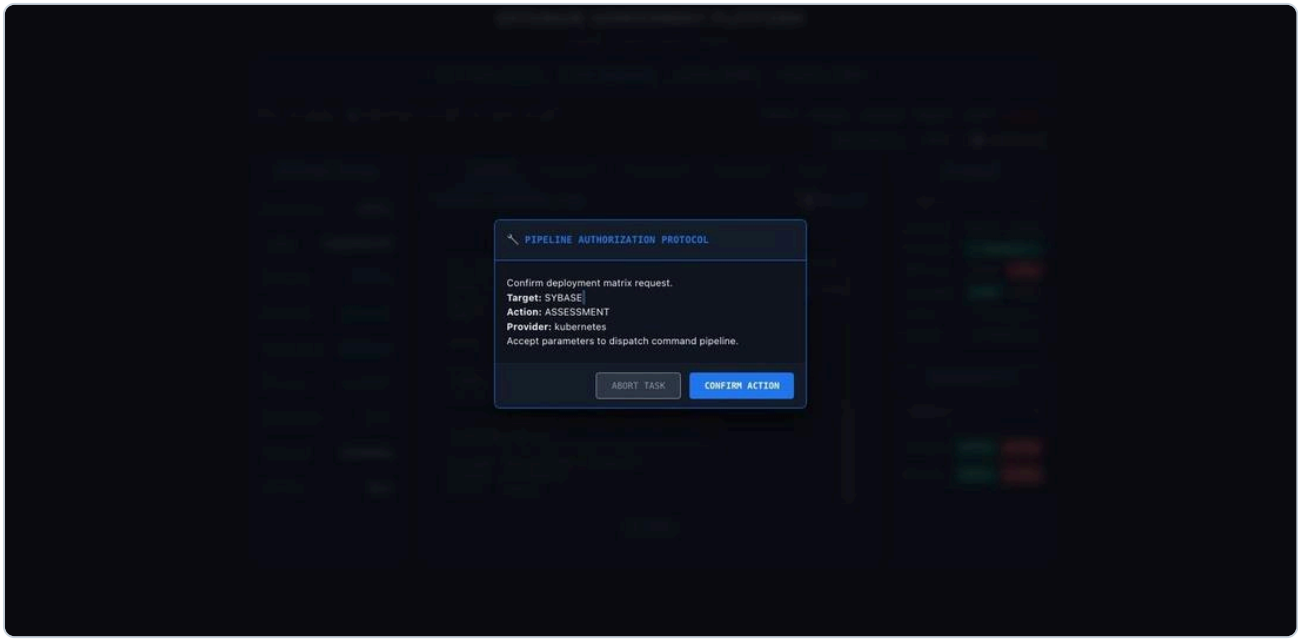


Figure 3.3 23 Oct 2026 · 08:16:27

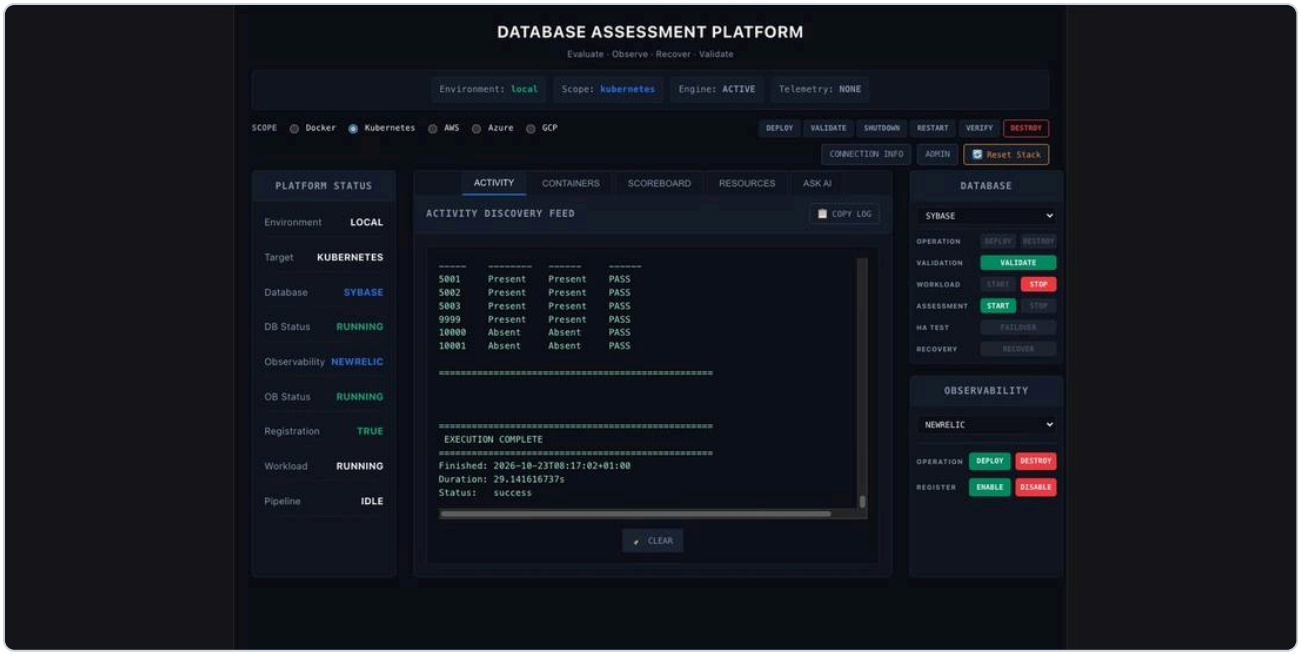


Figure 3.4 23 Oct 2026 · 08:17:25

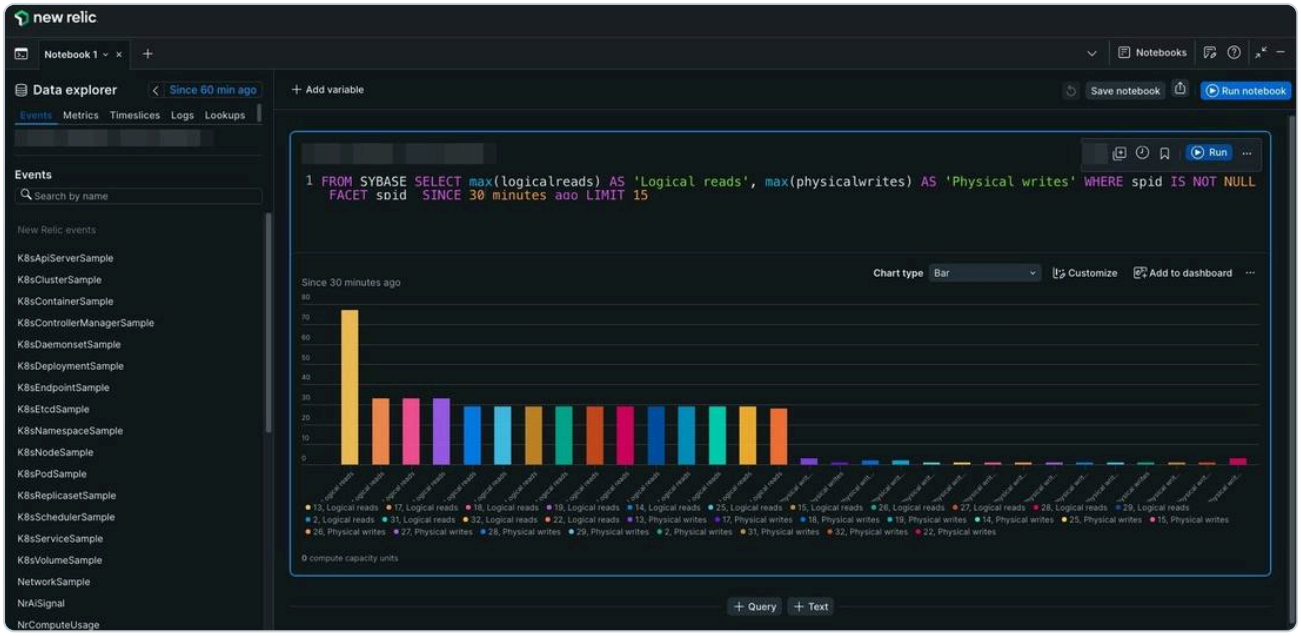


Figure 3.5 23 Oct 2026 · 08:30:33

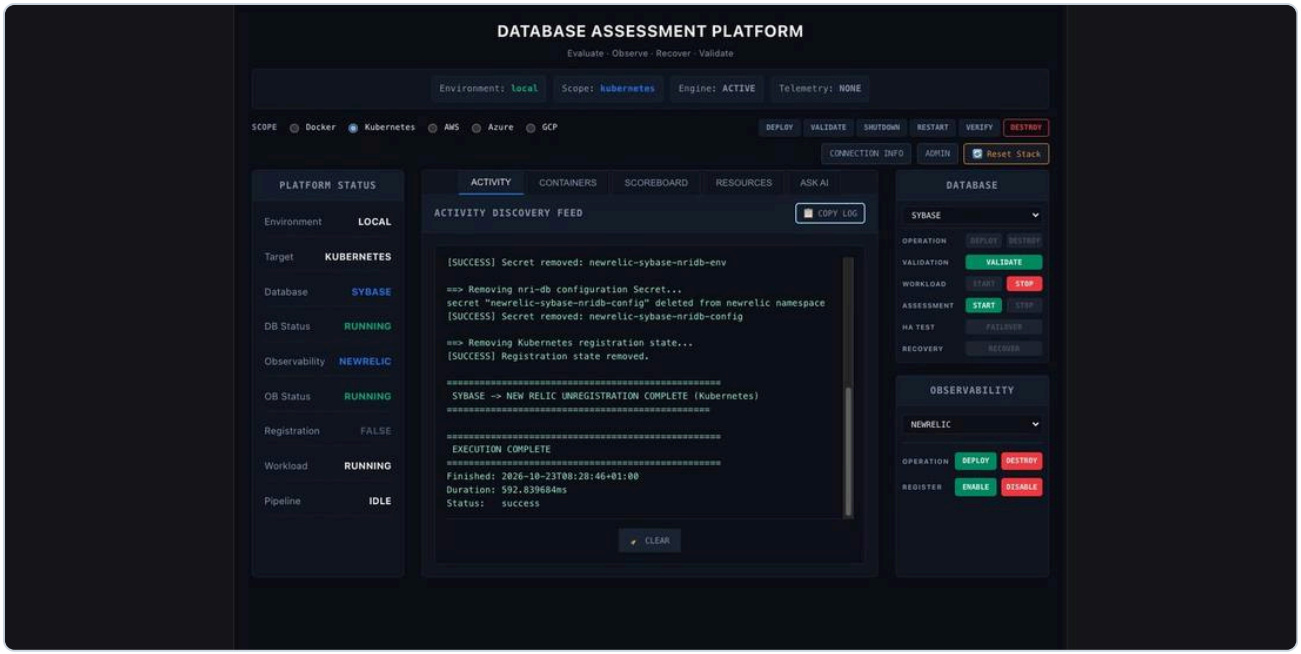


Figure 3.6 23 Oct 2026 · 08:36:16

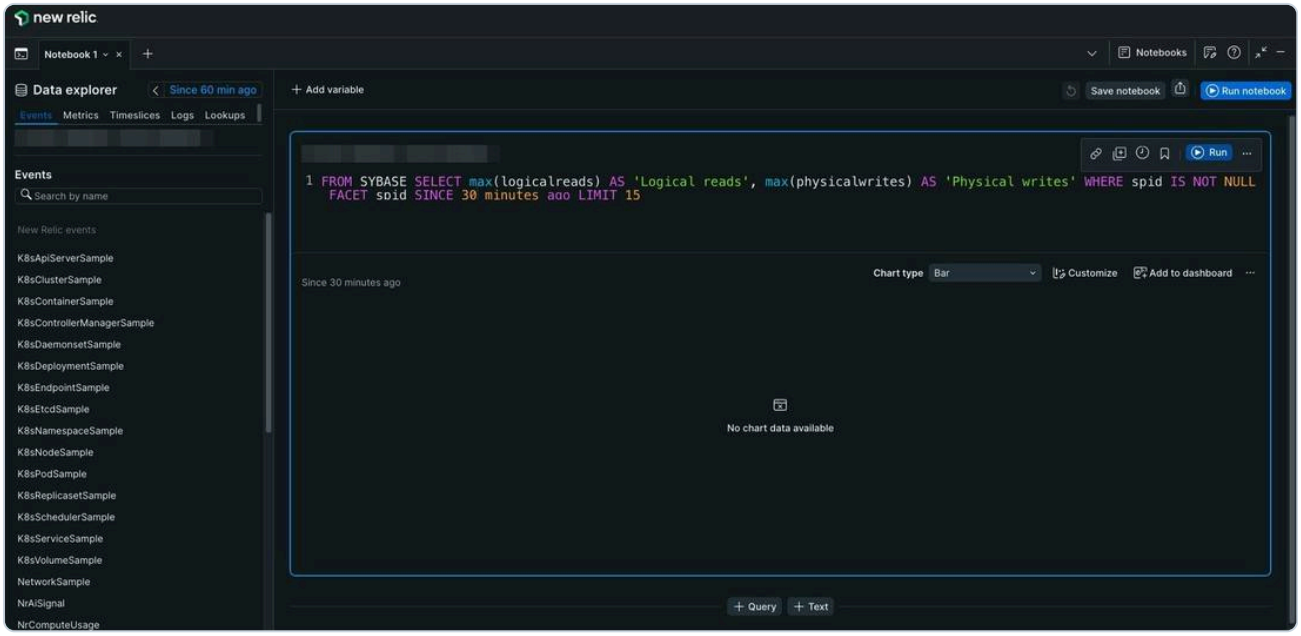


Figure 3.7 23 Oct 2026 · 10:37:02

3.3 Destroy SAP ASE

Destroy	SAP ASE	Completed

SAP ASE is removed and its storage released.

Started 23 Oct 2026, 08:36

COMMAND

```
SCOPE=kubernetes db-platform/sybase/deploy/kubernetes/destroy.sh
```

- ✔ Sybase image build cancelled (the next DEPLOY starts it again).
- ✔ Sybase pod Ready

3.4 Install Splunk

Deploy tool	Splunk	SAP ASE	Completed with warnings

Splunk is installed, ready to monitor the databases in this cluster.

Started 24 Oct 2026, 19:42 · took 8s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available
- ✔ Splunk Operator deployment already installed
- ✔ SPLUNK_GENERAL_TERMS configured on Operator
- ✔ No pods in a failed state
- ✔ Splunk accepted a HEC test event

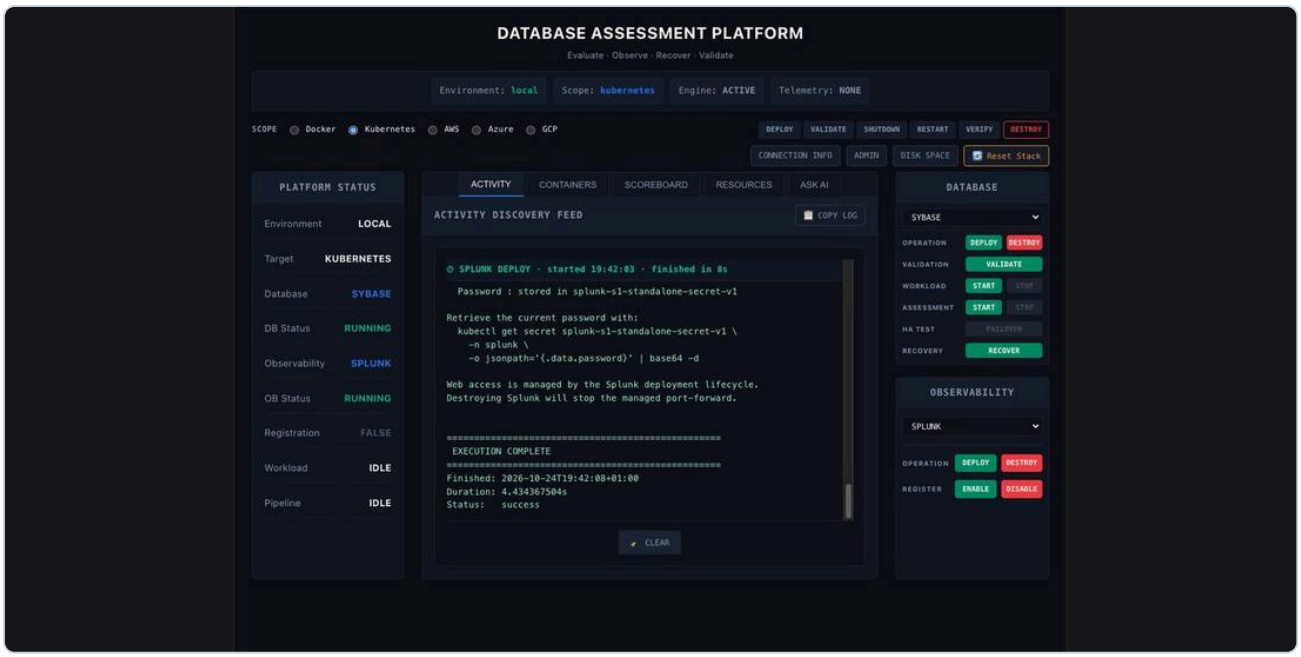


Figure 3.8 24 Oct 2026 · 19:44:26

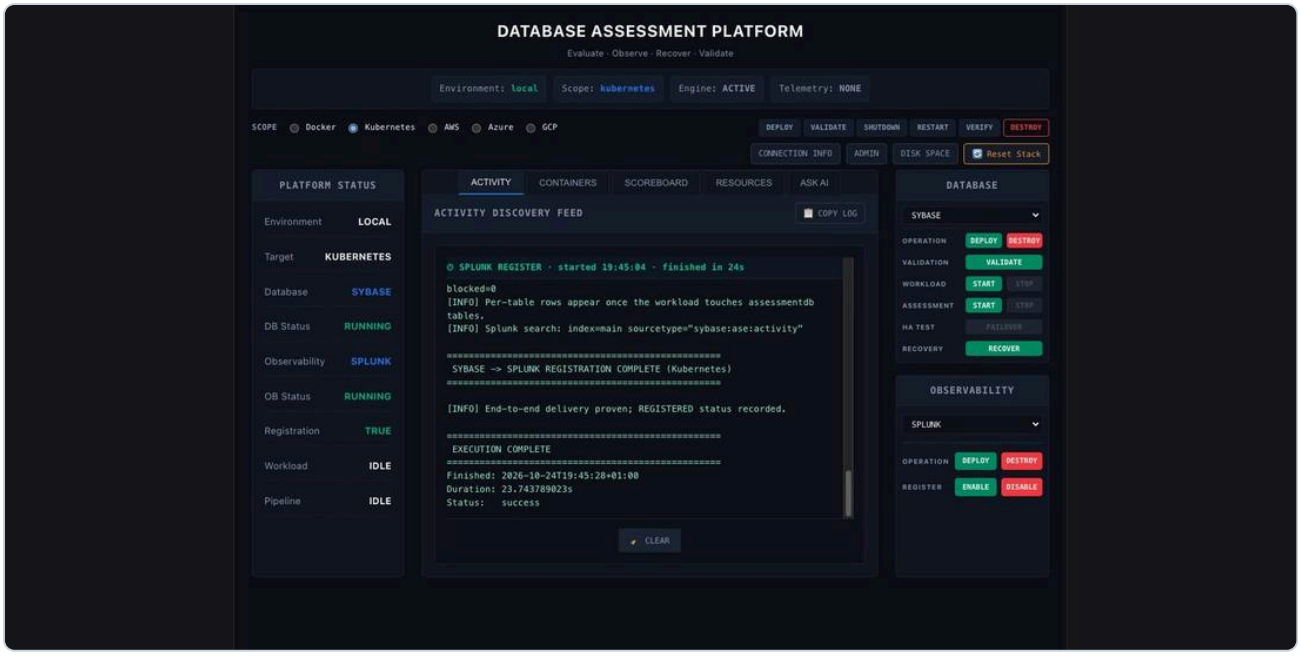


Figure 3.9 24 Oct 2026 · 19:45:38

3.5 Deploy SAP ASE

Deploy database | SAP ASE | Completed

RunShore deploys SAP ASE on Local Kubernetes and waits until every node is ready.

Started 25 Oct 2026, 07:03 · took 36s

COMMAND

```
SCOPE=kubernetes db-platform/sybase/deploy/kubernetes/deploy.sh
```

- ✔ Resolved Splunk admin credentials from secret: splunk-s1-standalone-secret-v1
- ✔ Existing HEC token 'sybase_hec_token' found
- ✔ Sybase log collector already uses the current Splunk token.

3.6 Point-in-time recovery test · SAP ASE

Assessment | SAP ASE | Completed

RunShore restores SAP ASE to a chosen moment from its backups and checks the restored data is correct.

Started 25 Oct 2026, 07:07 · took 33s

COMMAND

```
SCOPE=kubernetes db-platform/sybase/assessment/kubernetes/run_assessment.sh
```

3.7 Workload · SAP ASE

Workload | SAP ASE | Completed

A load generator runs against SAP ASE, so the monitoring shows real traffic. It stops by itself after a set time.

Started 25 Oct 2026, 07:11

COMMAND

```
bash db-platform/sybase/workload/kubernetes/start.sh
```

- ✔ Sybase pod Ready
- ✔ ASE connectivity confirmed
- ✔ assessmentdb exists -- left as it is
- ✔ Workload table recreated with zero rows
- ✔ Sybase pod Ready
- ✔ Sybase workload stopped.

3.8 Register SAP ASE with Splunk

Register | Splunk | SAP ASE | Completed

SAP ASE is connected to Splunk, so its metrics and dashboards appear there.

Started 25 Oct 2026, 07:19 · took 20s

COMMAND

```
bash observability/splunk/registration/kubernetes/sybase/register.sh
```

- ✔ Resolved Splunk admin credentials from secret: splunk-s1-standalone-secret-v1
- ✔ Existing HEC token 'sybase_hec_token' found
- ✔ Sybase log collector already uses the current Splunk token.
- ✔ Fluent Bit collector container Ready: platform-sybase-59f7668668-5wrzv
- ✔ Resolved Splunk admin credentials from secret: splunk-s1-standalone-secret-v1
- ✔ Validation event confirmed in Splunk (index=main sourcetype="sybase:ase")

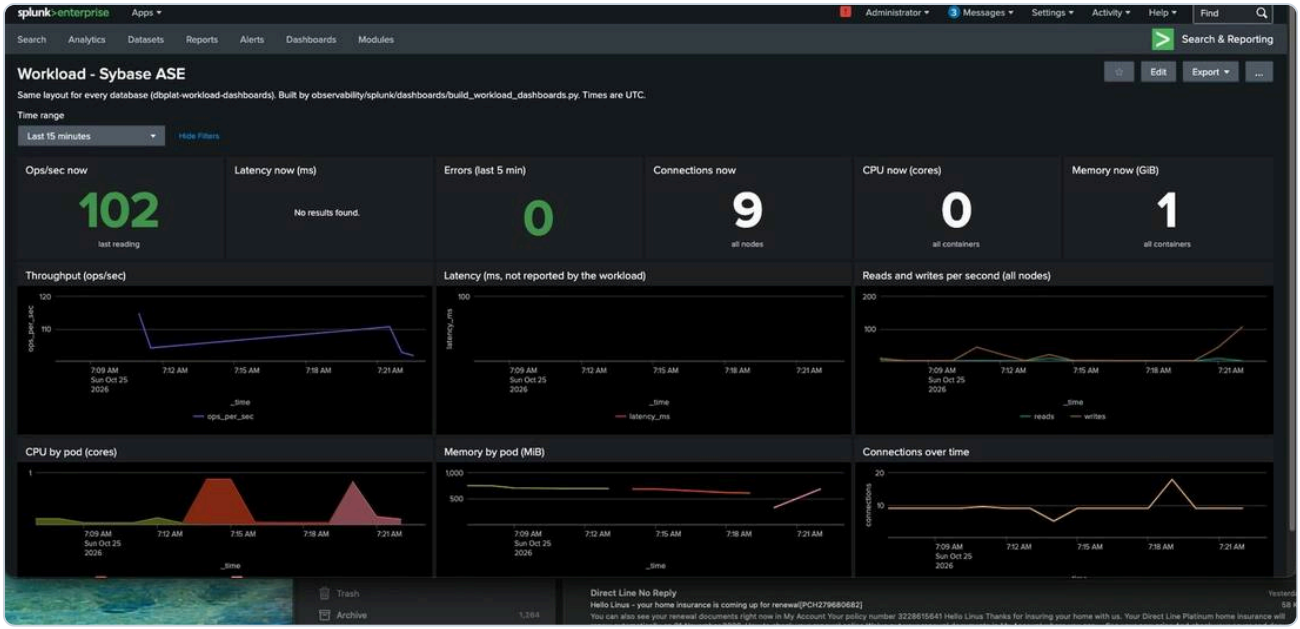


Figure 3.10 25 Oct 2026 · 07:23:11

04

MongoDB

Local Kubernetes

MongoDB is deployed as a Percona Server for MongoDB replica set, then monitored, loaded and tested for failover, replication and recovery.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Local Kubernetes	MongoDB	New Relic, PMM, Splunk	43 screens · 19 logs	23 Oct–25 Oct 2026

4.1 Screenshots

What the console and dashboards showed at this point.

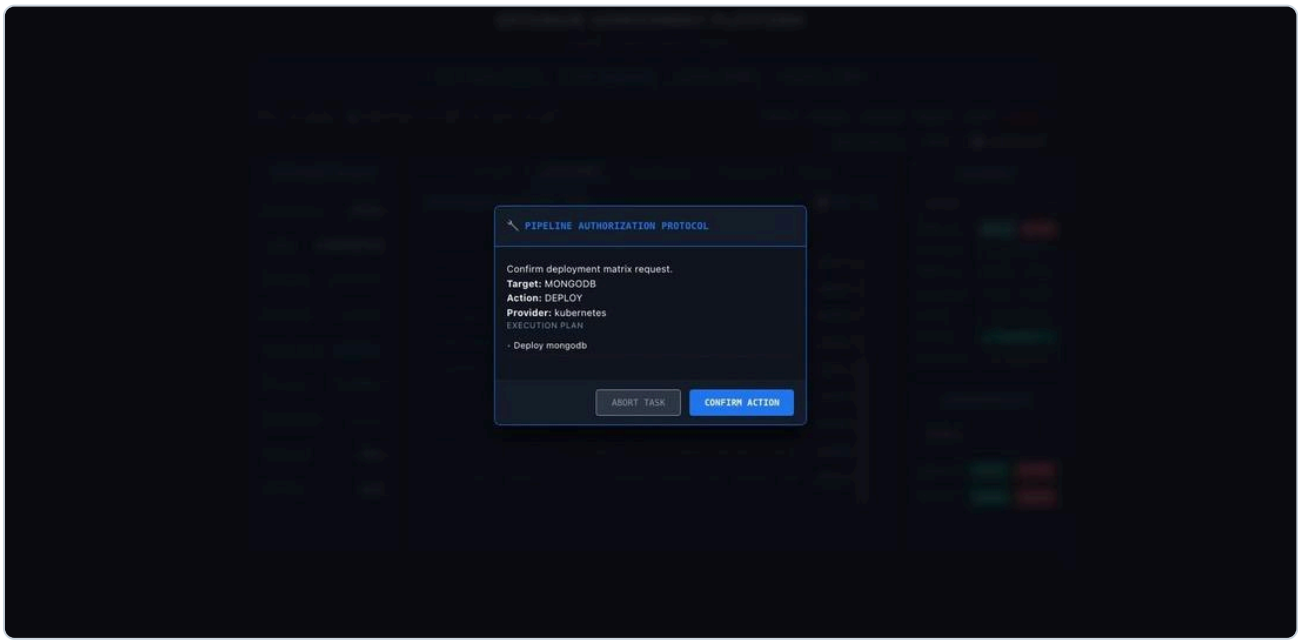


Figure 4.1 23 Oct 2026 · 10:40:20

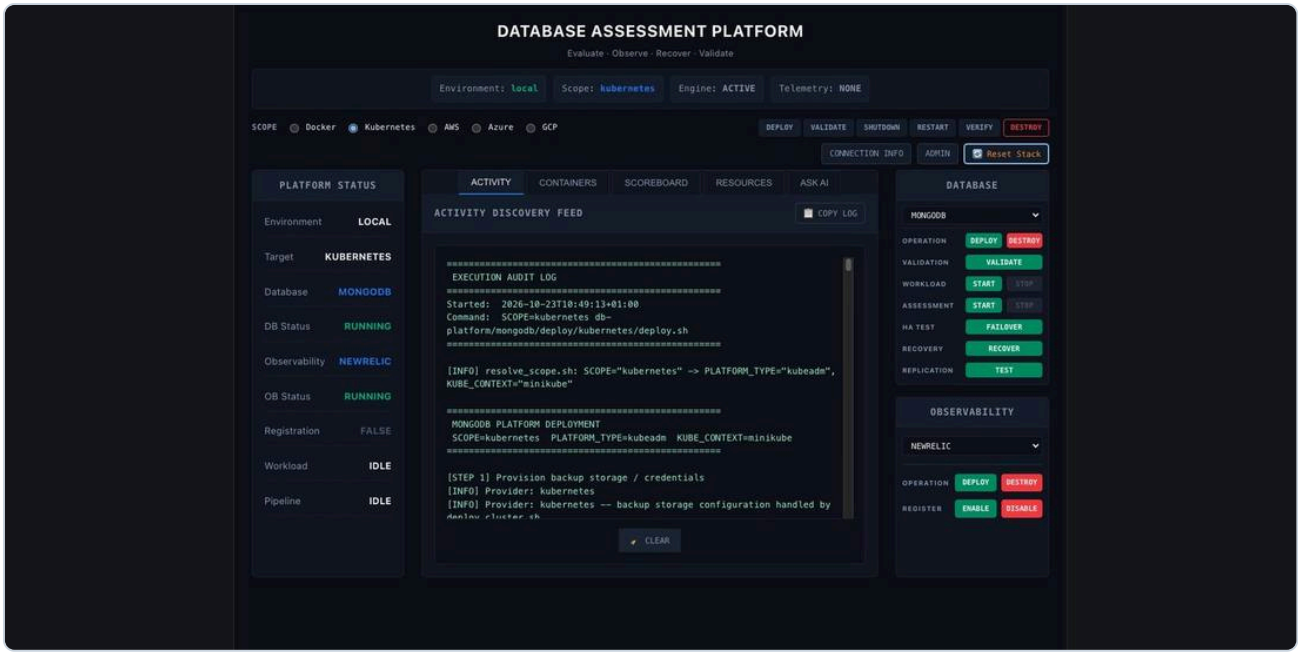


Figure 4.2 23 Oct 2026 · 10:53:56

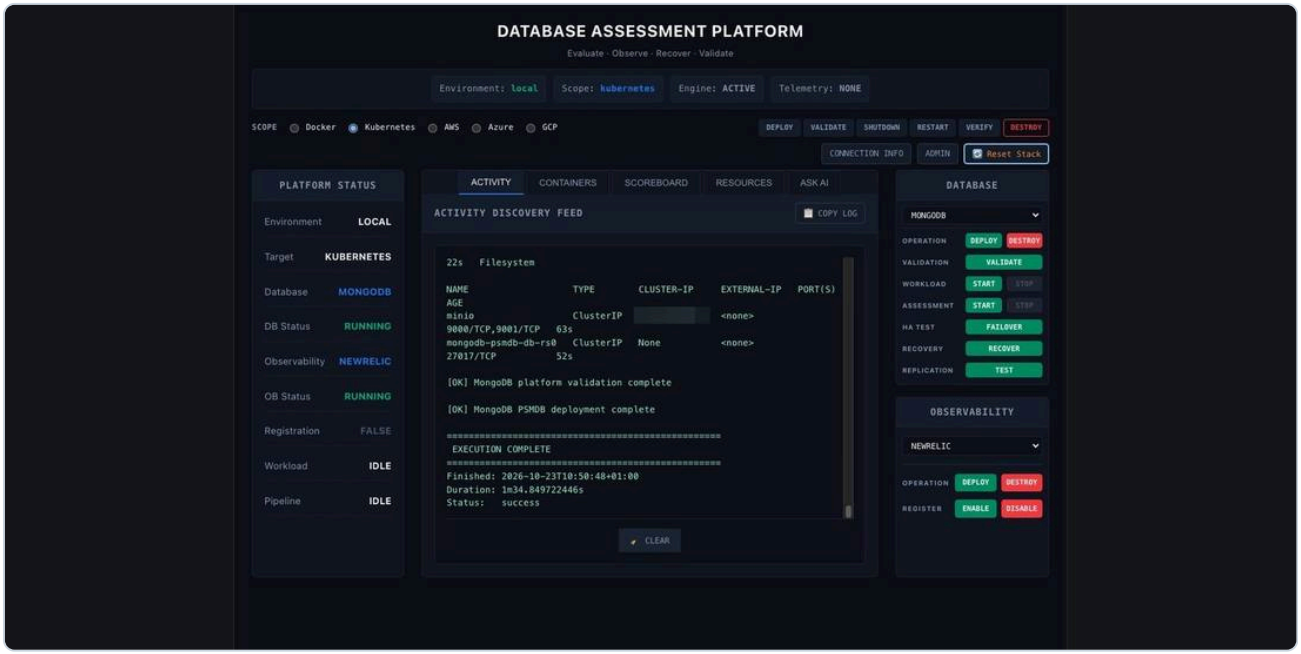


Figure 4.3 23 Oct 2026 · 10:54:06

4.2 Deploy MongoDB

Deploy database	MongoDB	Completed with warnings

RunShore deploys MongoDB on Local Kubernetes and waits until every node is ready.

Started 23 Oct 2026, 10:49

COMMAND

```
SCOPE=kubernetes db-platform/mongodb/deploy/kubernetes/deploy.sh
```

- ✔ Tooling available
- ✔ Kubernetes API reachable
- ✔ Required PSMDB charts available
- ✔ PSMDB operator ready
- ✔ Bound PVCs: 4
- ✔ MongoDB platform validation complete

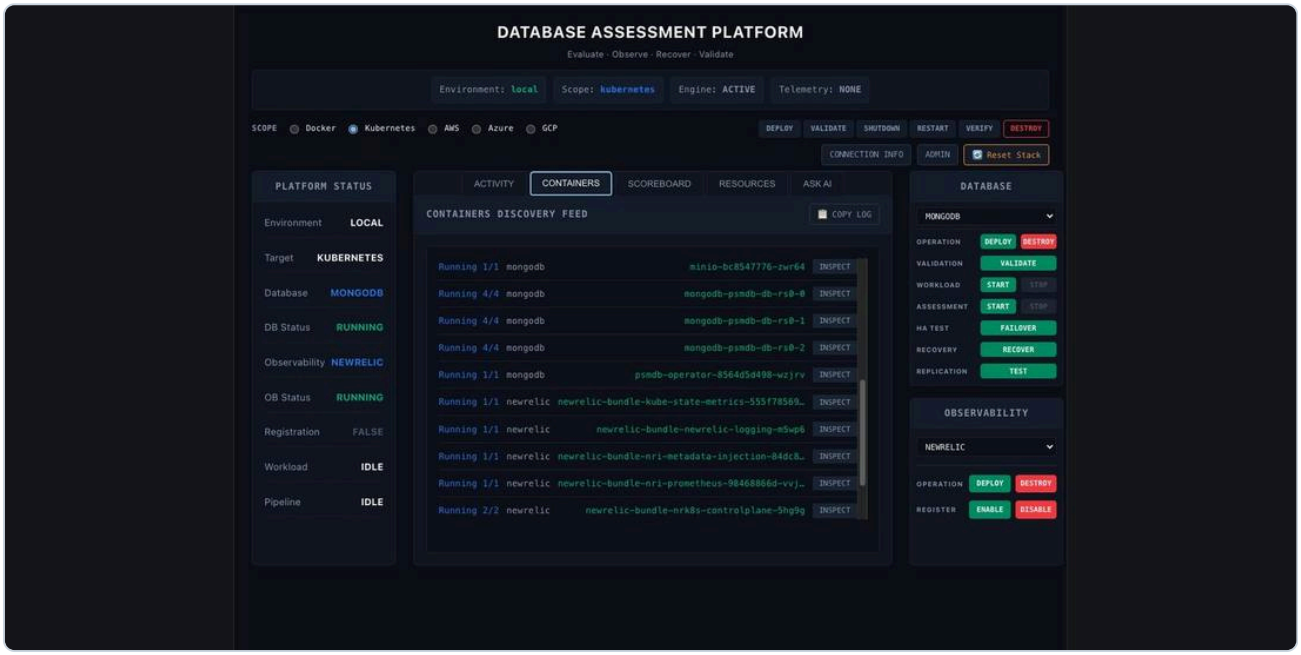


Figure 4.4 23 Oct 2026 · 10:55:05

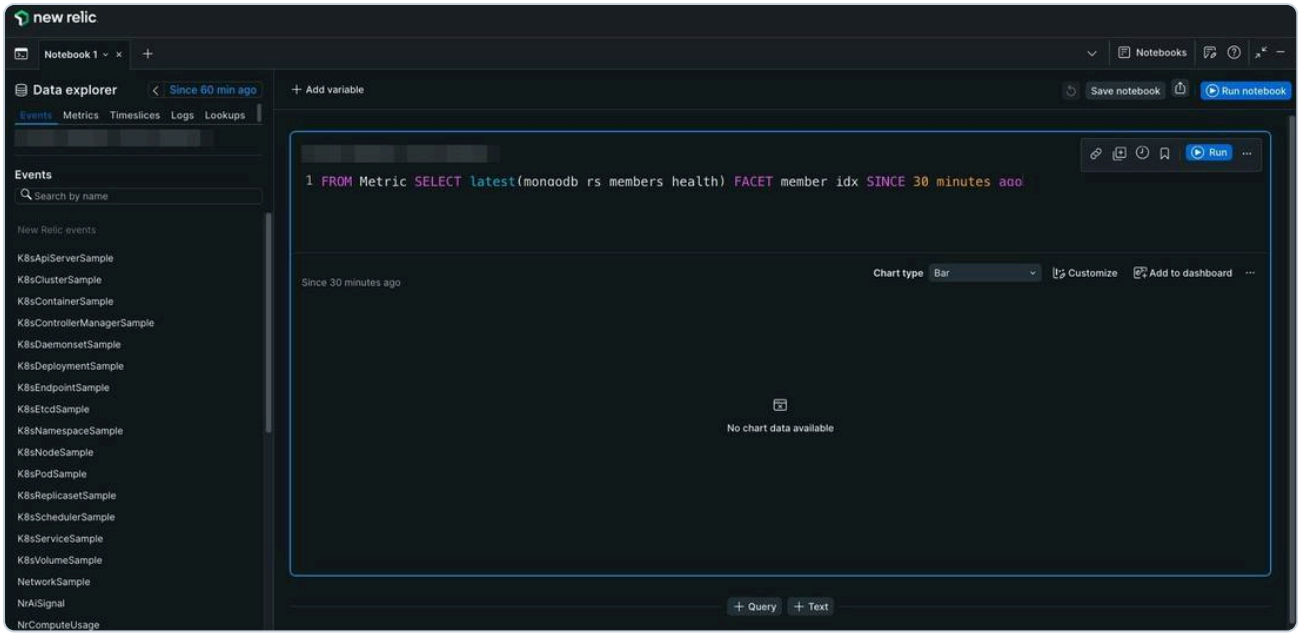


Figure 4.5 23 Oct 2026 · 11:06:56

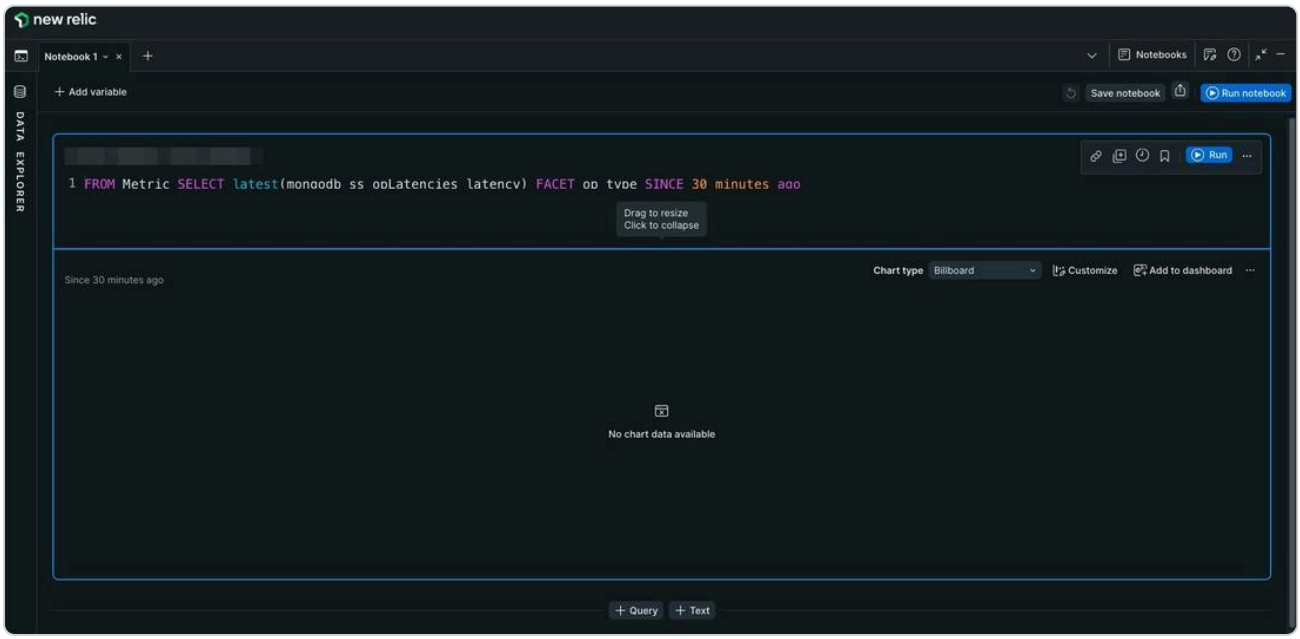


Figure 4.6 23 Oct 2026 · 11:13:45

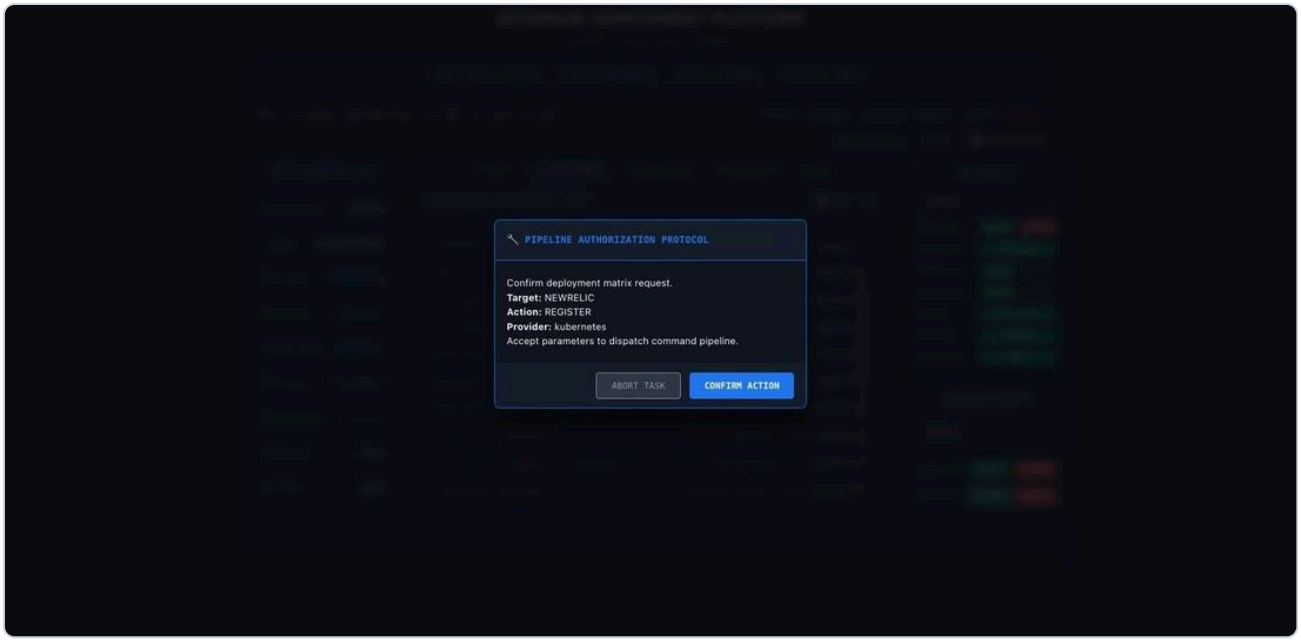


Figure 4.7 23 Oct 2026 · 11:14:07

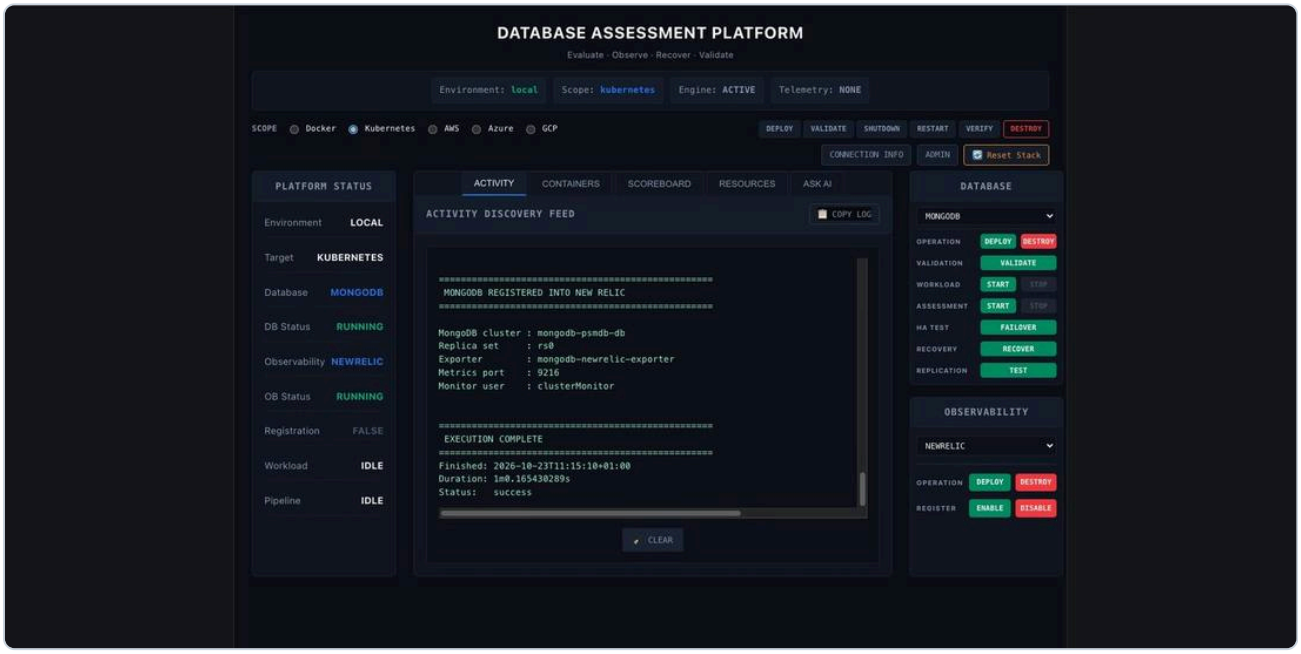


Figure 4.8 23 Oct 2026 · 11:15:34

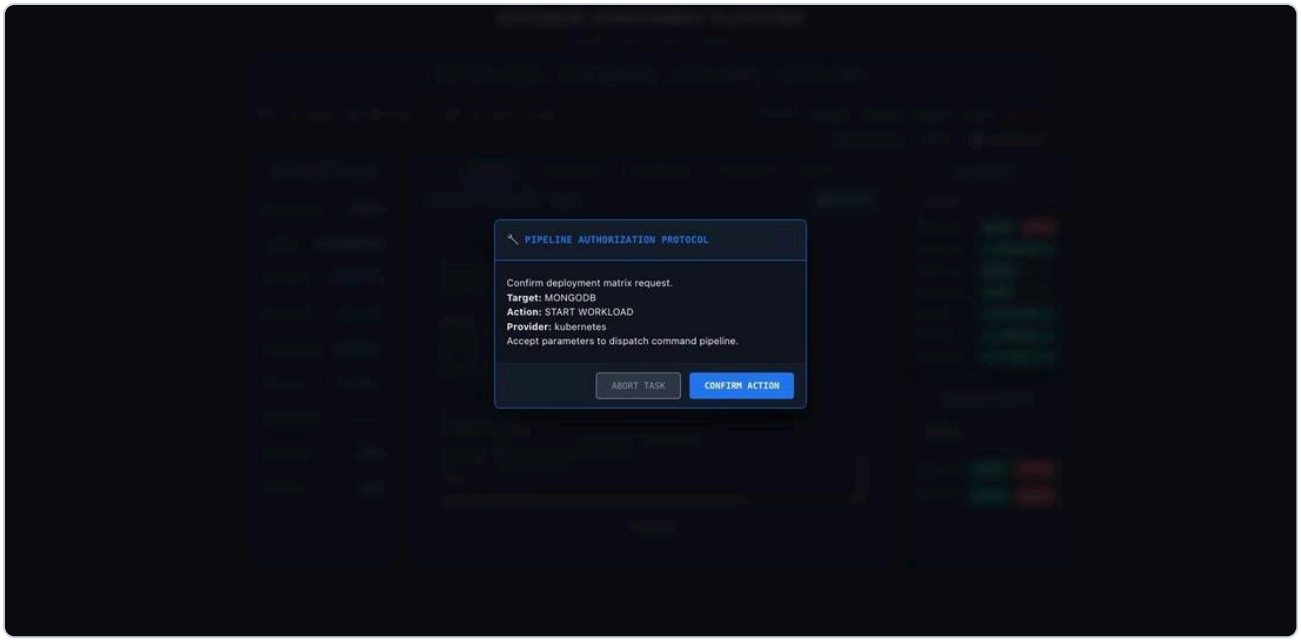


Figure 4.9 23 Oct 2026 · 11:15:56

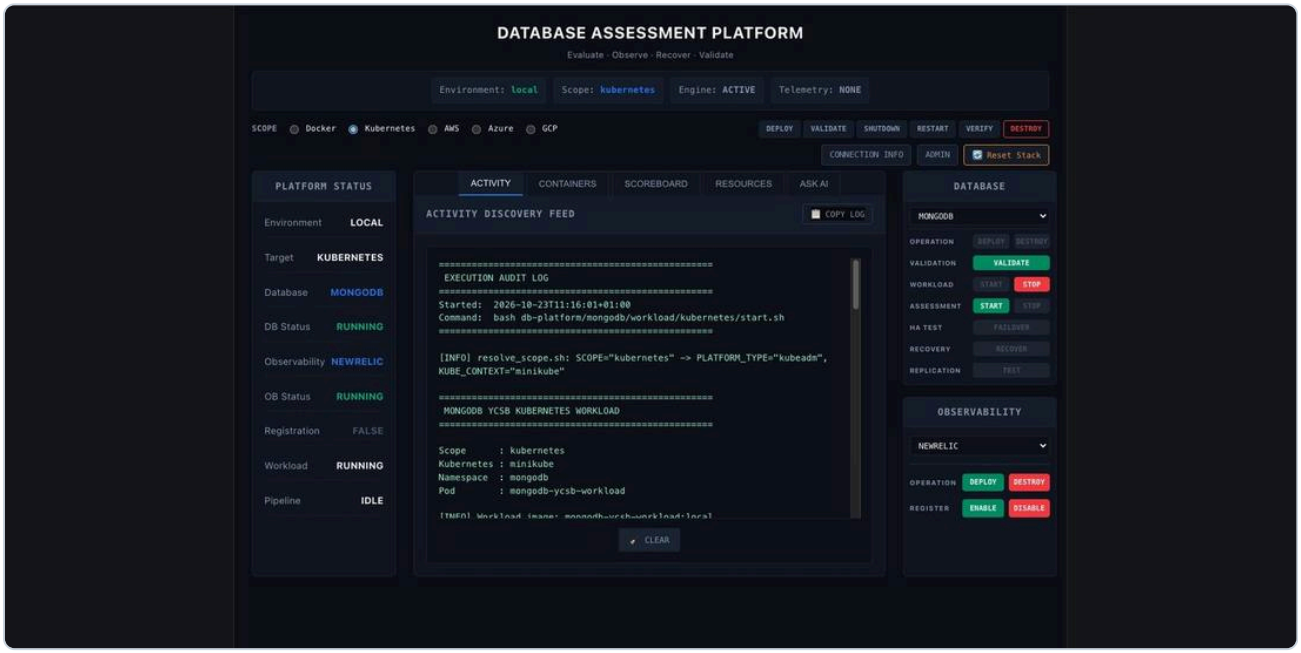


Figure 4.10 23 Oct 2026 · 11:17:23

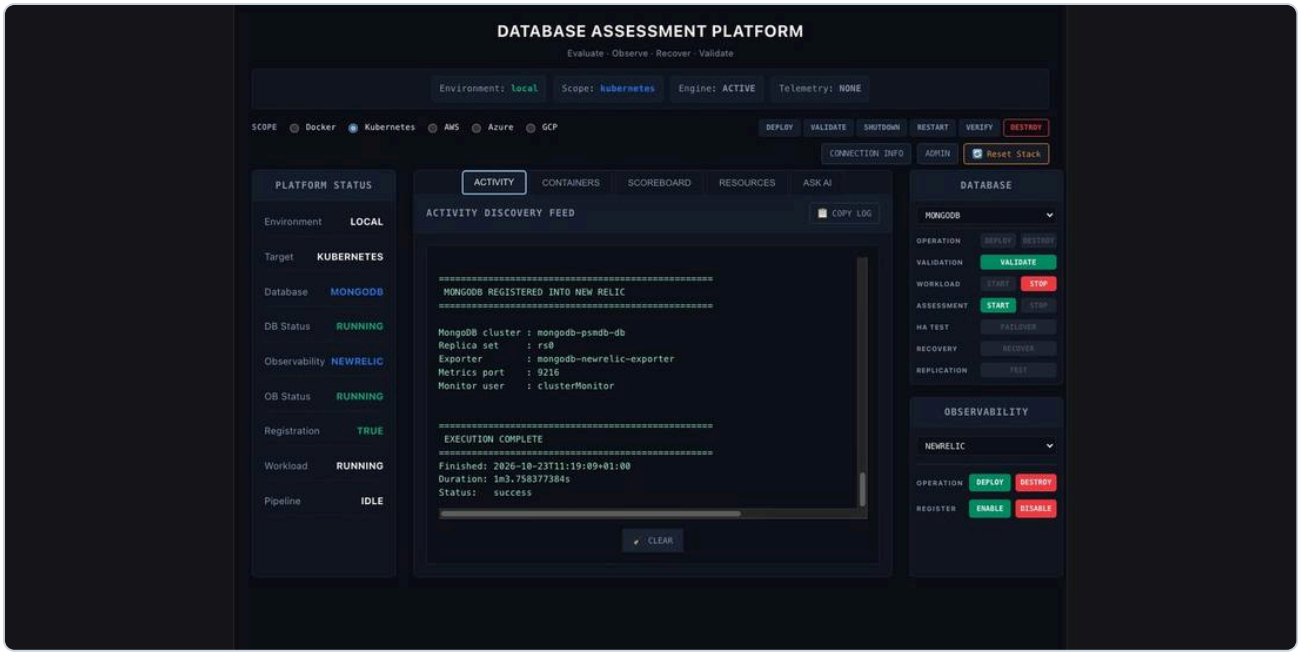


Figure 4.11 23 Oct 2026 · 11:29:46

4.3 More screenshots

Deploy database | **MongoDB**

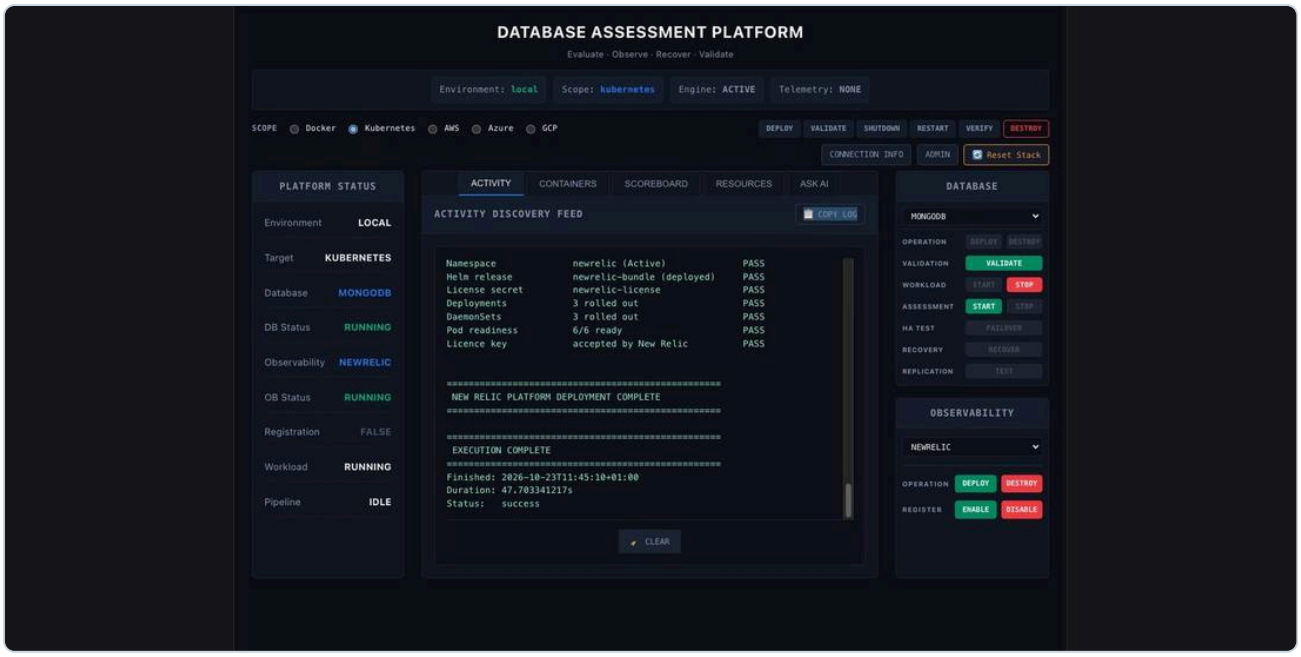


Figure 4.12 23 Oct 2026 · 11:46:42

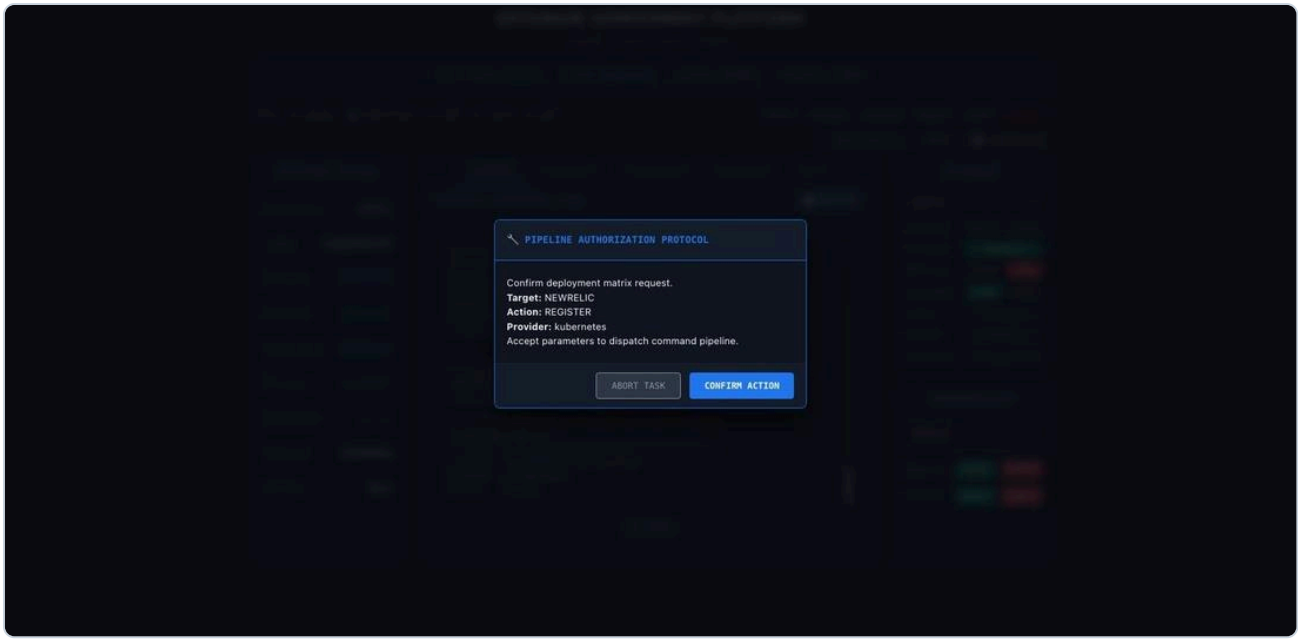


Figure 4.13 23 Oct 2026 · 11:46:57

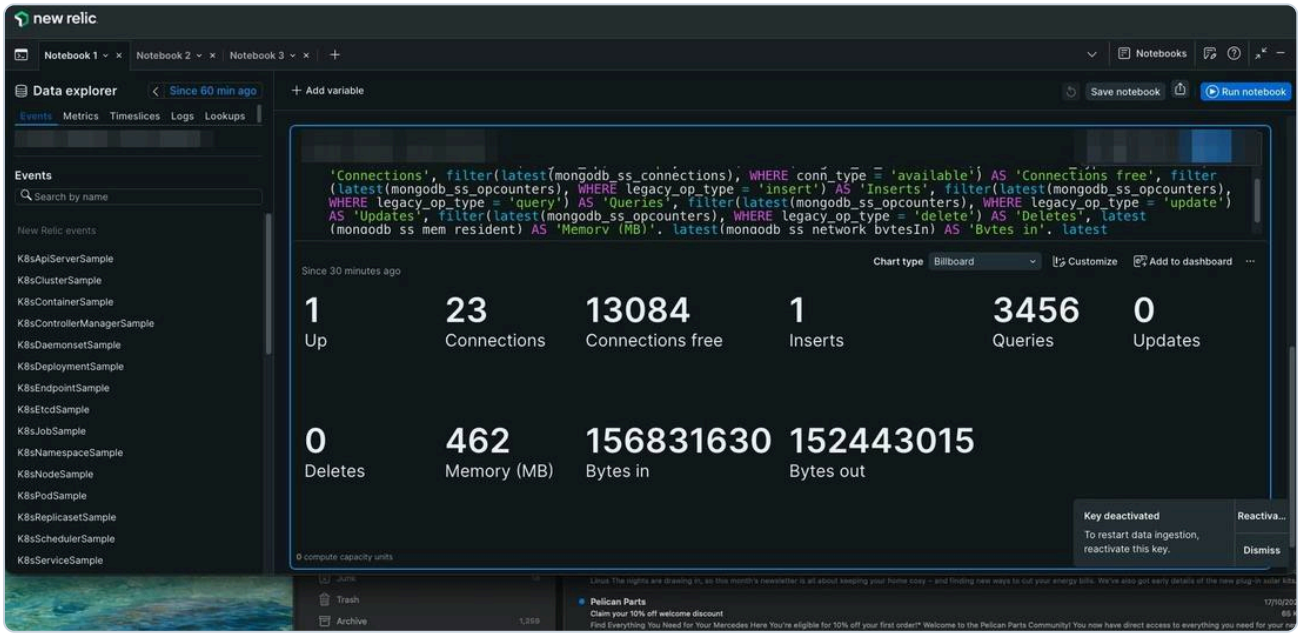


Figure 4.14 23 Oct 2026 · 11:59:10

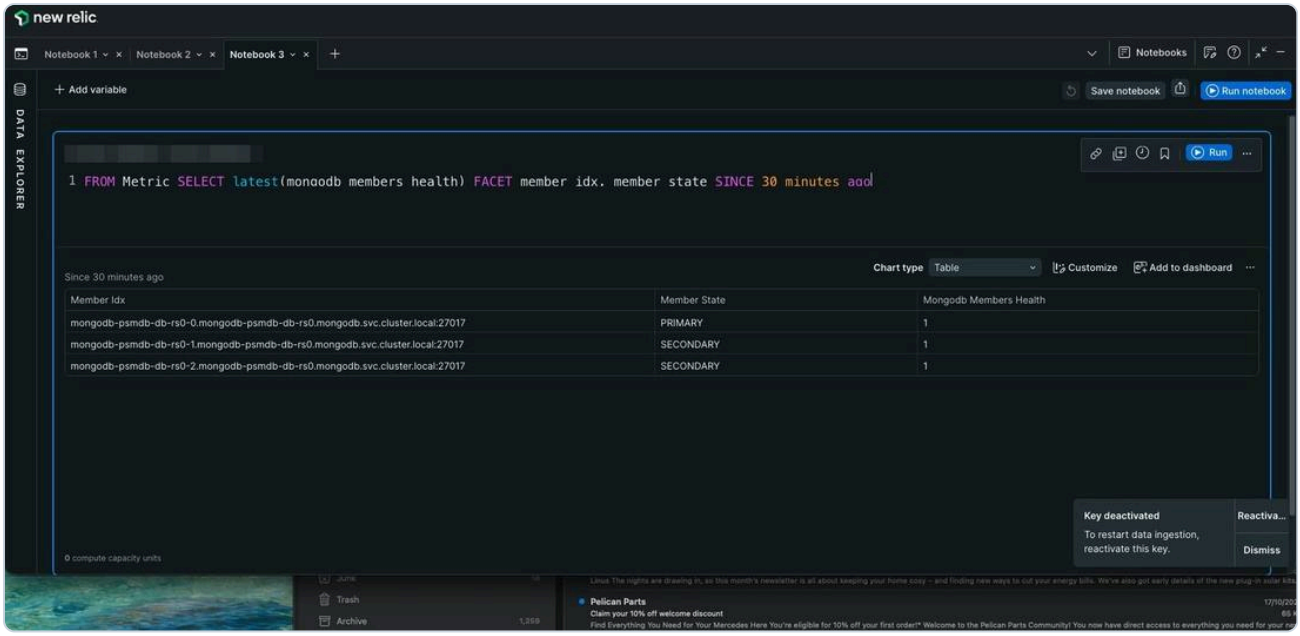


Figure 4.15 23 Oct 2026 · 12:17:05

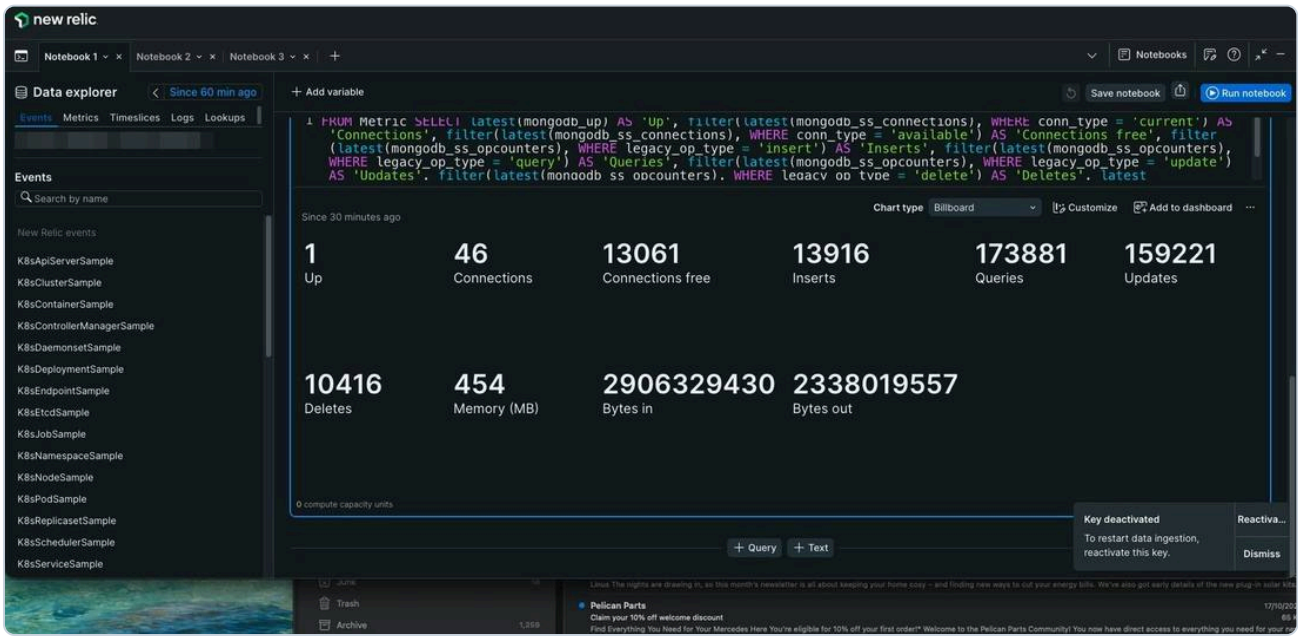


Figure 4.16 23 Oct 2026 · 12:17:32

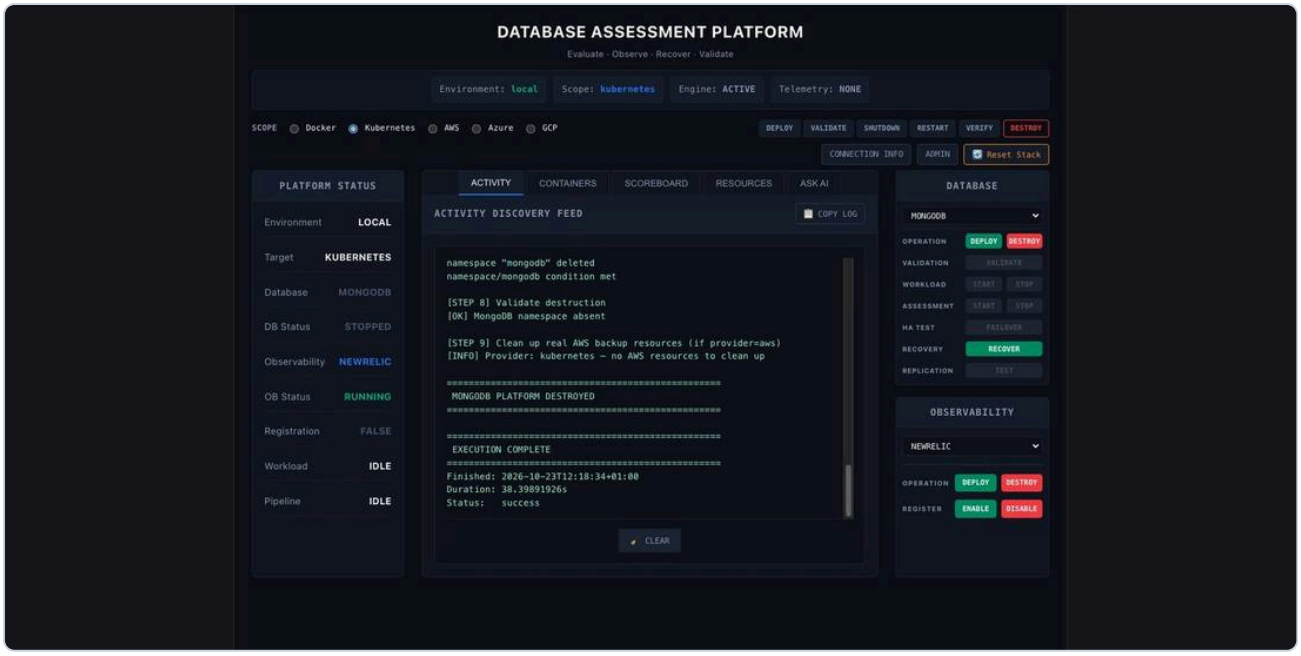


Figure 4.17 23 Oct 2026 · 12:20:08

4.4 Deploy MongoDB

Deploy database	MongoDB	Completed with warnings

RunShore deploys MongoDB on Local Kubernetes and waits until every node is ready.

Started 25 Oct 2026, 07:29 · took 1m 48s

COMMAND

```
SCOPE=kubernetes db-platform/mongodb/deploy/kubernetes/deploy.sh
```

- ✔ Tooling available
- ✔ Kubernetes API reachable
- ✔ Required PSMDB charts available
- ✔ PSMDB operator ready
- ✔ Bound PVCs: 4
- ✔ MongoDB platform validation complete

4.5 Install PMM

Deploy tool	PMM	MongoDB	Completed

PMM is installed, ready to monitor the databases in this cluster.

Started 25 Oct 2026, 08:12 · took 20s

COMMAND

```
SCOPE=kubernetes observability/pmm/deploy/kubernetes/deploy_pmm_stack.kubernetes.s
```

```
h
```

- ✔ Kubernetes API reachable
- ✔ Required lifecycle scripts found.
- ✔ Kubernetes API reachable
- ✔ Namespace pmm is Active
- ✔ The PMM token in ADMIN is accepted -- kept
- ✔ PMM server accepts the token from ADMIN

4.6 Failover test · MongoDB

Failover | MongoDB | Completed

The primary MongoDB node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 25 Oct 2026, 08:13 · took 28s

COMMAND

```
SCOPE=kubernetes db-platform/mongodb/failover/kubernetes/run_failover.sh
```

- ✔ PSMDB cluster is ready.
- ✔ Replica set: rs0
- ✔ Ready members: 3/3
- ✔ Cluster-admin credentials loaded from Kubernetes Secret.
- ✔ HA validation marker survived failover.
- ✔ MongoDB replica set recovered to 3/3 ready members.

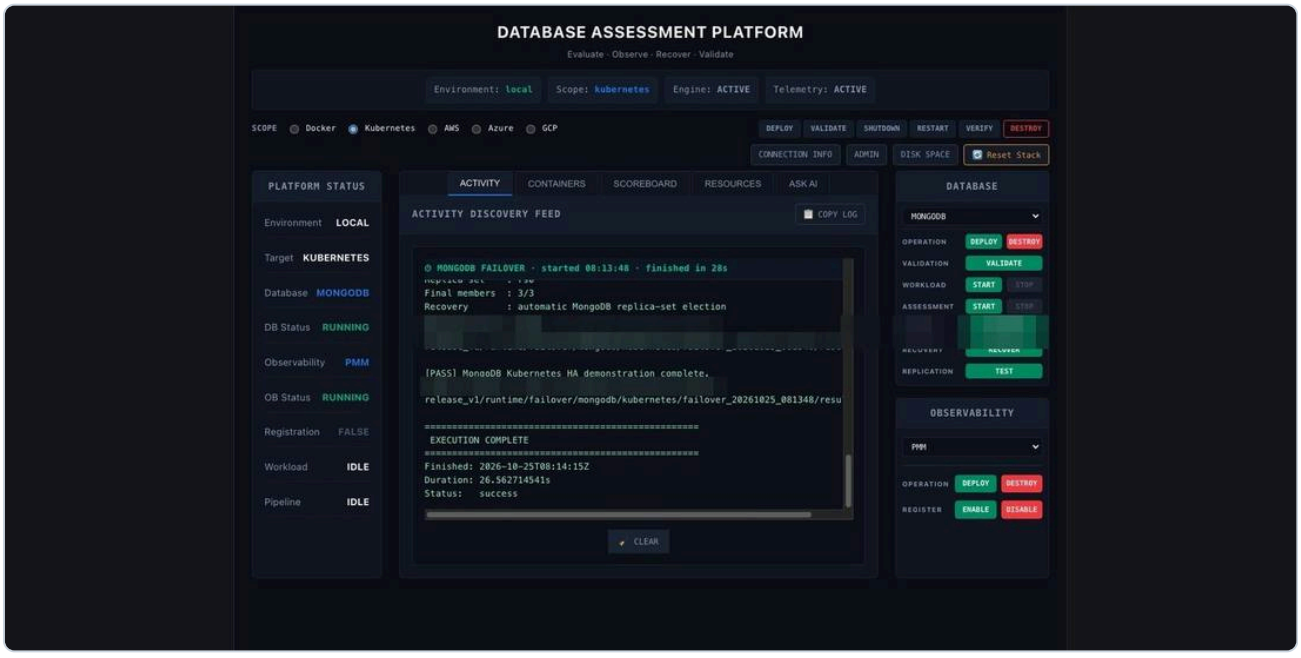


Figure 4.18 25 Oct 2026 · 08:14:48

4.7 Replication test · MongoDB

Replication | MongoDB | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 25 Oct 2026, 08:14 · took 20s

COMMAND

```
SCOPE=kubernetes db-platform/mongodb/replication/kubernetes/run_replication.sh
```

- ✔ Three-member replica-set topology verified.
- ✔ PRIMARY is writable.
- ✔ SECONDARY is in secondary state.
- ✔ Initial replica-set state captured.
- ✔ Replicated data matches.
- ✔ MONGODB REPLICATION: PASS

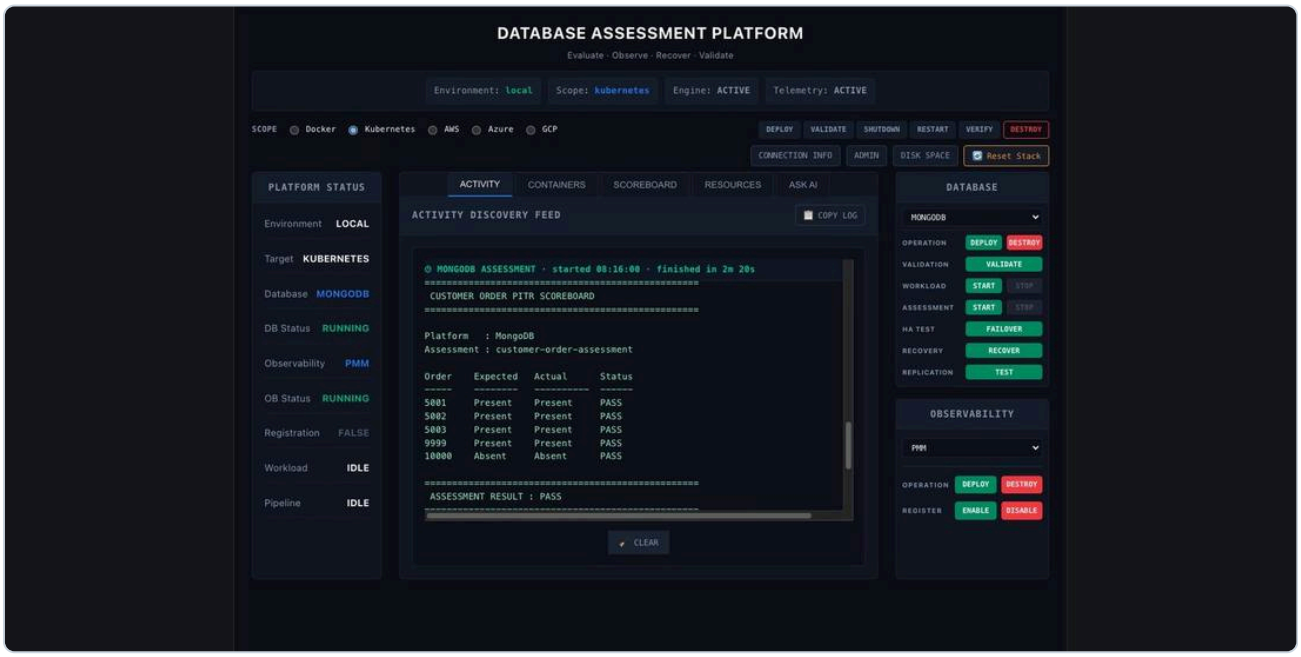


Figure 4.19 25 Oct 2026 · 08:18:37

4.8 Register MongoDB with PMM

Register | PMM | MongoDB | Completed

MongoDB is connected to PMM, so its metrics and dashboards appear there.

Started 25 Oct 2026, 08:20 · took 20s

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/register_mongodb_replicaset.sh
```

- ✔ MongoDB namespace : mongodb
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ Replica set : rs0
- ✔ Running MongoDB members: 3

- ✔ Cleanup complete
- ✔ MongoDB service present in PMM inventory

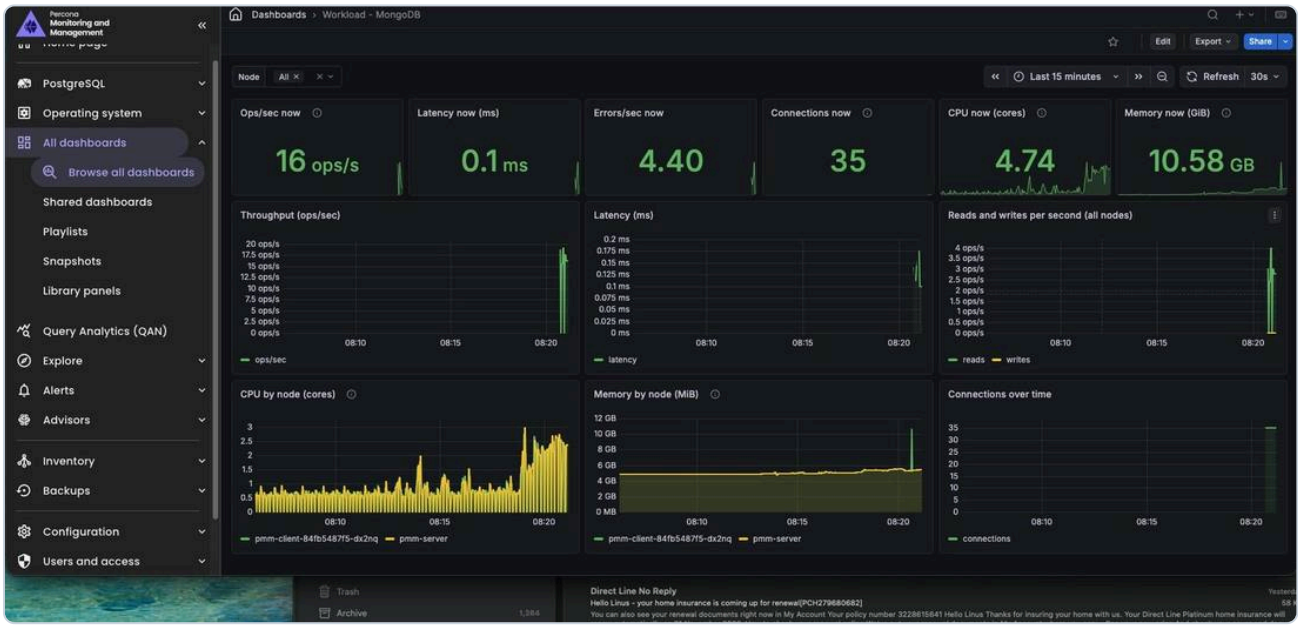


Figure 4.20 25 Oct 2026 · 08:21:19



Figure 4.21 25 Oct 2026 · 08:23:02

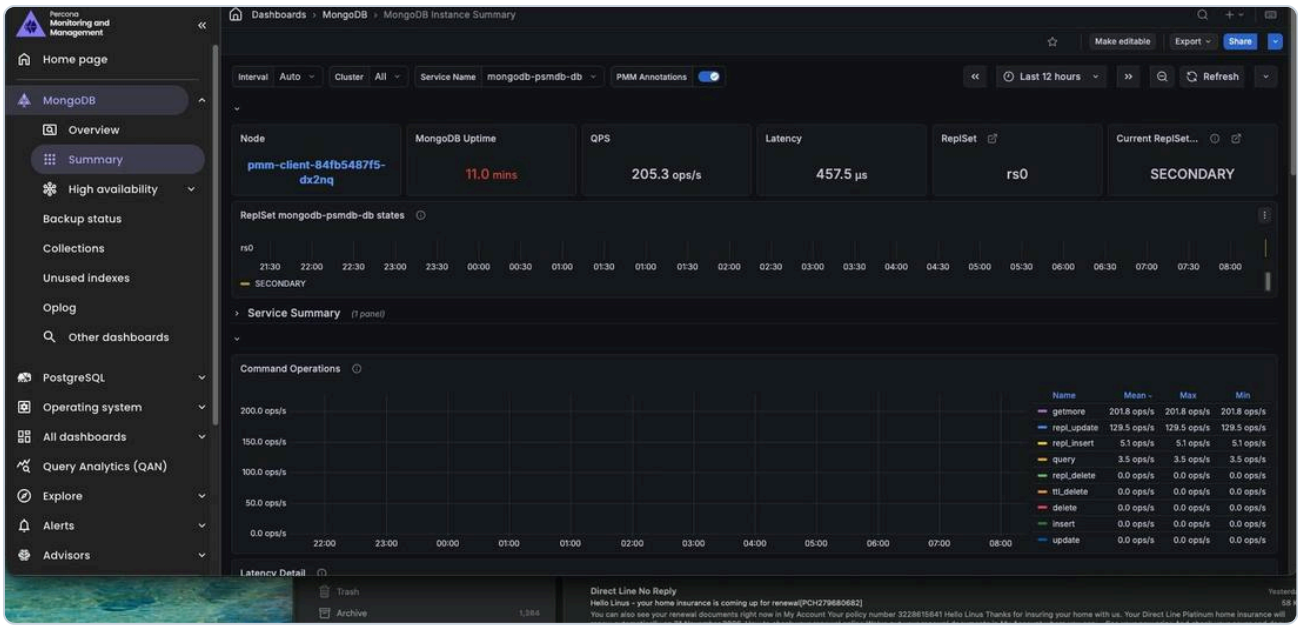


Figure 4.22 25 Oct 2026 · 08:25:43

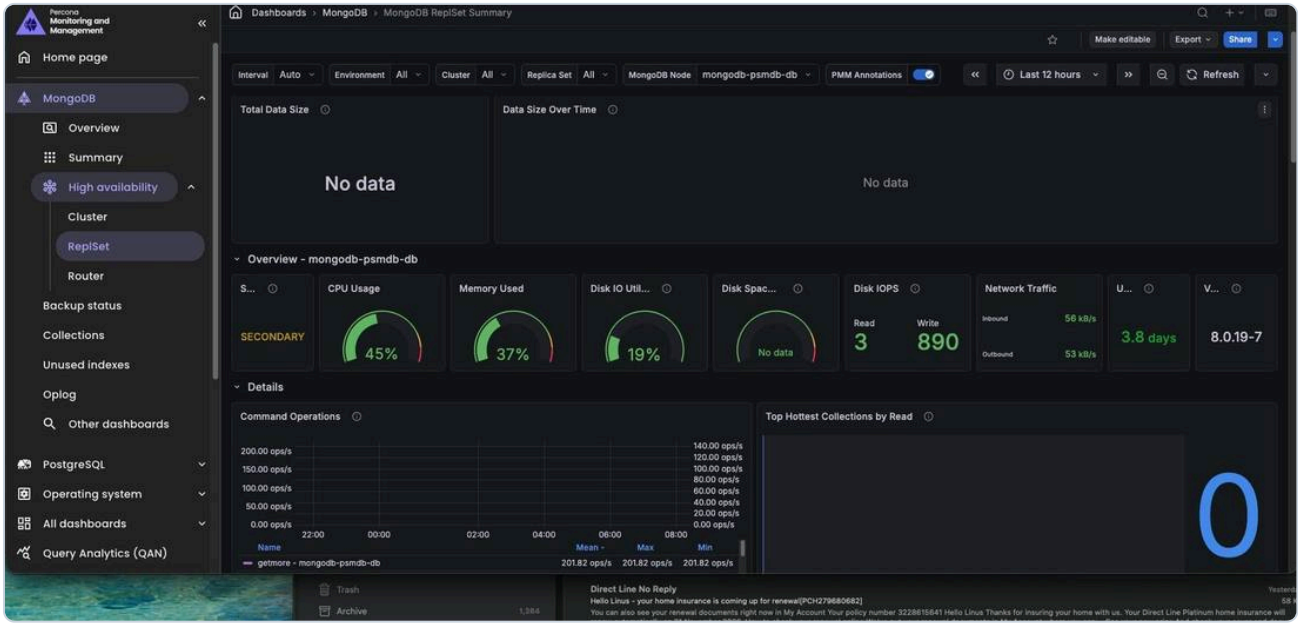


Figure 4.23 25 Oct 2026 · 08:26:26

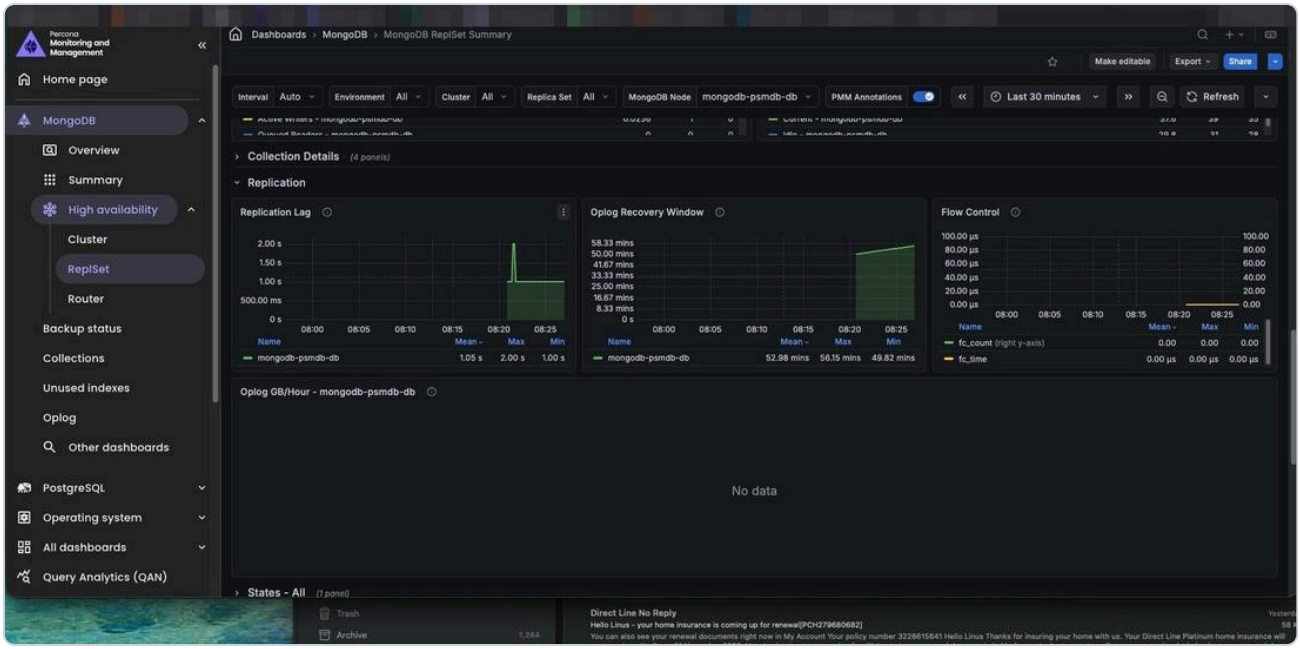


Figure 4.24 25 Oct 2026 · 08:28:34

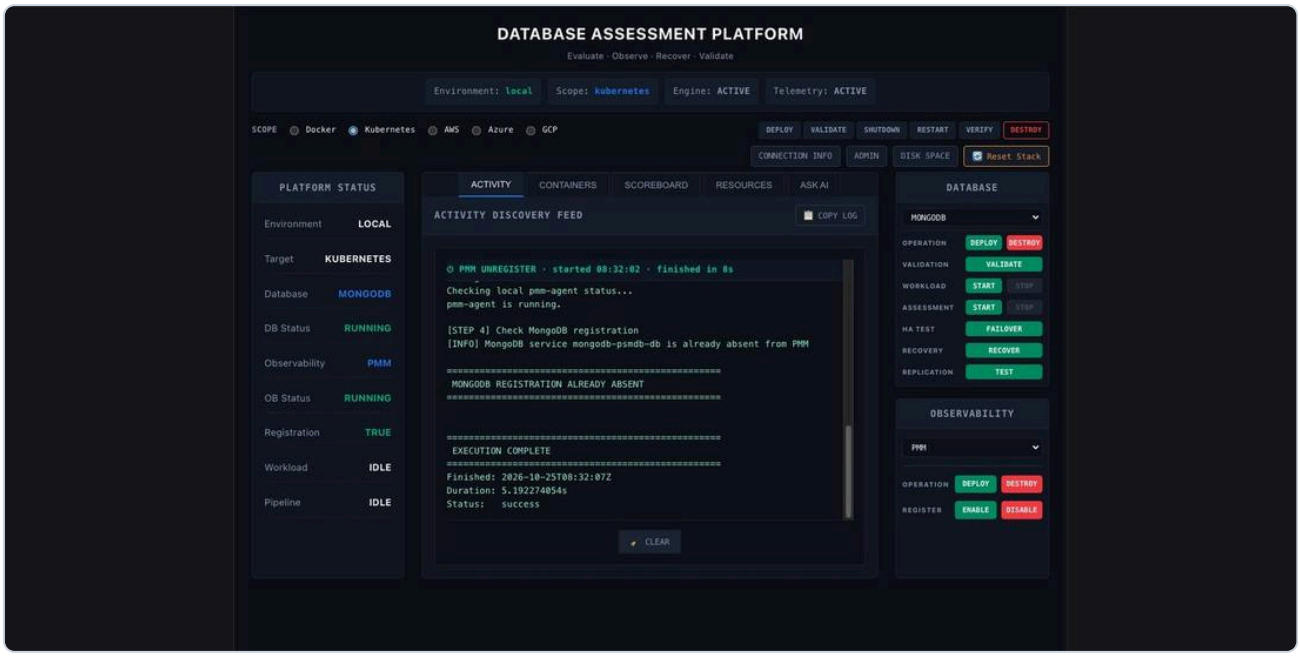


Figure 4.25 25 Oct 2026 · 08:32:36

4.9 Unregister MongoDB from PMM

Unregister	PMM	MongoDB	Completed

MongoDB is removed from PMM; the database itself keeps running.

Started 25 Oct 2026, 08:42 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/unregister.sh
```

✔ pmm-0

- ✔ nothing left in PMM inventory -- Registration shows FALSE
- ✔ MongoDB service absent from PMM: mongodb-psmdb-db

4.10 Remove PMM

Unregister	PMM	MongoDB	Completed

PMM is removed from the cluster once no database is registered with it.

Started 25 Oct 2026, 08:44 · took 16s

COMMAND

```
SCOPE=kubernetes observability/pmm/deploy/kubernetes/destroy.sh
```

- ✔ PMM Client deployment removed
- ✔ PMM namespace absent
- ✔ PMM Helm release absent

4.11 Register MongoDB with New Relic

Register	New Relic	MongoDB	Completed

MongoDB is connected to New Relic, so its metrics and dashboards appear there.

Started 25 Oct 2026, 08:46 · took 1m 17s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/register.sh
```

- ✔ Kubernetes API reachable
- ✔ New Relic deployment available
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ MongoDB namespace: mongodb
- ✔ Exporter reachable inside the cluster: http://mongodb-newrelic-exporter.mongodb.svc.cluster.local:9216/metri (mongodb_up 1)
- ✔ New Relic scraped MongoDB with no errors since 2026-10-25T08:47:27Z

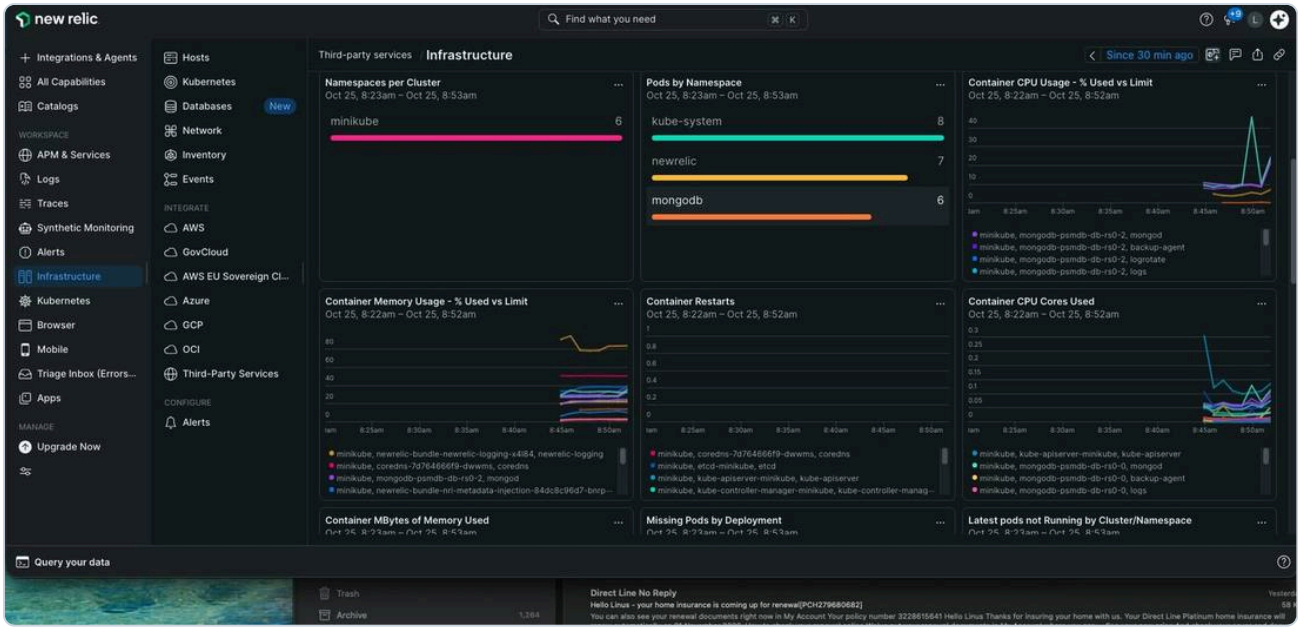


Figure 4.26 25 Oct 2026 · 08:53:36

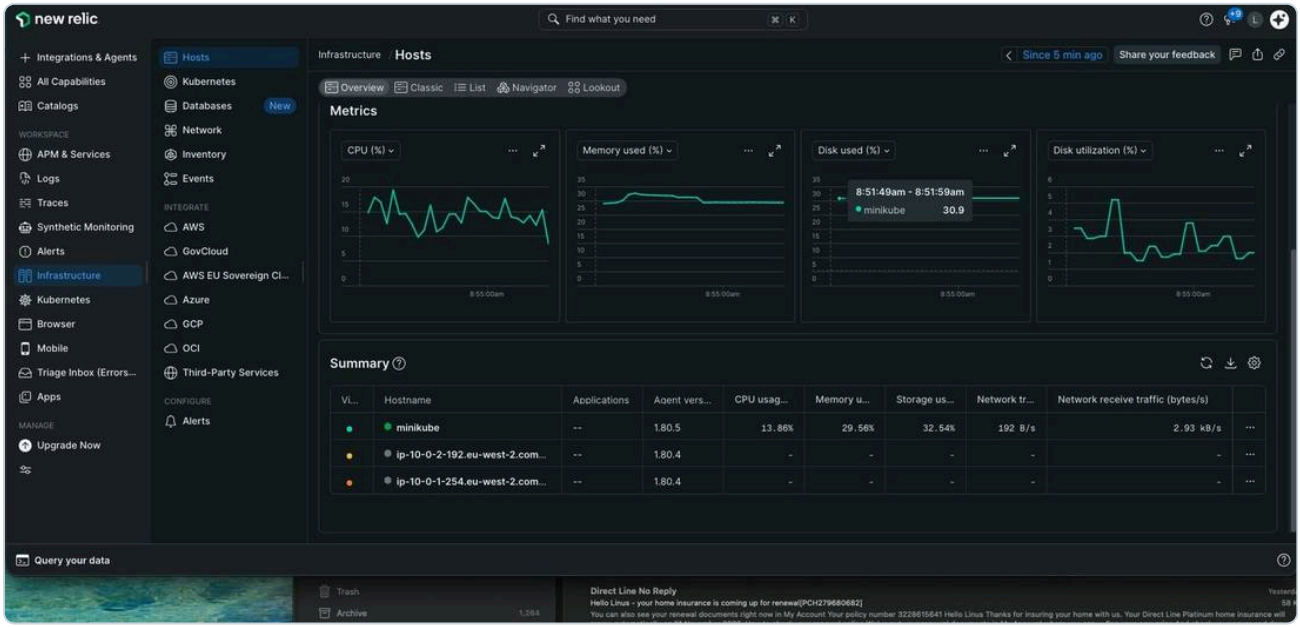


Figure 4.27 25 Oct 2026 · 08:56:57

4.12 Install New Relic

Deploy tool	New Relic	MongoDB	Completed with warnings

New Relic is installed, ready to monitor the databases in this cluster.

Started 25 Oct 2026, 08:59 · took 44s

COMMAND

```
SCOPE=kubernetes observability/newrelic/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available

- ✔ Cluster identity: minikube
- ✔ Namespace available: newrelic
- ! Dashboard Workload - Sybase ASE not installed: authentication required
- ✔ Dashboard links for CONNECTION INFO: runtime/state/newrelic/dashboards.json

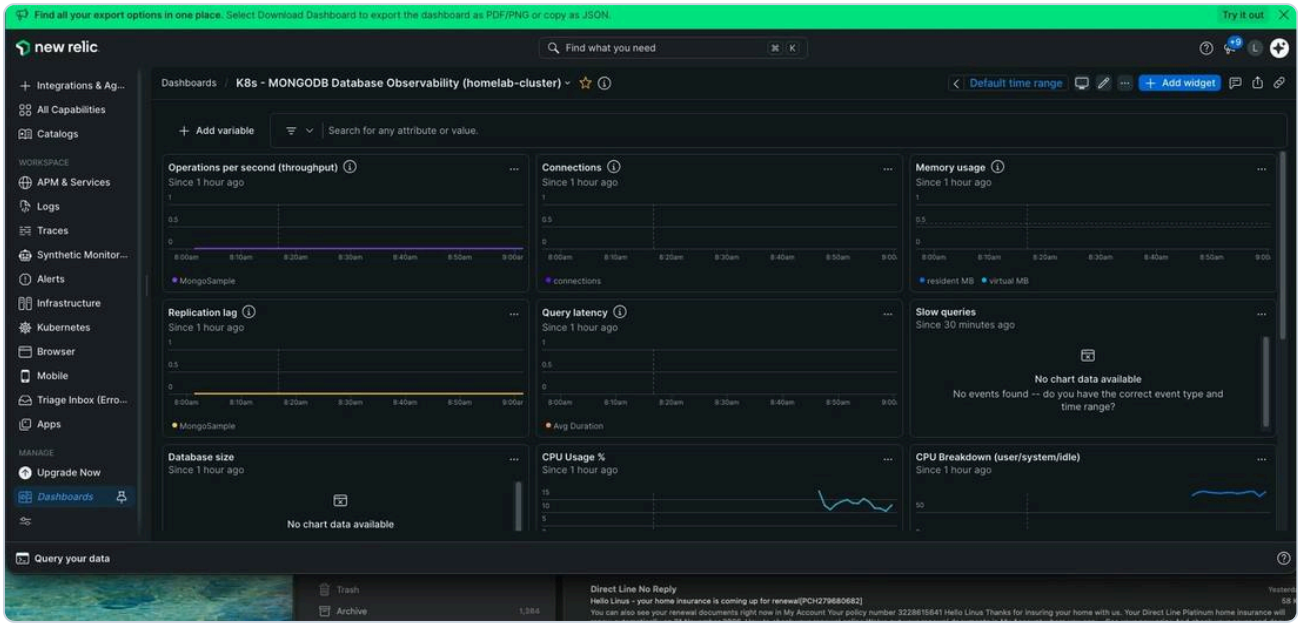


Figure 4.28 25 Oct 2026 · 09:02:01

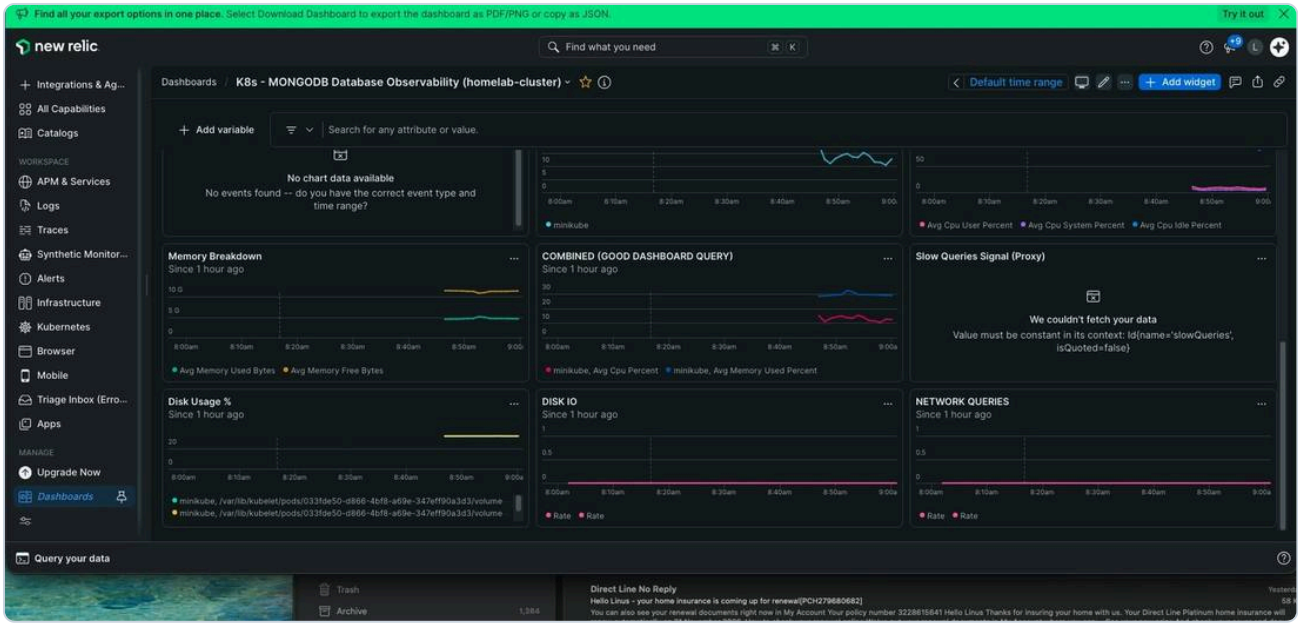


Figure 4.29 25 Oct 2026 · 09:02:10

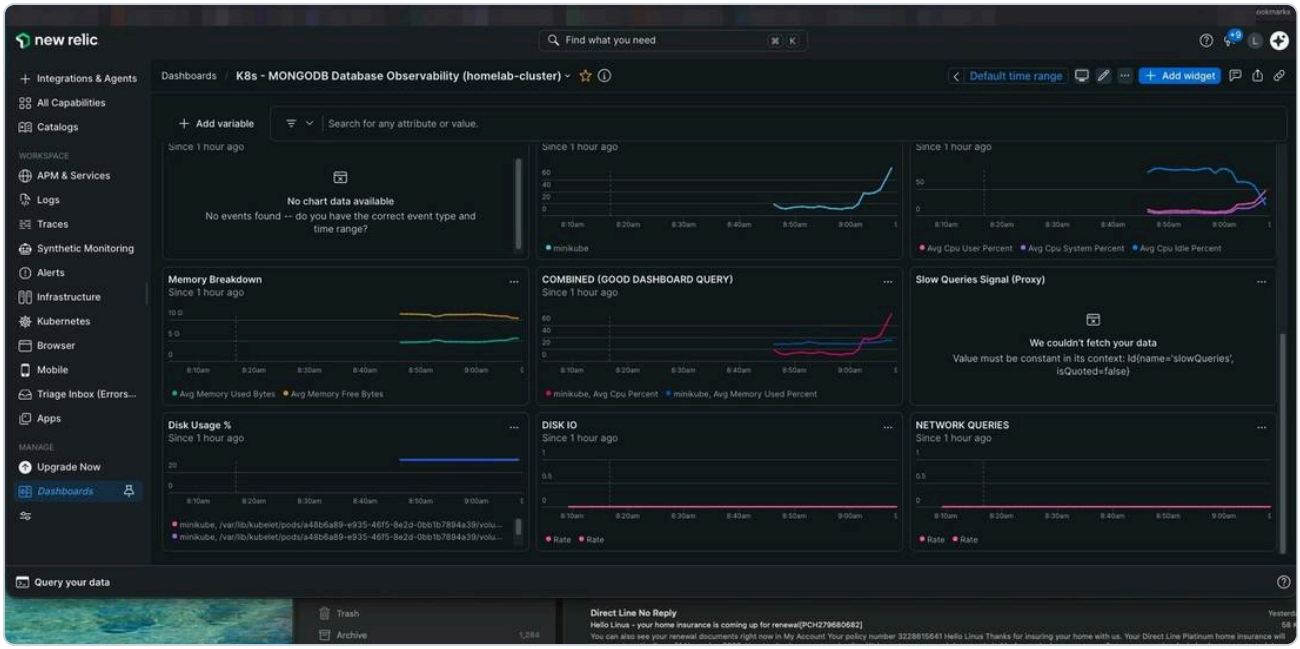


Figure 4.30 25 Oct 2026 · 09:09:34

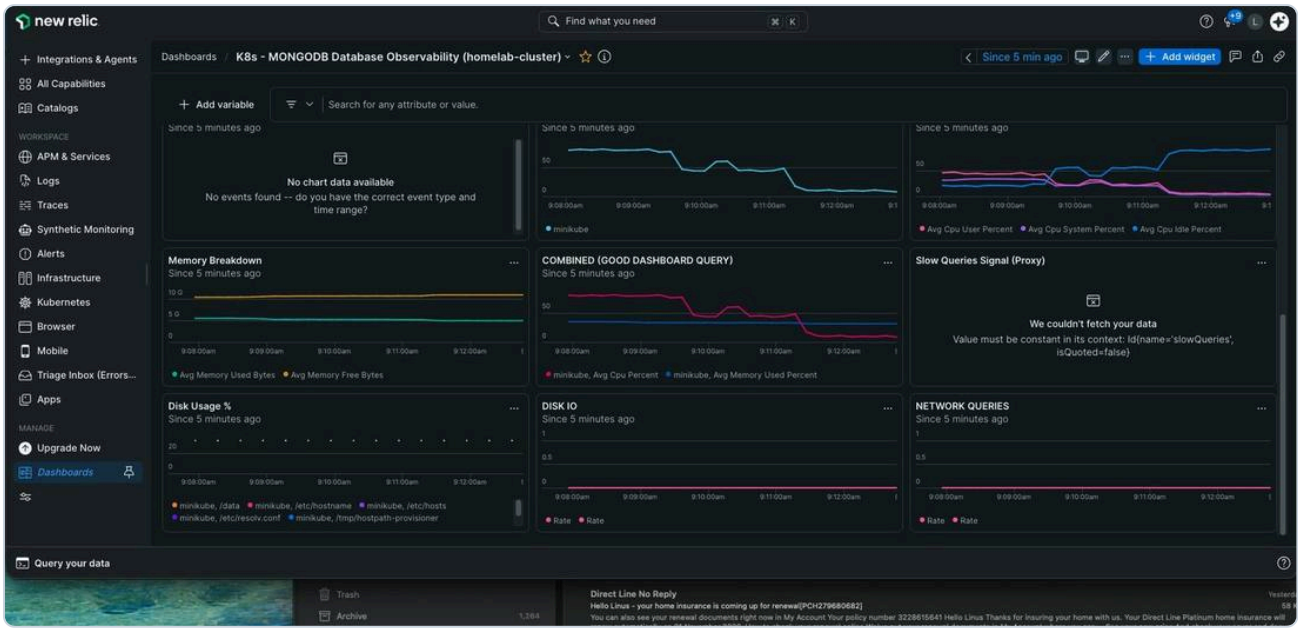


Figure 4.31 25 Oct 2026 · 09:13:05

4.13 Unregister MongoDB from New Relic

Unregister | New Relic | MongoDB | Completed

MongoDB is removed from New Relic; the database itself keeps running.

Started 25 Oct 2026, 09:25 · took 5s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/unregister.sh
```

- ✔ Kubernetes API reachable
- ✔ MongoDB New Relic registration state removed

- ✔ Removed Deployment/mongodb-newrelic-exporter (namespace mongodb)
- ✔ Removed Service/mongodb-newrelic-exporter (namespace mongodb)
- ✔ New Relic kubelet agents refreshed
- ✔ mongodb-config.yml absent

4.14 Remove New Relic

Unregister	New Relic	MongoDB	Completed

New Relic is removed from the cluster once no database is registered with it.

Started 25 Oct 2026, 09:26 · took 16s

COMMAND

```
SCOPE=kubernetes observability/newrelic/deploy/kubernetes/destroy.sh
```

- ✔ Kubernetes reachable
- ✔ Helm release removed
- ✔ Namespace removed
- ✔ New Relic platform removed

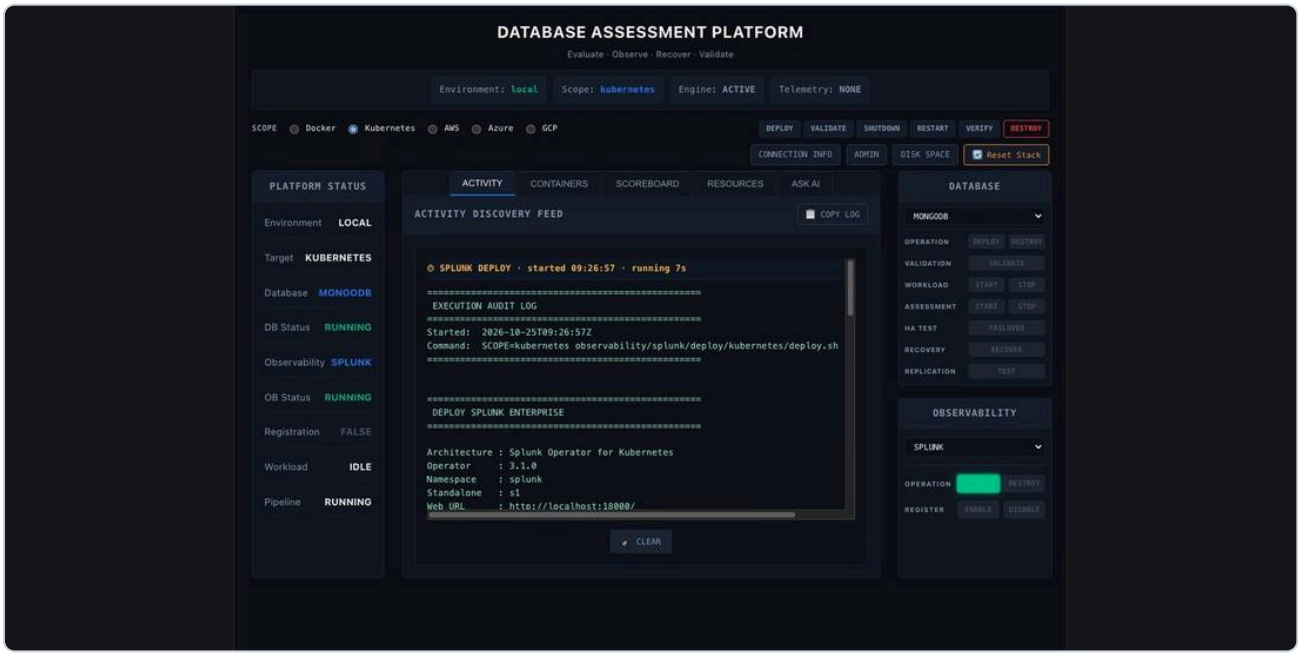


Figure 4.32 25 Oct 2026 · 09:27:05

4.15 Install Splunk

Deploy tool	Splunk	MongoDB	Completed with warnings

Splunk is installed, ready to monitor the databases in this cluster.

Started 25 Oct 2026, 09:26 · took 2m 28s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available
- ✔ Splunk Operator manifest applied
- ✔ SPLUNK_GENERAL_TERMS configured on Operator
- ✔ No pods in a failed state
- ✔ Splunk accepted a HEC test event

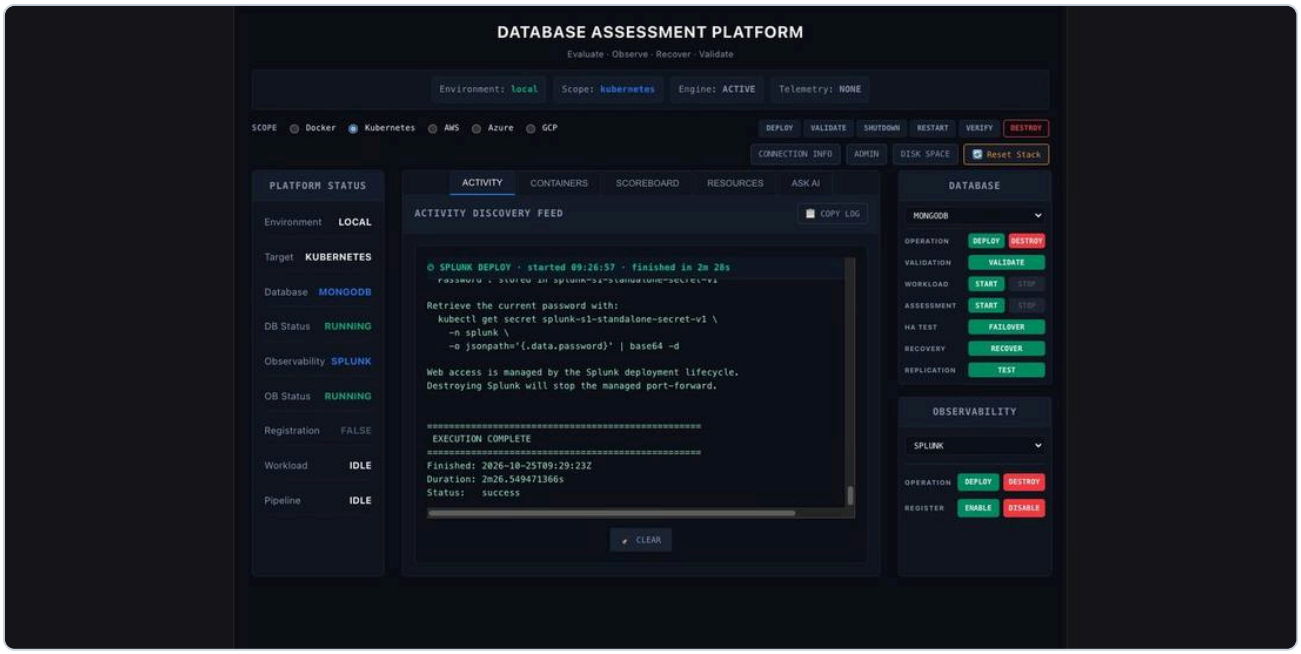


Figure 4.33 25 Oct 2026 · 09:30:30

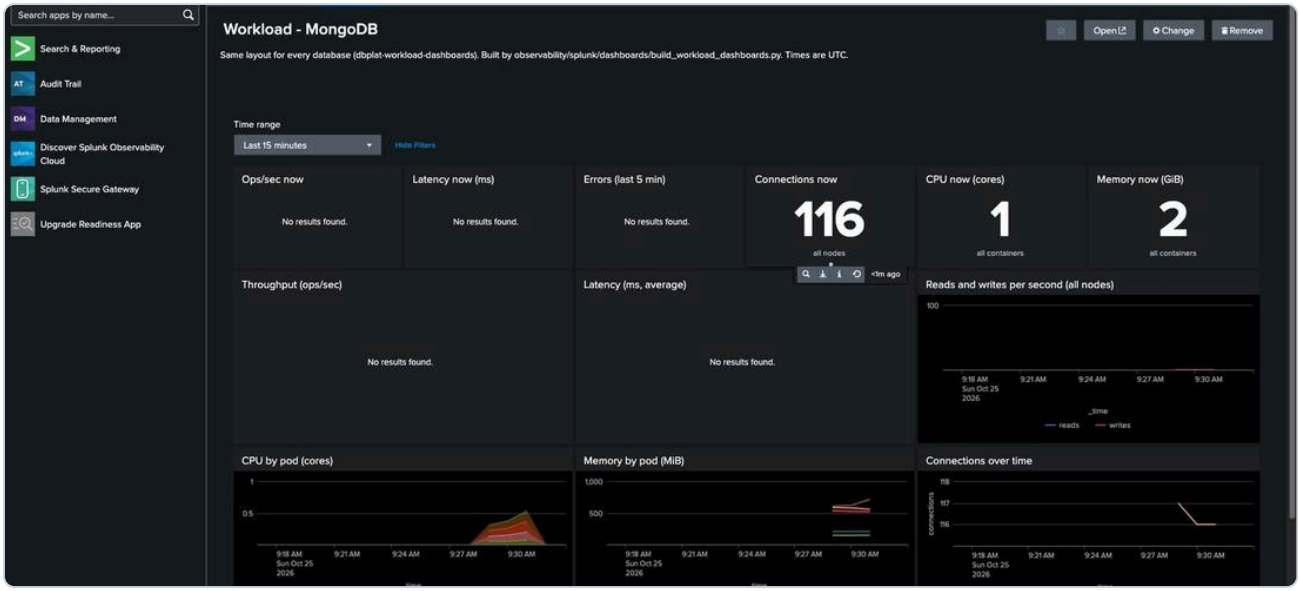


Figure 4.34 25 Oct 2026 · 09:32:50

4.16 Register MongoDB with Splunk

Register | Splunk | MongoDB | Completed

MongoDB is connected to Splunk, so its metrics and dashboards appear there.

Started 25 Oct 2026, 09:33 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/mongodb/register.sh
```

- ✔ PSMDB: mongodb-psmdb-db
- ✔ Splunk HEC service: splunk-s1-standalone-service
- ✔ Backup written:
- ✔ MongoDB Splunk HEC secret converged
- ✔ MongoDB Splunk HEC secret validated
- ✔ MongoDB logs successfully registered into Splunk

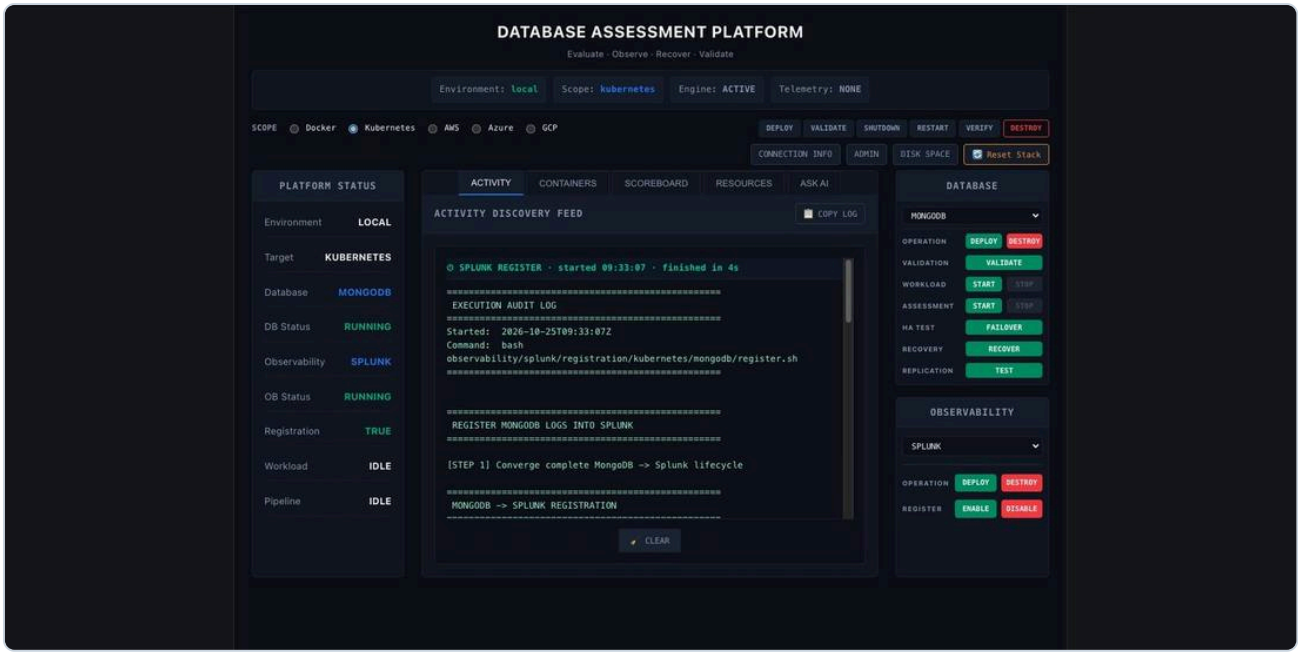


Figure 4.35 25 Oct 2026 · 09:35:12

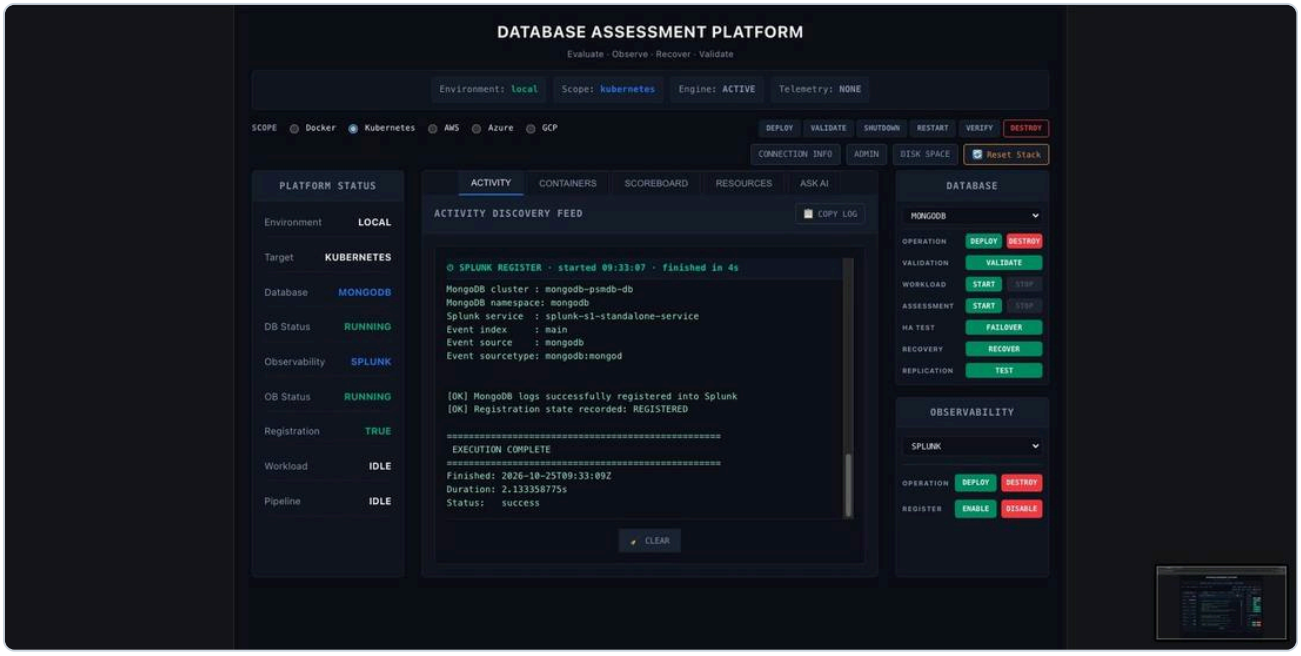


Figure 4.36 25 Oct 2026 · 09:35:17

4.17 Workload · MongoDB

Workload	MongoDB	Completed

A load generator runs against MongoDB, so the monitoring shows real traffic. It stops by itself after a

set time.

Started 25 Oct 2026, 09:35 · took 32s

COMMAND

```
bash db-platform/mongodb/workload/kubernetes/start.sh
```

- ✔ MongoDB YCSB workload Pod is Running.

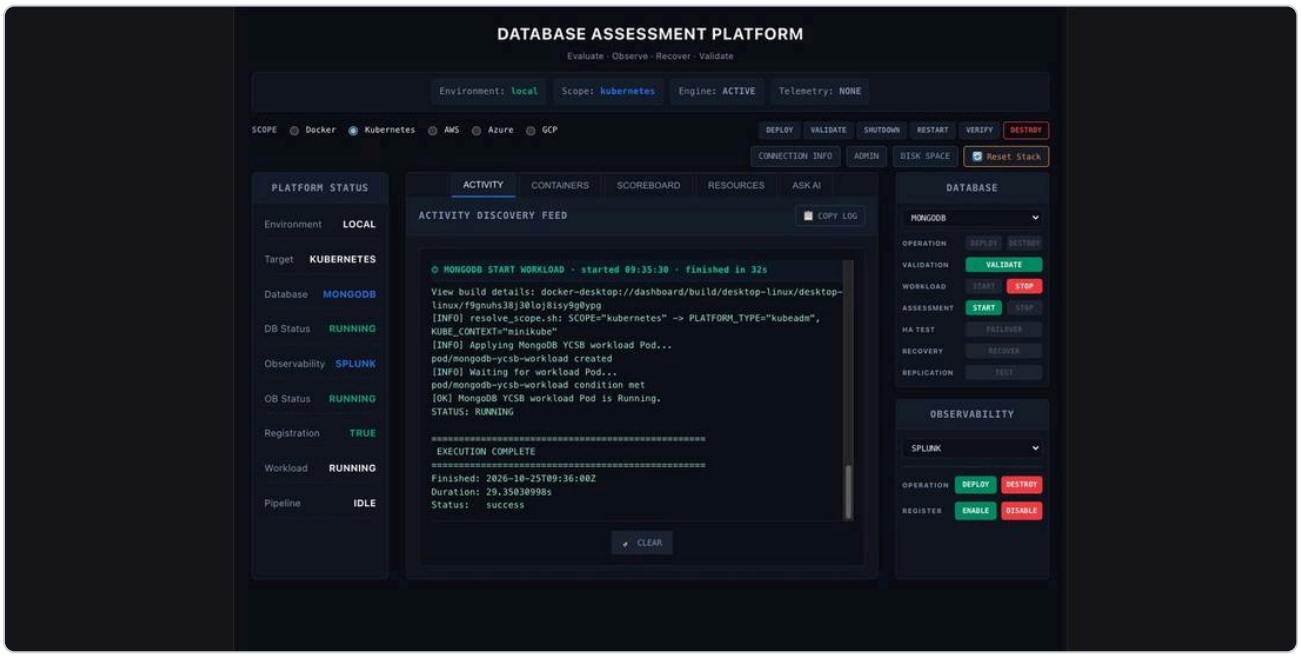


Figure 4.37 25 Oct 2026 · 09:36:54

4.18 Point-in-time recovery test · MongoDB

Assessment | MongoDB | Completed

RunShore restores MongoDB to a chosen moment from its backups and checks the restored data is correct.

Started 25 Oct 2026, 09:37 · took 2m 22s

COMMAND

```
SCOPE=kubernetes db-platform/mongodb/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ Cluster: mongodb-psmdb-db
- ✔ PITR archiving enabled
- ✔ Baseline seeded
- ✔ Backup completed: pitr-demo-1792921030
- ✔ Post-marker orders inserted
- ✔ Archived oplog covers the target (2026-10-25T09:37:20 - 2026-10-25T09:38:26) after 60s

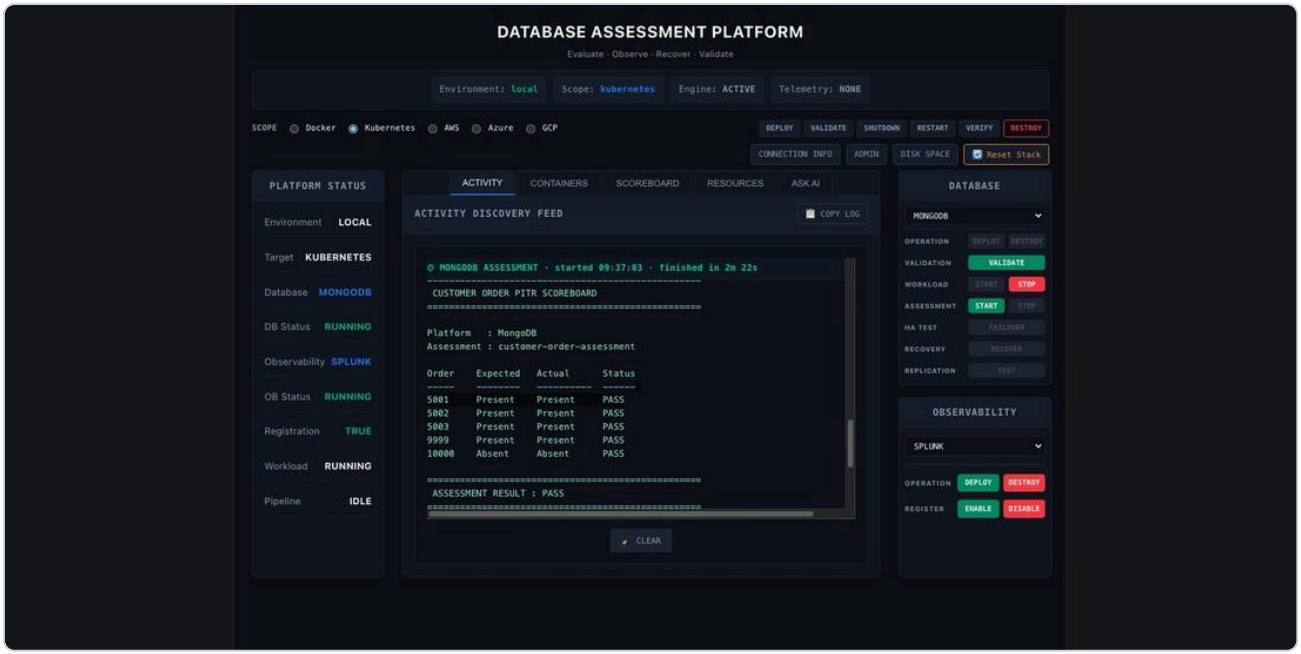


Figure 4.38 25 Oct 2026 · 09:41:51

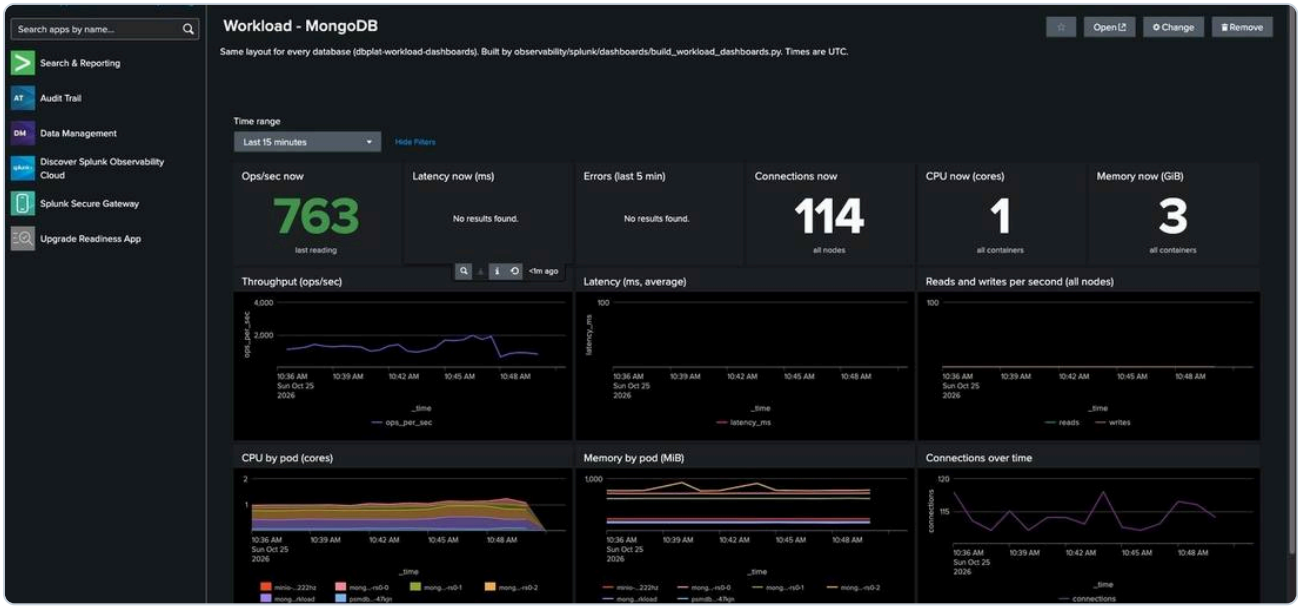


Figure 4.39 25 Oct 2026 · 10:51:54

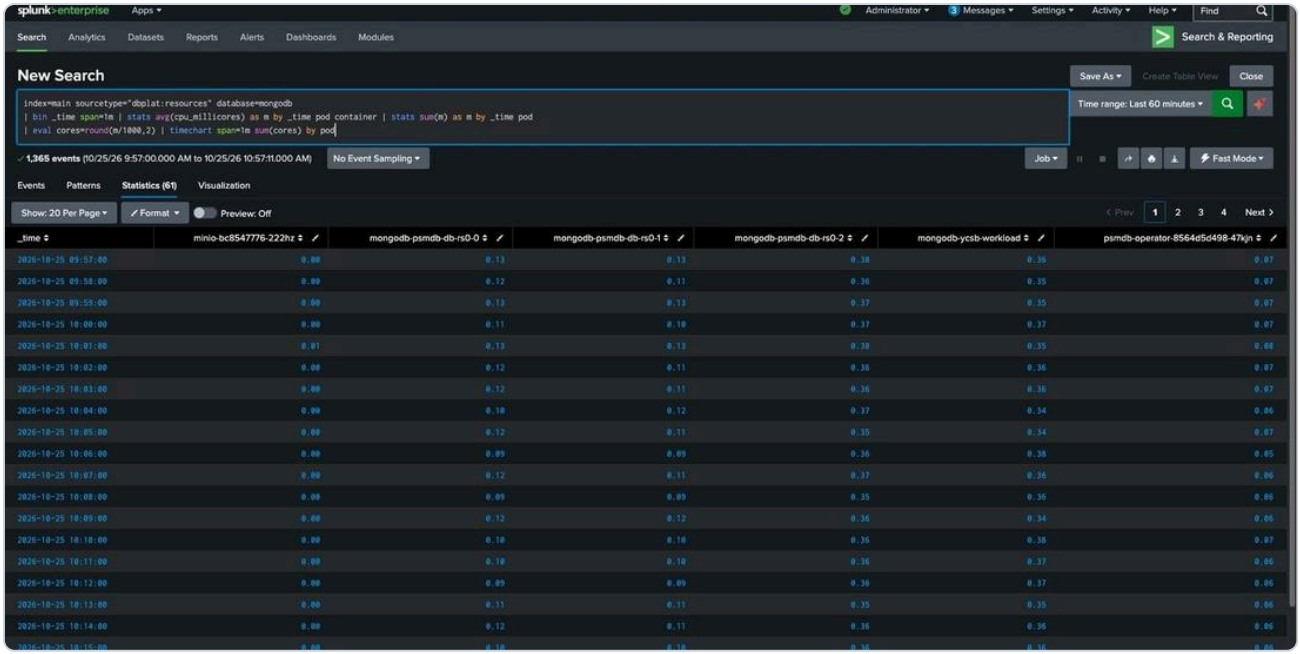


Figure 4.40 25 Oct 2026 · 10:57:21

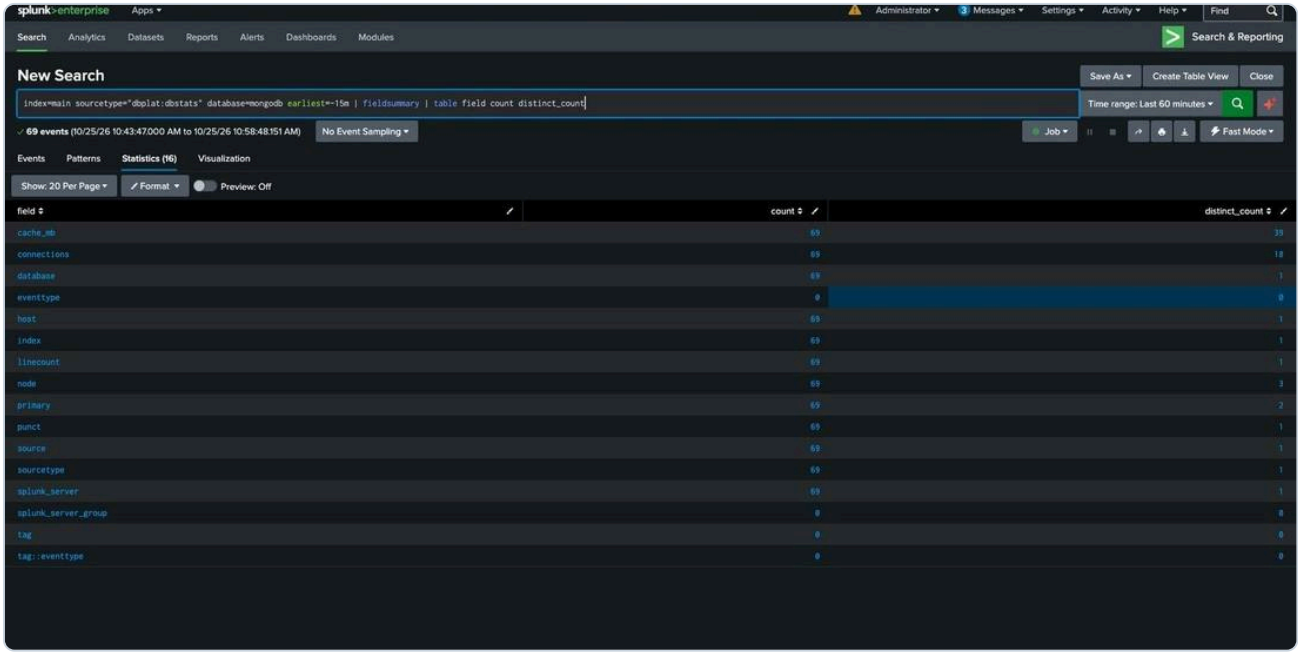


Figure 4.41 25 Oct 2026 · 10:58:58

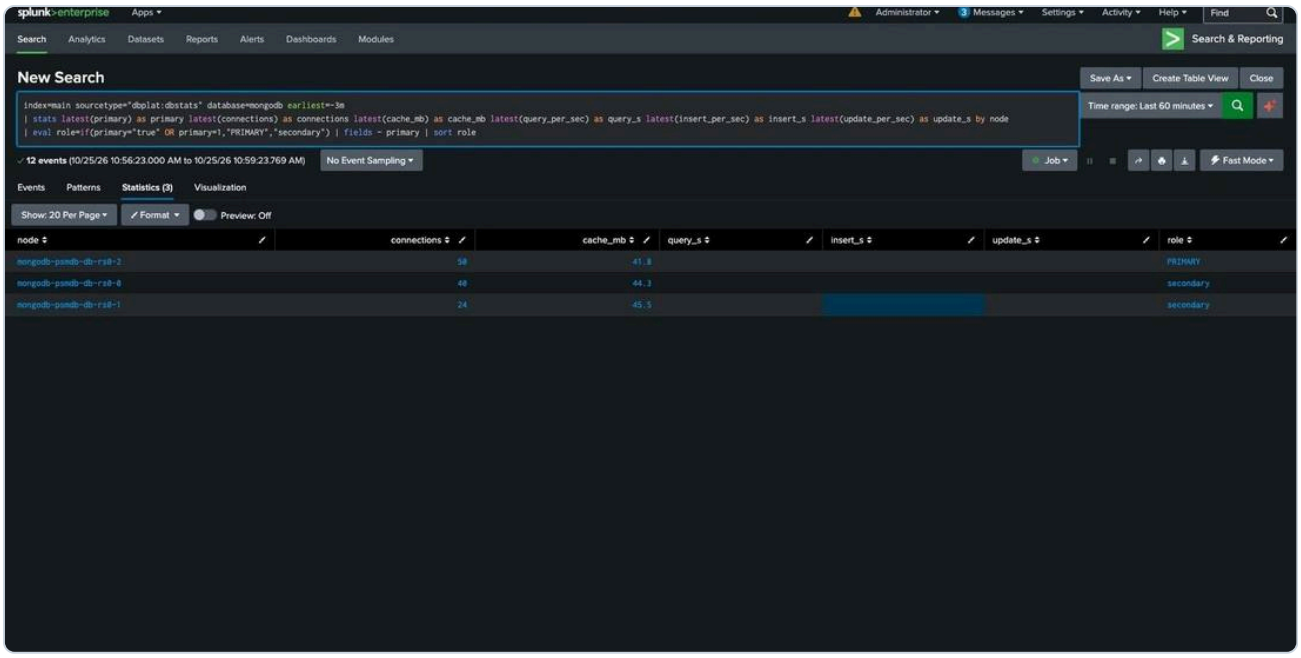


Figure 4.42 25 Oct 2026 · 10:59:28

4.19 Unregister MongoDB from Splunk

Unregister	Splunk	MongoDB	Completed

MongoDB is removed from Splunk; the database itself keeps running.

Started 25 Oct 2026, 11:00 · took 5s

COMMAND

```
bash observability/splunk/registration/kubernetes/mongodb/unregister.sh
```

- ✔ Splunk logcollector configuration cleared
- ✔ Confirmed: no Splunk configuration or envFrom remains on the CR
- ✔ All pods have rolled — Splunk forwarding should now be fully stopped
- ✔ MongoDB Splunk log forwarding removed
- ✔ Registration state cleared

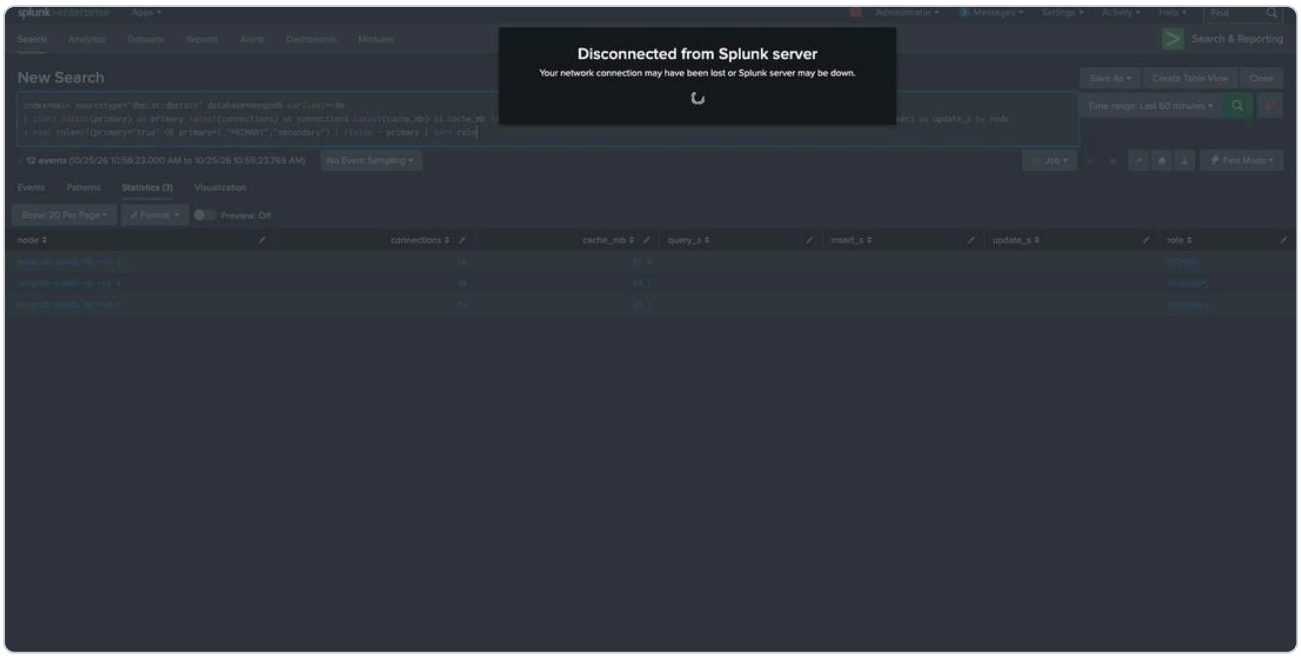


Figure 4.43 25 Oct 2026 · 11:03:24

4.20 Remove Splunk

Unregister | Splunk | MongoDB | Completed

Splunk is removed from the cluster once no database is registered with it.

Started 25 Oct 2026, 11:01 · took 1m 14s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/destroy.sh
```

- ✔ Managed Splunk Web port-forward stopped
- ✔ Splunk Web port 18000 released
- ✔ Workload feed to Splunk stopped
- ✔ Kubernetes API reachable
- ✔ Splunk resources removed
- ✔ Splunk Web access removed

4.21 Destroy MongoDB

Destroy | MongoDB | Completed

MongoDB is removed and its storage released.

Started 25 Oct 2026, 11:05 · took 4s

COMMAND

```
bash db-platform/mongodb/workload/kubernetes/stop.sh
```

05 MySQL

Local Kubernetes

MySQL is deployed as a Percona XtraDB (Galera) cluster behind HAProxy, then monitored, loaded and tested.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Local Kubernetes	MySQL	New Relic, PMM, Splunk	21 screens · 18 logs	26 Oct–27 Oct 2026

5.1 Deploy MySQL

Deploy database | MySQL | Completed

RunShore deploys MySQL on Local Kubernetes and waits until every node is ready.

Started 26 Oct 2026, 15:15 · took 4m 48s

COMMAND

```
SCOPE=kubernetes db-platform/mysql/deploy/kubernetes/deploy.sh
```

- ✔ PMM 3 token source validated
- ✔ Namespace ready
- ✔ PXC secret applied
- ✔ root present
- ✔ PXC cluster health converged
- ✔ PXC deployment complete

5.2 Point-in-time recovery test · MySQL

Assessment | MySQL | Completed with warnings

RunShore restores MySQL to a chosen moment from its backups and checks the restored data is correct.

Started 26 Oct 2026, 15:59 · took 6m 25s

COMMAND

```
SCOPE=kubernetes db-platform/mysql/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.
- ✔ Live root credential resolved from Kubernetes Secret pxc-db-secrets
- ✔ Prerequisites available
- ✔ Restore completed: pitr-demo-restore-1793030499
- ✔ PXC cluster ready

5.3 Failover test · MySQL

Failover | MySQL | Completed

The primary MySQL node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 26 Oct 2026, 16:55 · took 1m 18s

COMMAND

```
SCOPE=kubernetes db-platform/mysql/failover/kubernetes/run_failover.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.
- ✔ PXC administrative credential available
- ✔ All three PXC members are Kubernetes-Ready and Primary/Synced/Ready
- ✔ Assessment data verified on pxc-db-pxc-2
- ✔ Final three-member cluster is Kubernetes-Ready and Primary/Synced/Ready

5.4 Replication test · MySQL

Replication | MySQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 26 Oct 2026, 16:59 · took 44s

COMMAND

```
SCOPE=kubernetes db-platform/mysql/replication/kubernetes/run_replication.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.

5.5 Register MySQL with New Relic

Register | New Relic | MySQL | Completed

MySQL is connected to New Relic, so its metrics and dashboards appear there.

Started 26 Oct 2026, 17:02 · took 8s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/register.sh
```

- ✔ Kubernetes, PXC and New Relic available
- ✔ PXC cluster ready

- ✔ MySQL endpoint discovered
- ✔ PXC administrative credential available
- ✔ PXC HAProxy endpoint present
- ✔ MySQL telemetry secret present

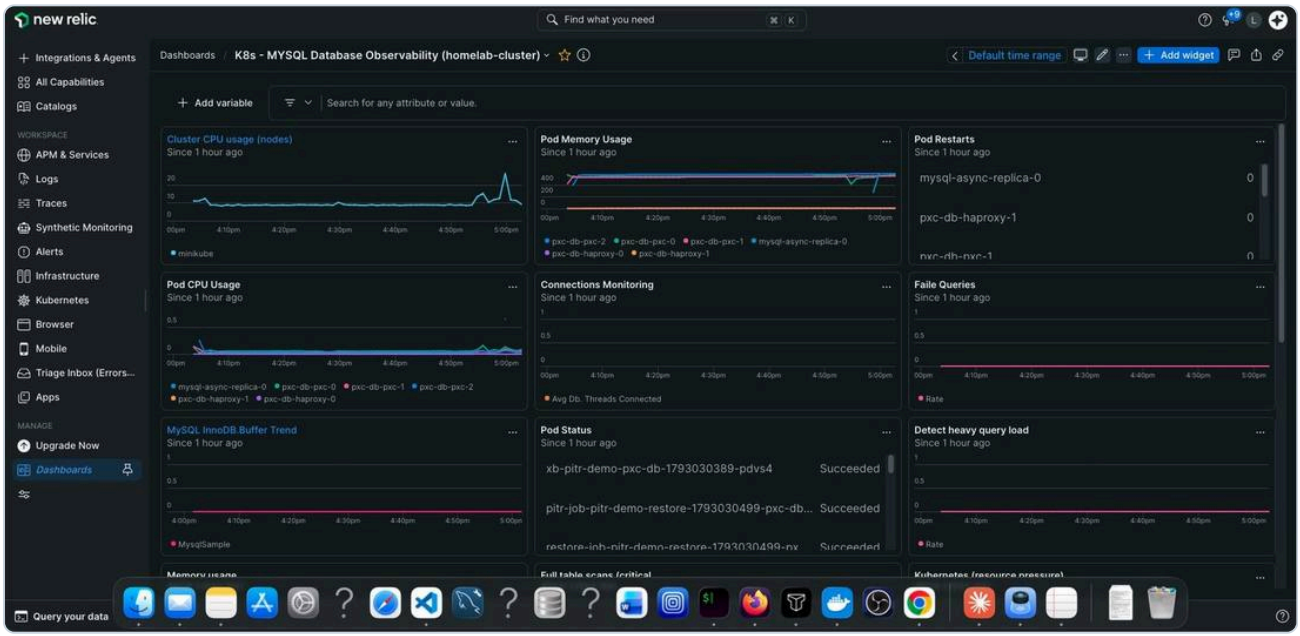


Figure 5.1 26 Oct 2026 · 17:03:54

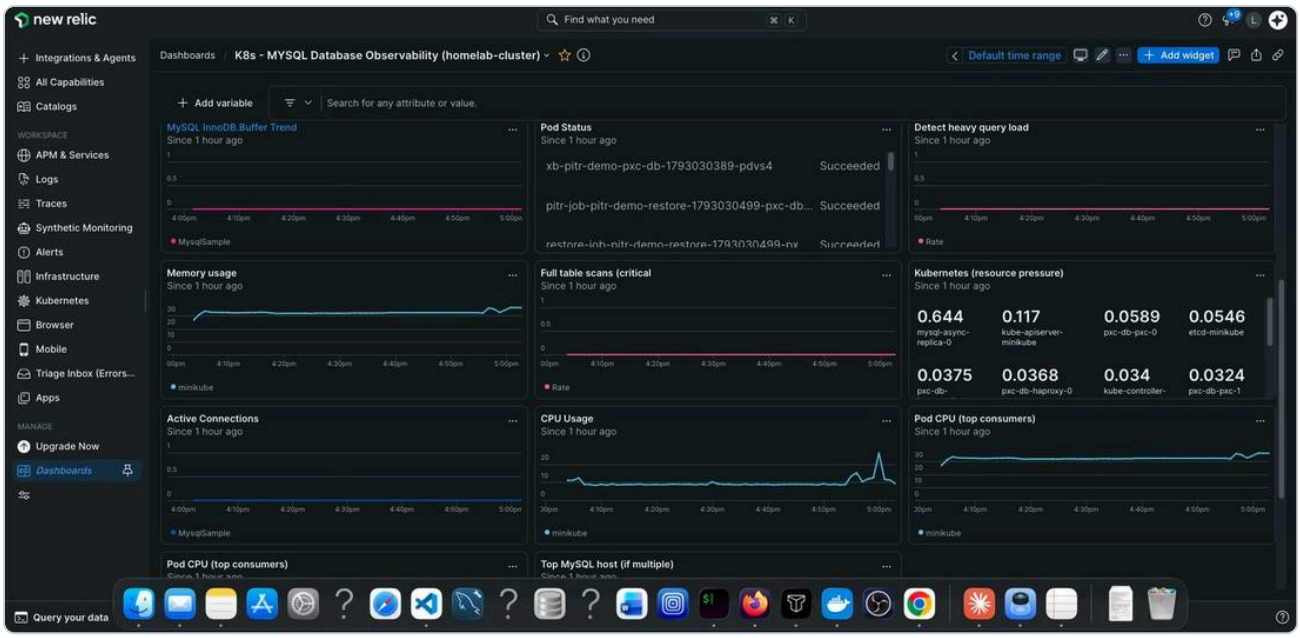


Figure 5.2 26 Oct 2026 · 17:04:08

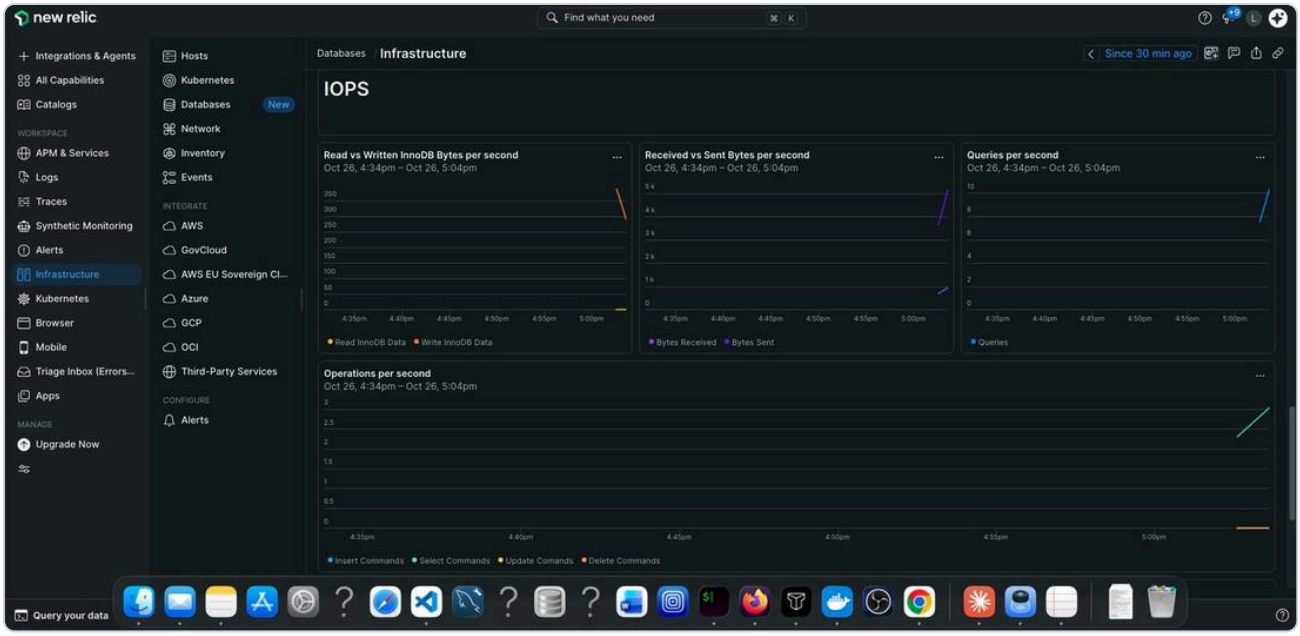


Figure 5.3 26 Oct 2026 · 17:05:16

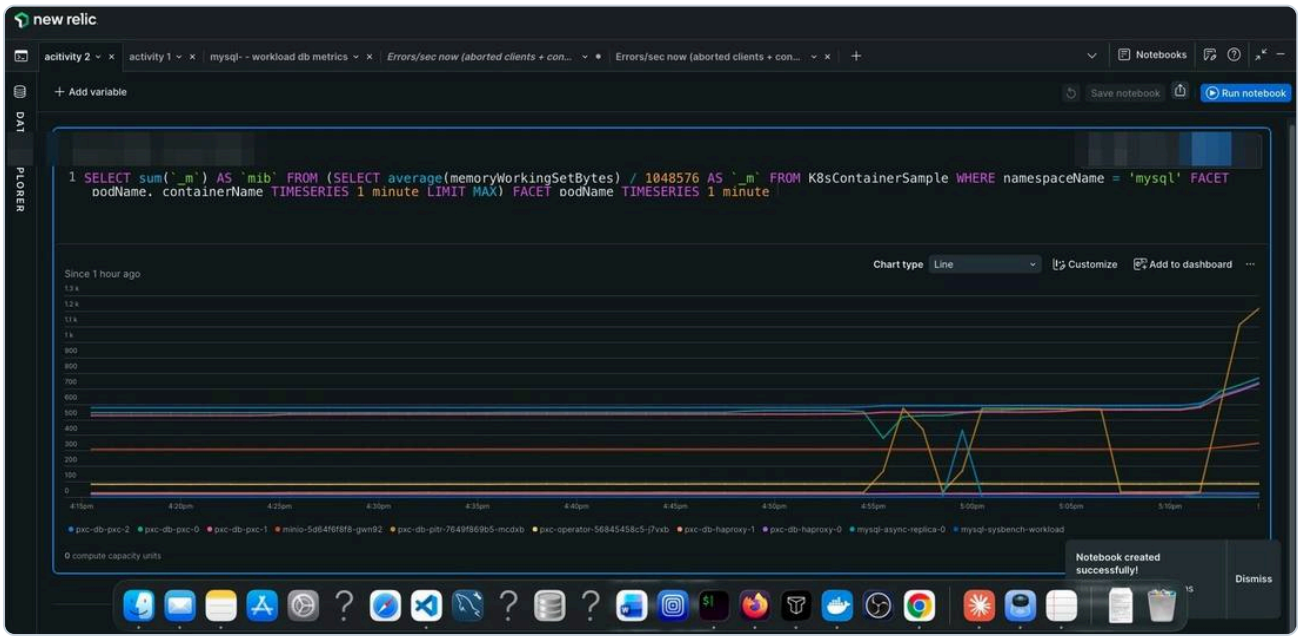


Figure 5.4 26 Oct 2026 · 17:16:58

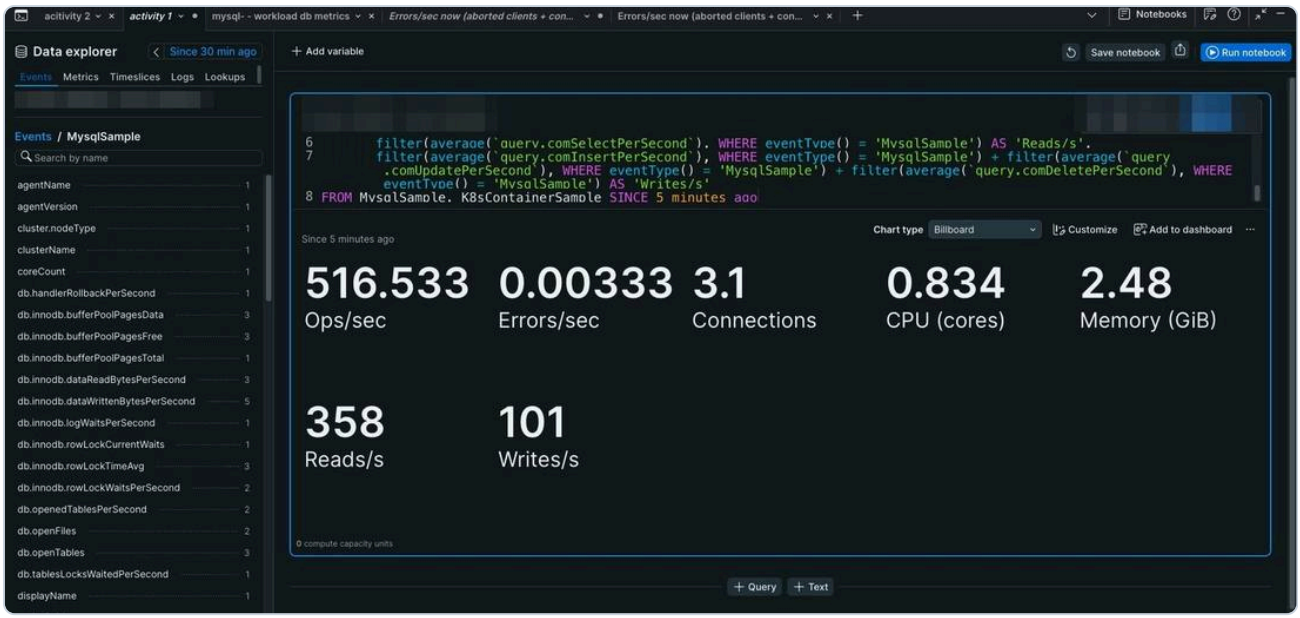


Figure 5.5 26 Oct 2026 · 17:17:58

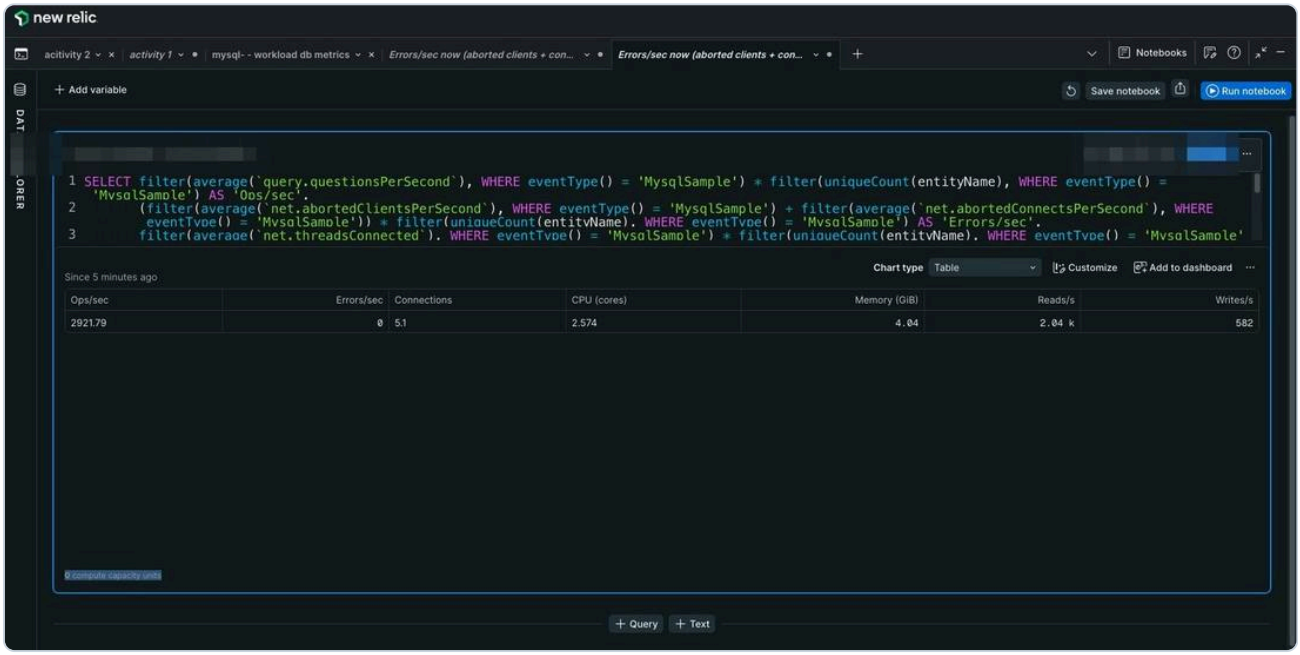


Figure 5.6 26 Oct 2026 · 17:19:32

5.6 Unregister MySQL from New Relic

Unregister	New Relic	MySQL	Completed

MySQL is removed from New Relic; the database itself keeps running.

Started 26 Oct 2026, 17:20 · took 8s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/unregister.sh
```

- ✔ Kubernetes and New Relic available
- ✔ MySQL registration state cleared
- ✔ Removed mysql-config.yml

- ✔ Removed newrelic-mysql-secret
- ✔ mysql-config.yml absent
- ✔ newrelic-mysql-secret absent

5.7 Install New Relic

Deploy tool	New Relic	MySQL	Completed

New Relic is installed, ready to monitor the databases in this cluster.

Started 26 Oct 2026, 17:21 · took 2m 24s

COMMAND

```
SCOPE=kubernetes observability/newrelic/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available
- ✔ Cluster identity: minikube
- ✔ Namespace available: newrelic
- ✔ Dashboard links for CONNECTION INFO: runtime/state/newrelic/dashboards.json
- ✔ results ledger: 0 new check(s), 1061 in total

5.8 Remove New Relic

Unregister	New Relic	MySQL	Completed

New Relic is removed from the cluster once no database is registered with it.

Started 26 Oct 2026, 17:24 · took 16s

COMMAND

```
SCOPE=kubernetes observability/newrelic/deploy/kubernetes/destroy.sh
```

- ✔ Kubernetes reachable
- ✔ Helm release removed
- ✔ Namespace removed
- ✔ New Relic platform removed

5.9 Install Splunk

Deploy tool	Splunk	MySQL	Completed with warnings

Splunk is installed, ready to monitor the databases in this cluster.

Started 26 Oct 2026, 17:25 · took 5m 05s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/deploy.sh
```

- ✔ Kubernetes API reachable
- ✔ Required tooling available
- ✔ Splunk Operator manifest applied
- ✔ SPLUNK_GENERAL_TERMS configured on Operator
- ✔ No pods in a failed state
- ✔ Splunk accepted a HEC test event

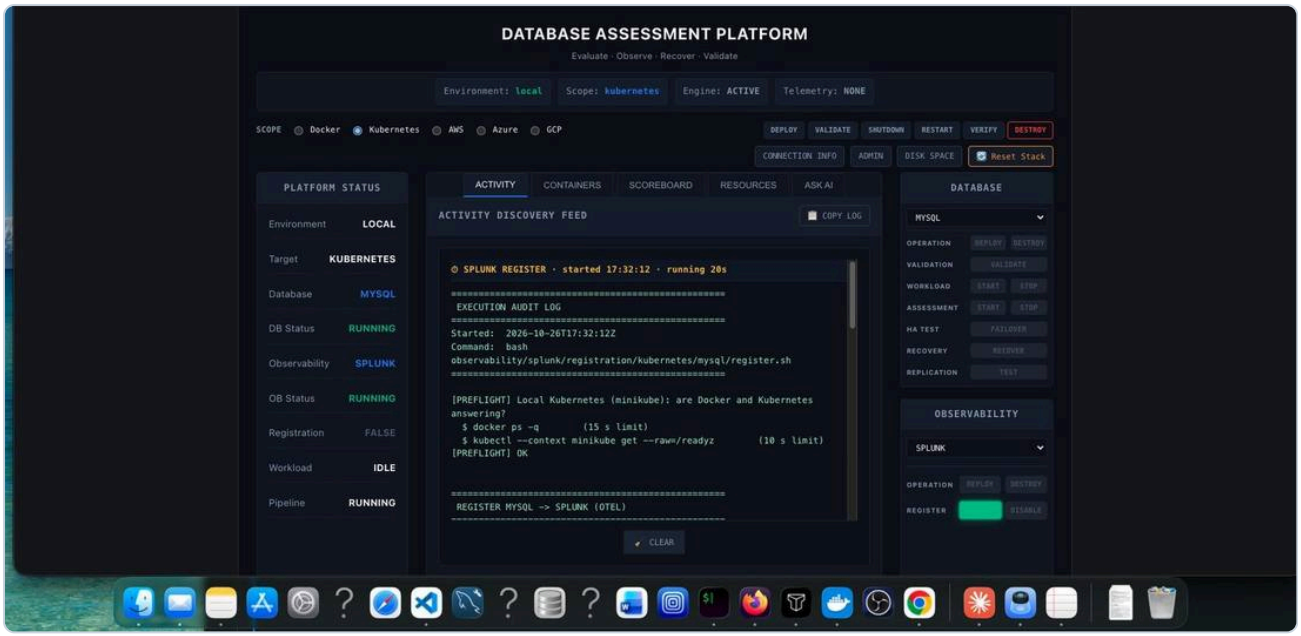


Figure 5.7 26 Oct 2026 · 17:32:32

5.10 Register MySQL with Splunk

Register	Splunk	MySQL	Completed

MySQL is connected to Splunk, so its metrics and dashboards appear there.

Started 26 Oct 2026, 17:32 · took 4m 37s

COMMAND

```
bash observability/splunk/registration/kubernetes/mysql/register.sh
```

- ✔ PXC: pxc-db
- ✔ Splunk HEC service: splunk-s1-standalone-service

- ✔ Splunk runtime Secret: splunk-splunk-secret
- ✔ mysql/mysql-splunk-hec converged
- ✔ Percona logrotate container present
- ✔ MySQL → Splunk OTel registration complete

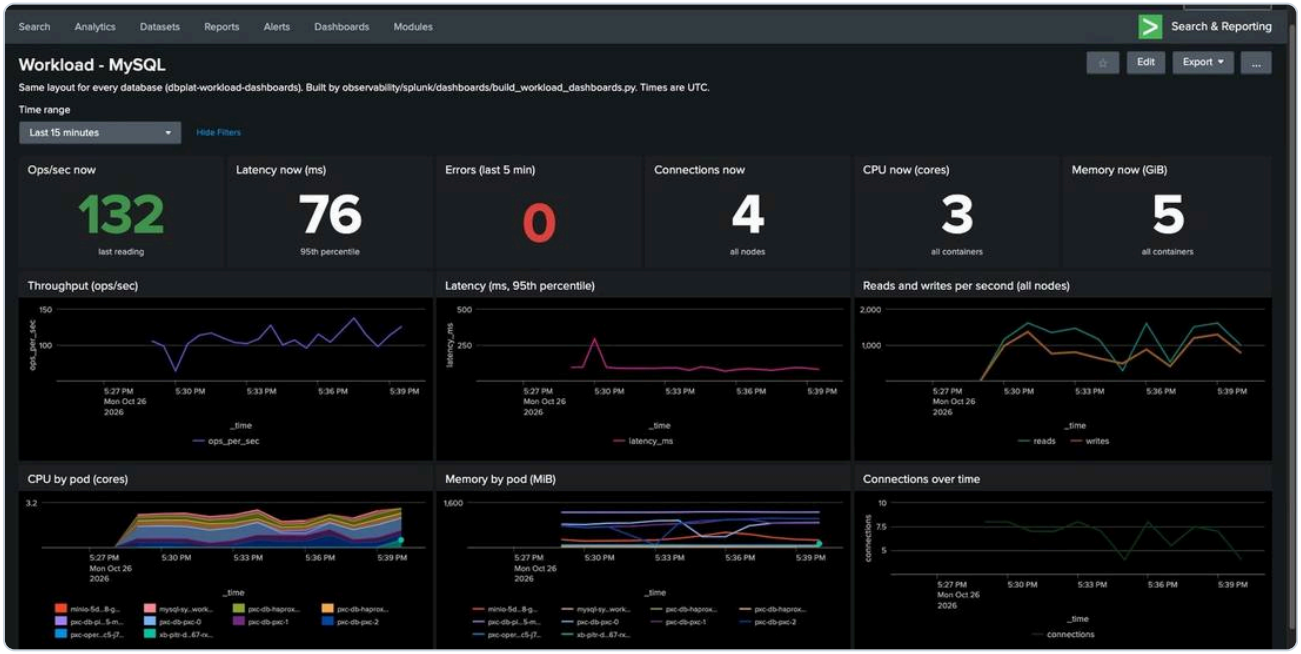


Figure 5.8 26 Oct 2026 · 17:40:34

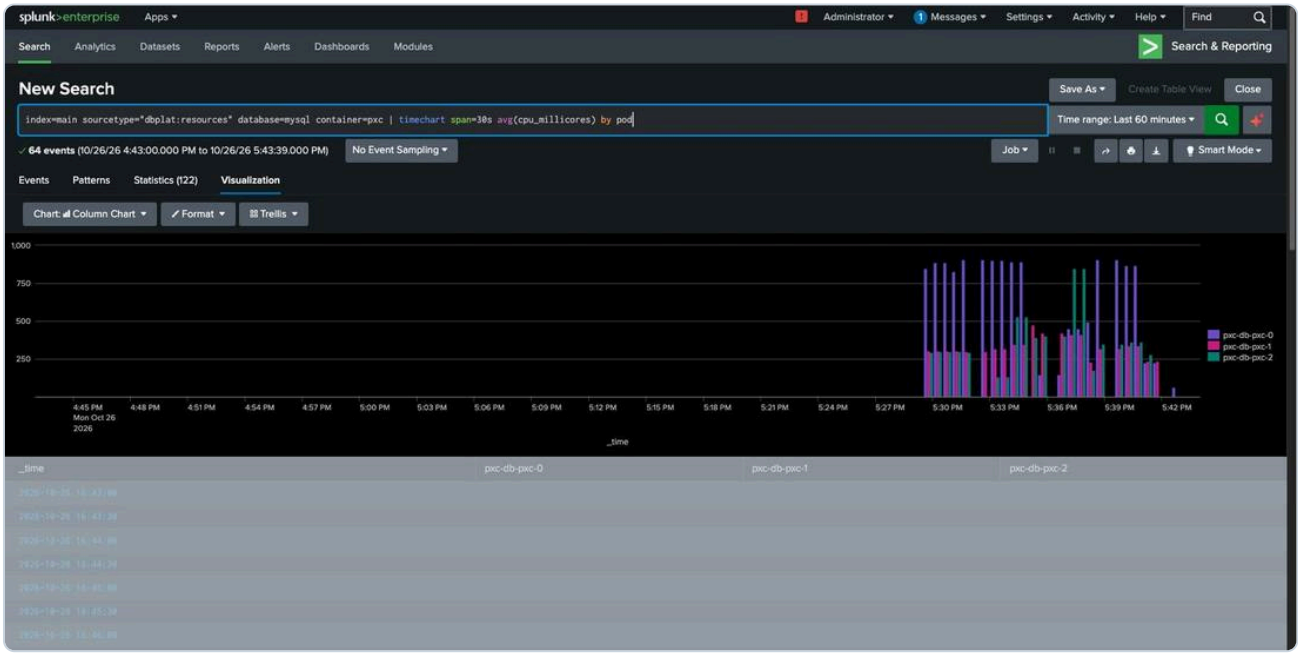


Figure 5.9 26 Oct 2026 · 17:45:28



Figure 5.10 26 Oct 2026 · 17:46:09

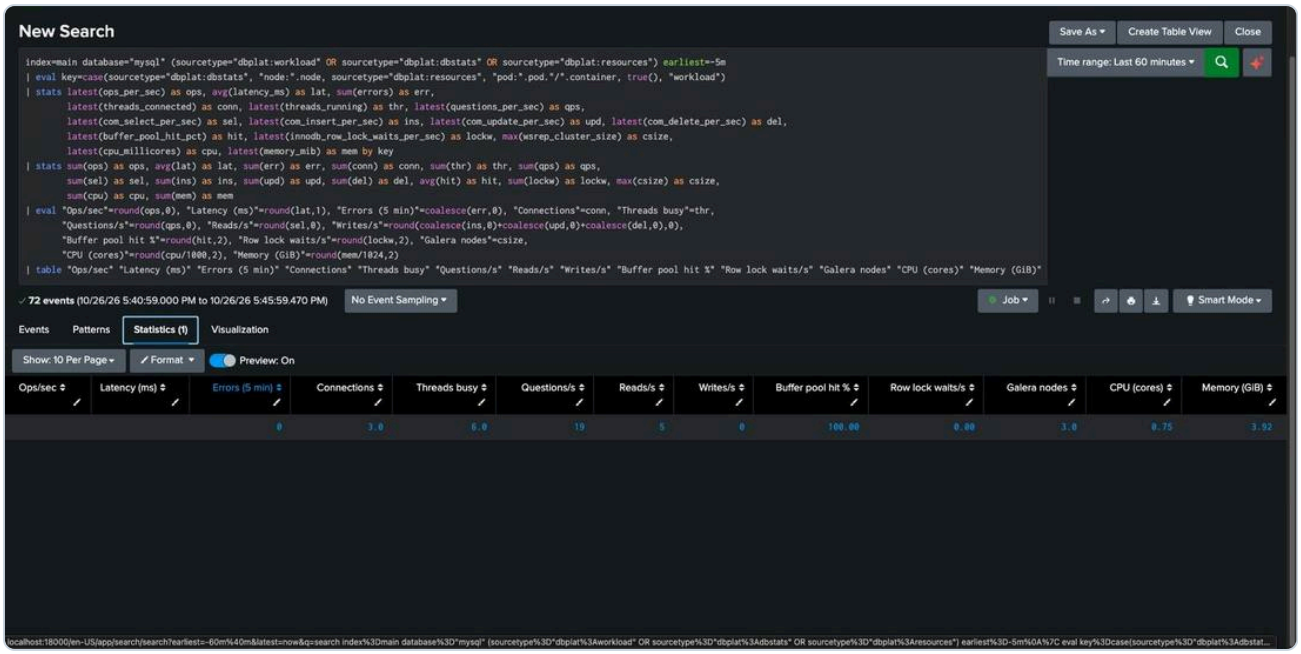


Figure 5.11 26 Oct 2026 · 17:46:23

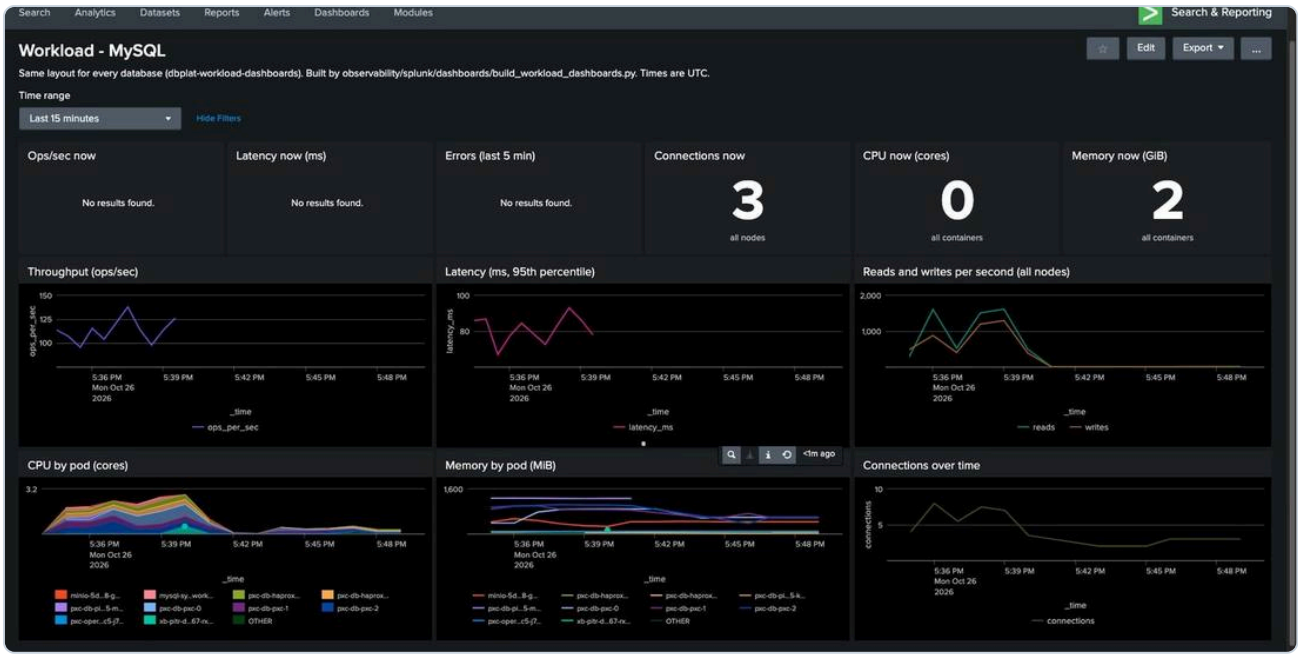


Figure 5.12 26 Oct 2026 · 17:50:14

5.11 Unregister MySQL from Splunk

Unregister | Splunk | MySQL | Completed

MySQL is removed from Splunk; the database itself keeps running.

Started 26 Oct 2026, 17:50 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/mysql/unregister.sh
```

- ✔ Percona LogCollector configuration disabled
- ✔ MySQL-local Splunk HEC Secret removed
- ✔ MySQL → Splunk OTel registration removed
- ✔ Registration state cleared

5.12 Remove Splunk

Unregister | Splunk | MySQL | Completed

Splunk is removed from the cluster once no database is registered with it.

Started 26 Oct 2026, 17:51 · took 1m 13s

COMMAND

```
SCOPE=kubernetes observability/splunk/deploy/kubernetes/destroy.sh
```

- ✔ Managed Splunk Web port-forward stopped
- ✔ Splunk Web port 18000 released
- ✔ Workload feed to Splunk stopped
- ✔ Kubernetes API reachable
- ✔ Splunk resources removed
- ✔ Splunk Web access removed

5.13 Install PMM

Deploy tool	PMM	MySQL	Completed with warnings

PMM is installed, ready to monitor the databases in this cluster.

Started 26 Oct 2026, 17:52 · took 2m 48s

COMMAND

```
SCOPE=kubernetes observability/pmm/deploy/kubernetes/deploy_pmm_stack.kubernetes.sh
```

- ✔ Kubernetes API reachable
- ✔ Required lifecycle scripts found.
- ✔ Kubernetes API reachable
- ✔ Namespace pmm is Active
- ✔ The PMM token in ADMIN is accepted -- kept
- ✔ PMM server accepts the token from ADMIN

5.14 Register MySQL with PMM

Register	PMM	MySQL	Completed

MySQL is connected to PMM, so its metrics and dashboards appear there.

Started 27 Oct 2026, 01:42 · took 16s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/register_pxc_service.sh
```

- ✔ pmm-0
- ✔ Credentials loaded
- ✔ PXC successfully registered into PMM



Figure 5.13 27 Oct 2026 · 01:46:45

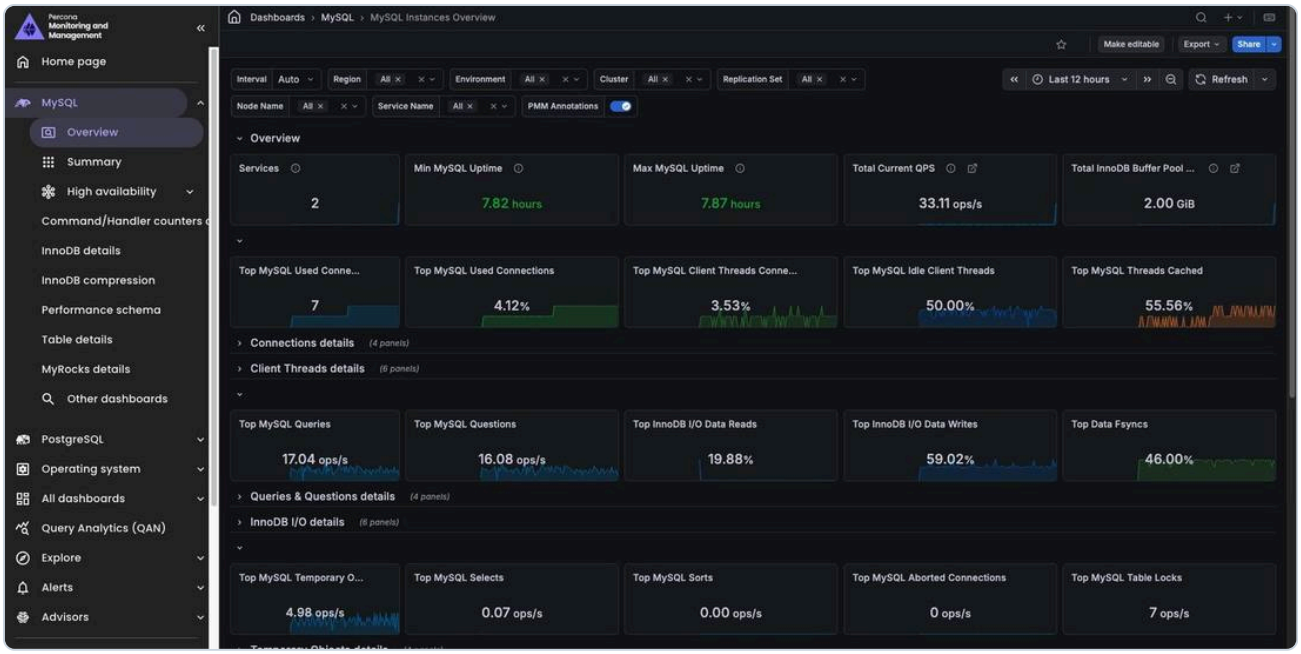


Figure 5.14 27 Oct 2026 · 01:47:01

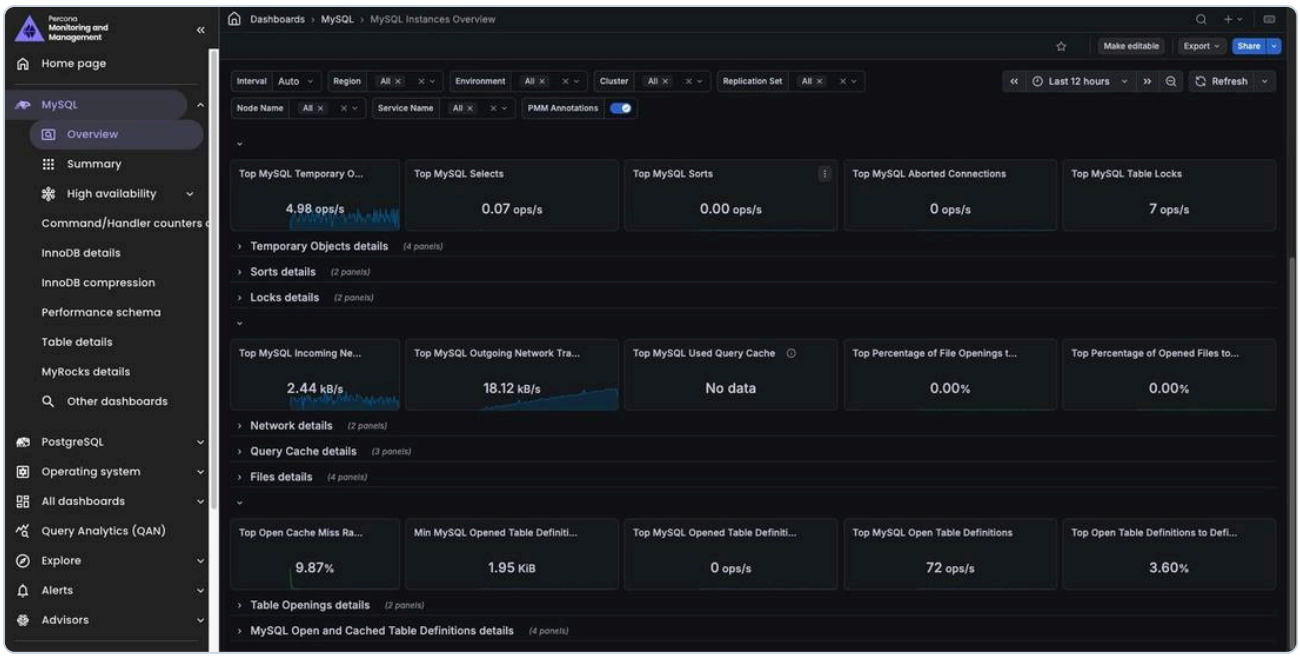


Figure 5.15 27 Oct 2026 · 01:47:11

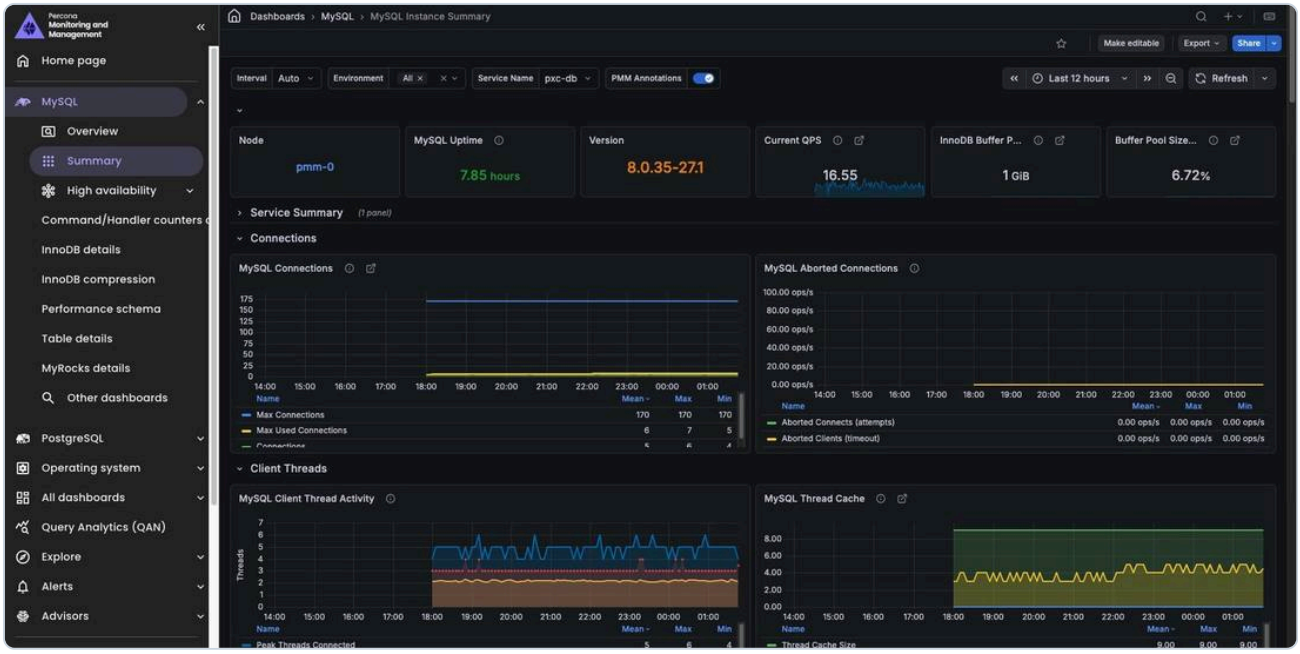


Figure 5.16 27 Oct 2026 · 01:47:19

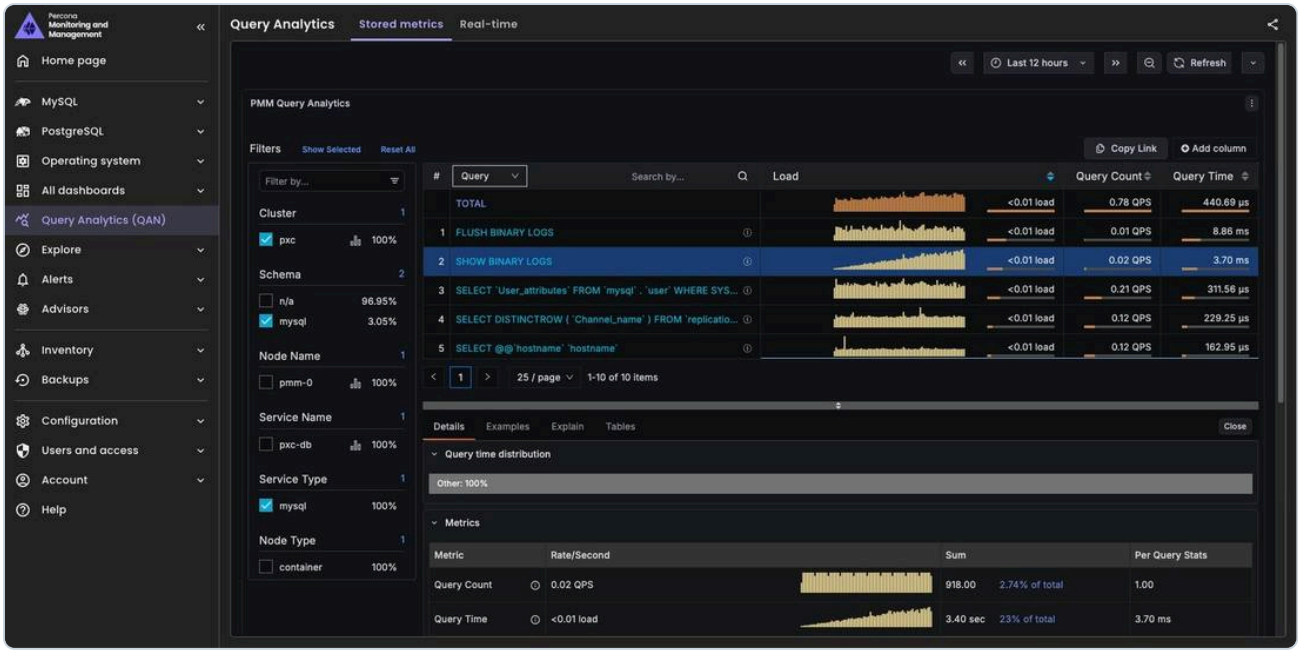


Figure 5.17 27 Oct 2026 · 01:48:25

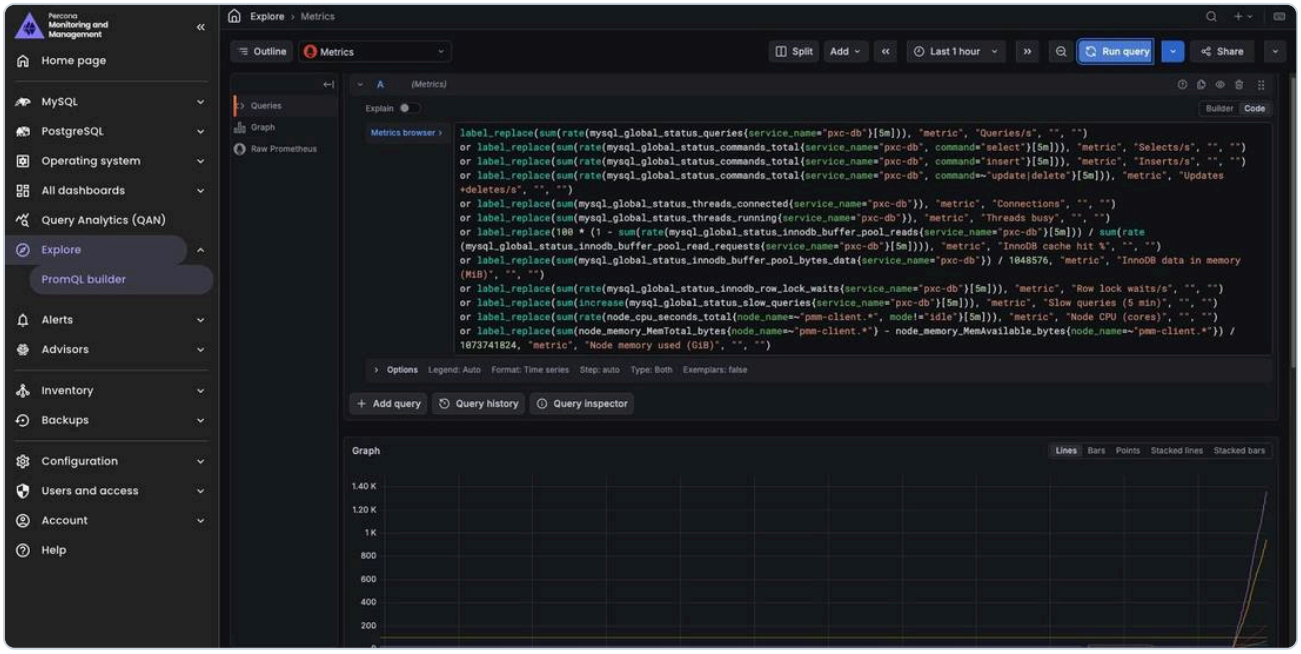


Figure 5.18 27 Oct 2026 · 01:55:29

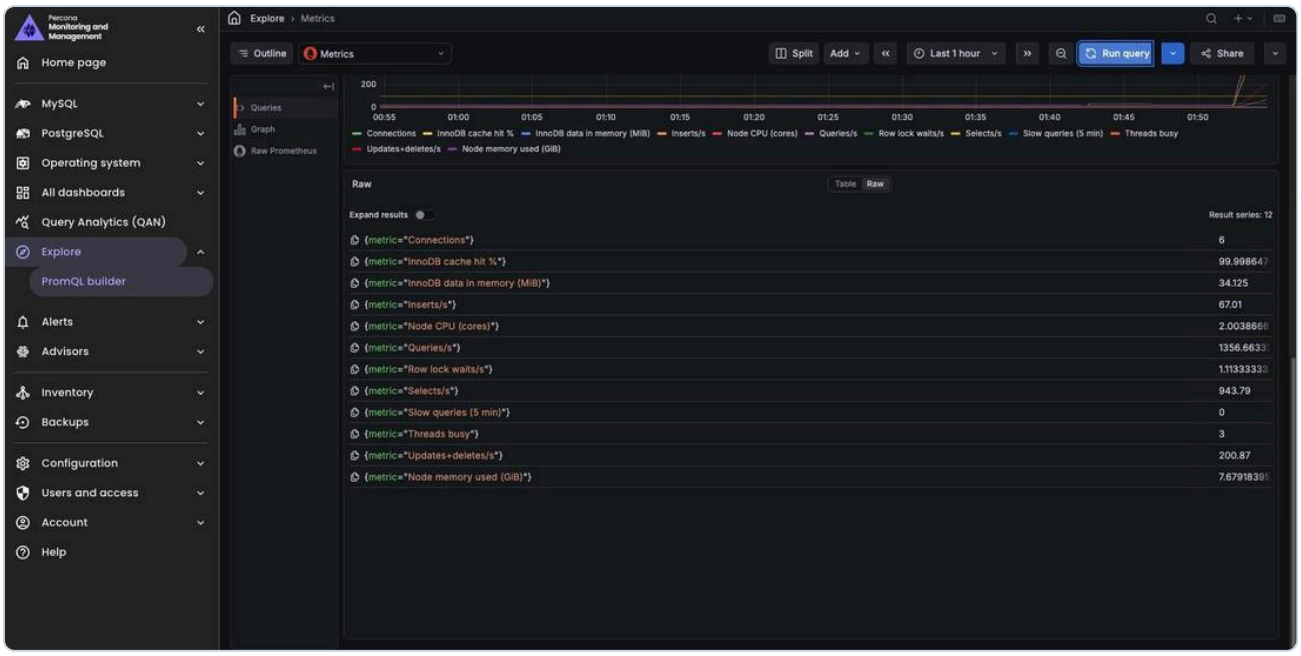


Figure 5.19 27 Oct 2026 · 01:55:41

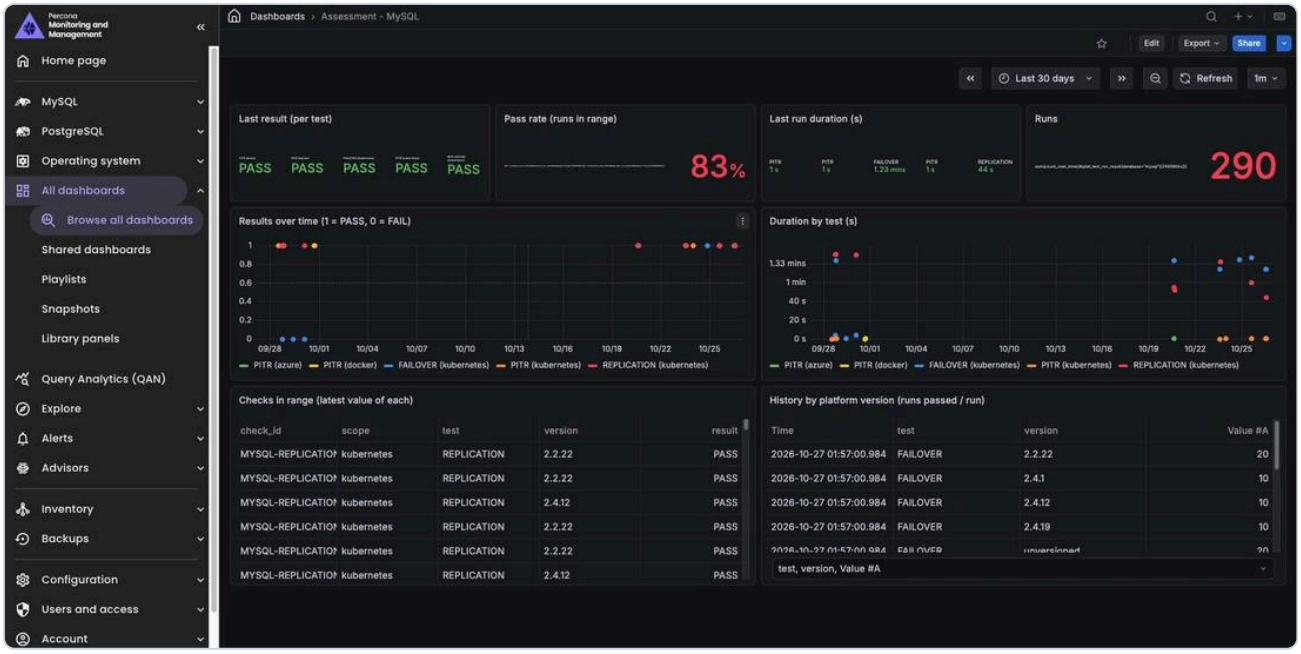


Figure 5.20 27 Oct 2026 · 01:57:19

5.15 More screenshots

Register | PMM | MySQL

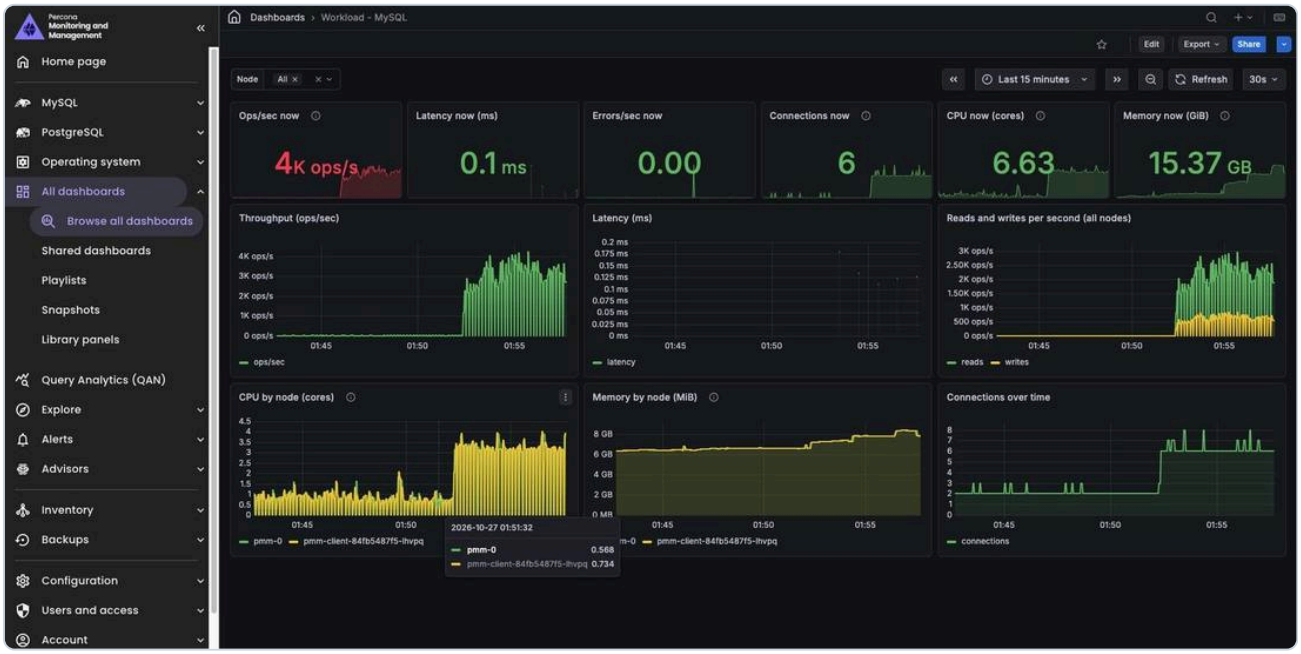


Figure 5.21 27 Oct 2026 · 01:57:51

5.16 Unregister MySQL from PMM

Unregister | PMM | MySQL | Completed

MySQL is removed from PMM; the database itself keeps running.

Started 27 Oct 2026, 01:58 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE

5.17 Remove PMM

Unregister | PMM | MySQL | Completed

PMM is removed from the cluster once no database is registered with it.

Started 27 Oct 2026, 01:59 · took 20s

COMMAND

```
SCOPE=kubernetes observability/pmm/deploy/kubernetes/destroy.sh
```

- ✔ PMM local web access stopped.
- ✔ PMM local web access stopped.
- ✔ PMM Client deployment removed
- ✔ PMM namespace absent
- ✔ PMM Helm release absent

5.18 Workload · MySQL

Workload | MySQL | Completed

A load generator runs against MySQL, so the monitoring shows real traffic. It stops by itself after a set time.

Started 27 Oct 2026, 01:59 · took 4s

COMMAND

```
bash db-platform/mysql/workload/stop.sh
```

5.19 Destroy MySQL

Destroy | MySQL | Completed

MySQL is removed and its storage released.

Started 27 Oct 2026, 02:00 · took 40s

COMMAND

```
SCOPE=kubernetes db-platform/mysql/deploy/kubernetes/destroy.sh
```

- ✔ kubectl available
- ✔ helm available
- ✔ Kubernetes API reachable
- ✔ Backup resources cleared
- ✔ All namespace Pods terminated
- ✔ Namespace mysql fully removed

06

Destroy local Kubernetes

Local Kubernetes

The local environment is shut down and removed, leaving the machine clean for the next run.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Local Kubernetes	—	—	3 screens · 1 logs	27 Oct 2026

6.1 Screenshots

What the console and dashboards showed at this point.

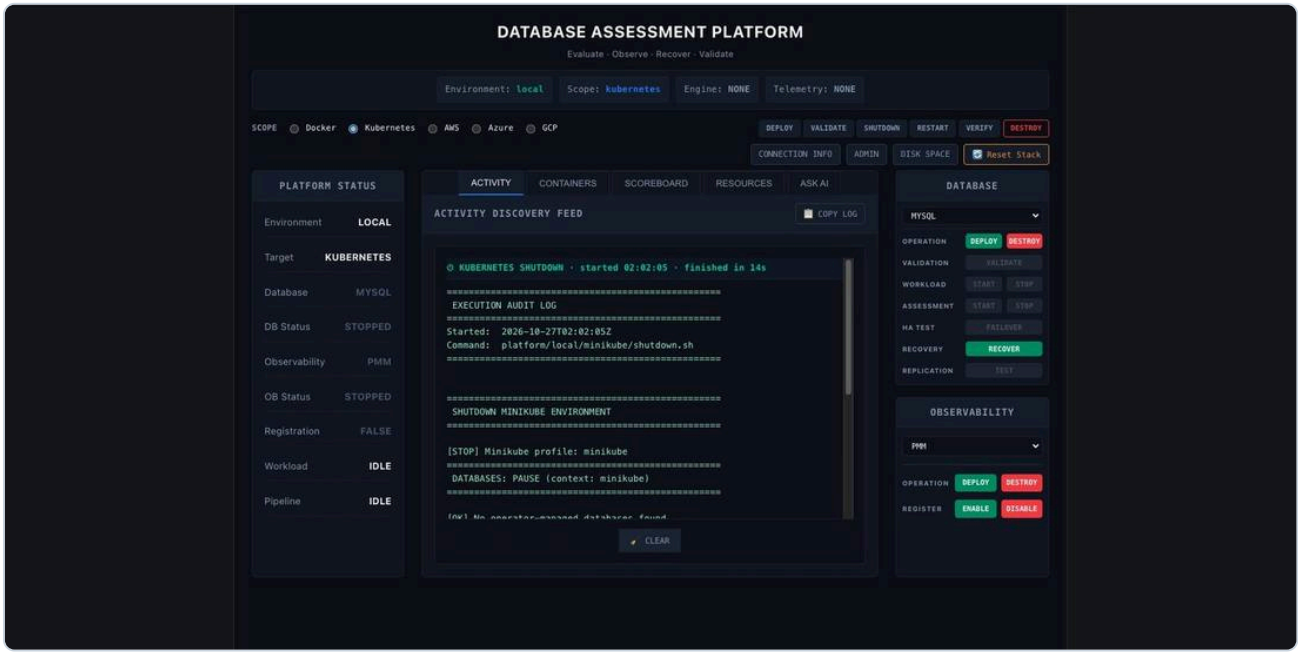


Figure 6.1 27 Oct 2026 · 02:03:17

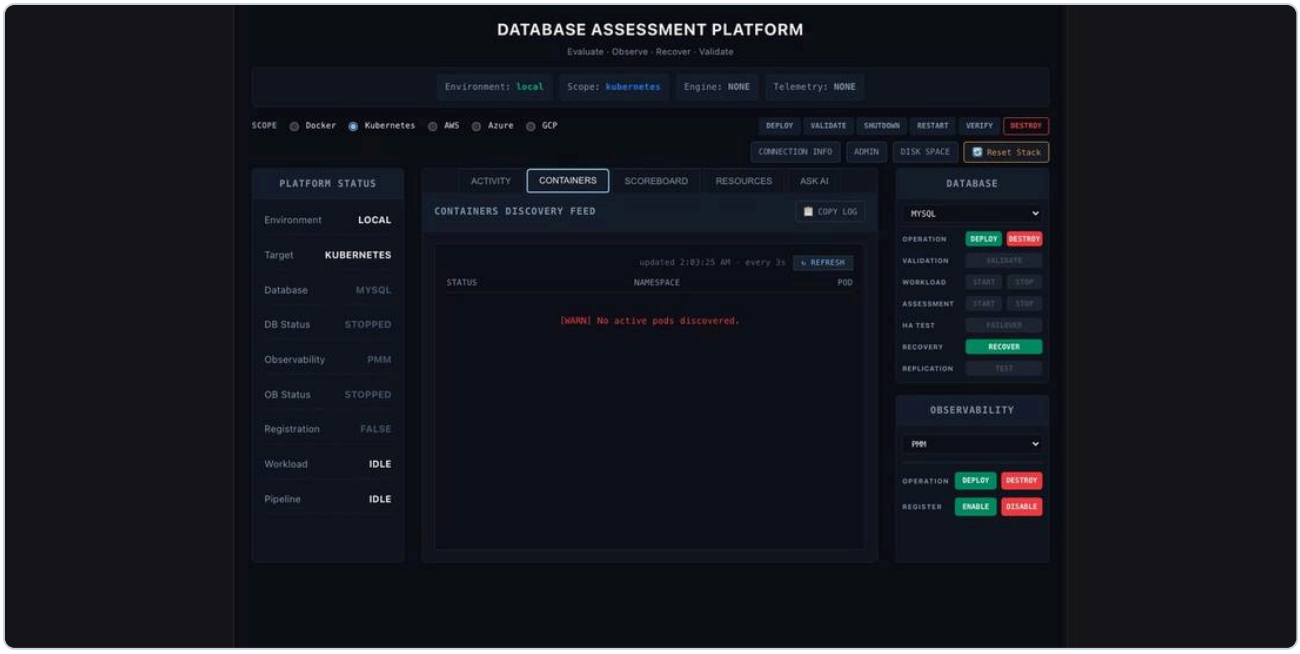


Figure 6.2 27 Oct 2026 · 02:03:27

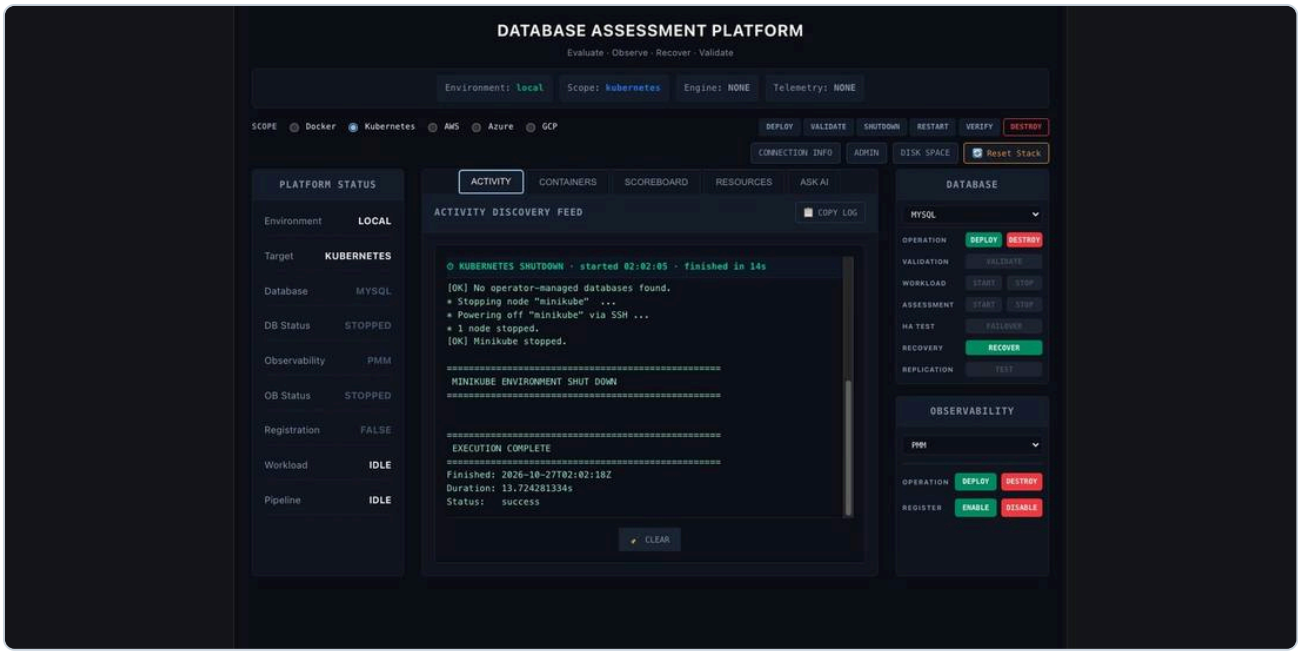


Figure 6.3 27 Oct 2026 · 02:03:40

6.2 Destroy

Destroy | **Completed**

The deployment is removed and its storage released.

Started 27 Oct 2026, 02:02 · took 14s

COMMAND

```
platform/local/minikube/shutdown.sh
```

- ✔ No operator-managed databases found.
- ✔ Minikube stopped.

What this shows

- The same RunShore workflow deploys, monitors, tests and destroys every database on local Kubernetes and on all three clouds.
- Every action leaves a log; each step above links to it.
- Recovery is tested, not assumed: failover, replication and point-in-time recovery runs are recorded with their results.
- Cloud environments are destroyed at the end of each chapter, and the destroy is verified.

RunShore · Database assessment and observability platform